Dual-View Native Time-Series Prompting for Few-Shot Time Series Classification

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Abstract

Few-shot time series classification remains challenging because extremely limited supervision is often insufficient to learn robust class-discriminative patterns from raw sequences. While large language models (LLMs) offer strong pretrained priors, directly applying them to time series is nontrivial: continuous temporal signals must first be converted into representations that are both compatible with frozen language models and stable under scarce labels. In this work, we present DuoTSP, a generative framework for few-shot time series classification based on dual-view native time-series prompting. Instead of relying on a single time-series-to-language pathway, DuoTSP encodes each sequence from two complementary views: a temporal branch captures local patch relations and long-range dynamics, while a visual branch transforms the sequence into a pseudo-image and leverages a pretrained vision transformer to model global morphology and cross-local structures. The resulting token-level representations are fused into soft time-series prompts and inserted into the input of a frozen LLM. To bridge the gap between time-series representations and language modeling, we further introduce a three-stage curriculum pretraining strategy that progressively improves dual-view representation learning, time-series-to-language generation transfer, and semantic alignment. Finally, for downstream few-shot classification, we formulate prediction as constrained generative label selection with dataset-specific class tokens, avoiding an additional classifier head and restricting decoding to a compact label subspace. This design yields a parameter-efficient and LLM-consistent adaptation mechanism for few-shot classification. Experiments on few-shot UCR benchmarks show that DuoTSP provides a strong and stable transfer paradigm for time series classification under limited supervision.

1 Introduction

Time series classification (TSC) is a fundamental problem in machine learning, with applications spanning healthcare monitoring, industrial diagnosis, sensor analytics, and scientific discovery Ismail Fawaz et al. [2019]. In many real-world scenarios, however, labeled time series are scarce: obtaining expert annotations can be expensive, class distributions are often imbalanced, and newly introduced categories may have only a handful of examples. These challenges make few-shot time series classification particularly difficult, as the model must generalize from very limited supervision while still capturing subtle temporal patterns that distinguish classes Tang et al. [2020]. Although recent deep architectures have substantially improved TSC performance, their effectiveness often degrades in the low-data regime, where robust inductive biases and transferable representations become especially important Pan et al. [2024a].

Existing few-shot TSC methods have mainly explored meta-learning, prototype- or shapelet-based reasoning, and spectral or contrastive representation learning Barreda et al. [2024], Tang et al. [2020],

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Yang et al. [2022]. In parallel, a growing line of work has attempted to exploit pretrained foundation models for time-series understanding, especially through large language models (LLMs), which offer strong sequence modeling priors and a unified generative interface Zhang et al. [2024], Jin et al. [2024], Cheng et al. [2024]. However, applying LLMs to time series is far from straightforward. Unlike text, time series are continuous numerical signals whose discriminative cues may appear simultaneously in local temporal dependencies, long-range dynamics, and global waveform morphology. Existing approaches often address only part of this mismatch. Some methods focus on a single temporal encoding pathway and project time-series features into the language space, while others discretize time series into textual prototypes or convert them into images for visual encoders Zhang et al. [2024], Jin et al. [2024], Cheng et al. [2024]. These strategies are useful, but they leave three key issues insufficiently resolved for few-shot classification: incomplete representation, unstable cross-modal transfer, and a decision mechanism that is not fully aligned with the generative nature of LLMs.

First, representation is often incomplete when time series are modeled from only one perspective. The class identity of a sequence may depend not only on local segments and temporal evolution, but also on holistic shape, texture-like patterns, and cross-local structural cues. Recent work increasingly suggests that visualized time-series signals can expose morphology-aware information that is complementary to sequence-space modeling Ni et al. [2025], Liu et al. [2025]. Under few-shot supervision, missing any of these signals can be especially harmful, because the model has little data to compensate for a weak inductive bias. Second, transferring time-series features into a frozen LLM is itself a difficult optimization problem. A small labeled support set is rarely sufficient to simultaneously learn task-specific classification and bridge the distribution gap between continuous time-series representations and the hidden space of a language model Chang et al. [2025], Cheng et al. [2024]. Third, even when time-series features are successfully injected into an LLM, many approaches still rely on a conventional classifier head for downstream prediction or on unconstrained open-vocabulary generation. This makes the final decision process only weakly aligned with the native autoregressive mechanism of the pretrained LLM. Recent evidence that prompt outputs can underestimate the time-series classification capability already contained in LLM representations further reinforces this concern Schumacher et al. [2026].

In this paper, we argue that effective few-shot TSC with frozen LLMs requires a complete transfer pipeline that jointly addresses these three challenges. Based on this view, we propose **DuoTSP**, short for **Du**al-view native time-series **T**ime-**S**eries **P**rompting, a generative framework that turns time series into native soft prompts for LLMs. Instead of mapping a sequence to language through a single pathway, DuoTSP builds token-level time-series prompts from two complementary views. A temporal branch models patch-level relations and long-range dependencies directly on the raw sequence, preserving the native sequential structure of time series. In parallel, a visual branch converts the sequence into a pseudo-image and uses a pretrained vision transformer to capture global morphology, texture-like patterns, and cross-local structures that are difficult to express through a purely temporal pathway. The temporal encoder is inspired by patchified time-series Transformers Nie et al. [2022]. The two branches are fused at the token level, producing soft time-series tokens that are inserted together with textual tokens into the input sequence of a frozen causal LLM. In this way, the model conditions generation on a richer and more complete time-series representation before any downstream supervision is applied.

To make this dual-view prompting mechanism transferable under limited labels, we further introduce a three-stage curriculum pretraining strategy. The first stage focuses on branch-faithful self-supervised representation learning, encouraging both branches and their fused representation to learn stable features while using branch dropout to reduce collapse toward a single dominant view. The second stage establishes the time-series-to-language generation pathway by training the model to answer textual questions conditioned on time-series soft prompts, with a projector warm-up strategy that stabilizes the transition from encoder space to LLM space. The third stage enhances semantic bridgeability through captioning, synthetic attribute supervision, and alignment objectives, encouraging the learned time-series representations to move closer to language-expressible semantic structures. These stages are not merely additional pretraining for scale; rather, they progressively address three different obstacles in transfer: representation learning, generative interfacing, and semantic alignment.

Finally, for downstream few-shot classification, we avoid introducing an extra classifier head and instead formulate prediction as constrained generative label selection. For each dataset, we dynamically add class-specific tokens to the vocabulary and only expose the corresponding embedding and output rows for learning, thereby confining task adaptation to a compact label subspace. During inference, decoding is explicitly restricted to the candidate class tokens and the end-of-sequence token, which turns open-vocabulary generation into a controlled label-space decision process. This design preserves consistency with the autoregressive objective of the frozen LLM, reduces unnecessary output freedom under scarce labels, and provides a parameter-efficient adaptation mechanism well suited to few-shot classification.

Our work makes three main contributions. First, we propose a dual-view native time-series prompting framework that jointly preserves temporal dependency patterns and morphology-level structures before conditioning a frozen LLM. Second, we develop a three-stage curriculum pretraining strategy that progressively bridges time-series representation learning, language generation transfer, and semantic alignment. Third, we cast few-shot time series classification as constrained generative prediction in a dataset-specific label space, avoiding an additional discriminative head and yielding a stable parameter-efficient adaptation paradigm. Extensive experiments on few-shot UCR classification benchmarks demonstrate the effectiveness of the proposed framework.

2 Related Work

Few-shot time series classification. Few-shot time series classification (TSC) has mainly been studied through meta-learning, metric learning, and prototype-based inductive biases. Narwariya et al. Narwariya et al. [2020] introduced a meta-learning framework that trains residual networks across heterogeneous UCR tasks for rapid adaptation under scarce supervision. Tang et al. Tang et al. [2020] proposed Dual Prototypical Shapelet Networks (DPSN), which combine representative class prototypes with discriminative local shapelets to improve both interpretability and data efficiency. Yang et al. Yang et al. [2022] further explored spectral relations and introduced the Spectral Propagation Graph Network (SPGN) to propagate class information through spectrum-wise comparisons. More recently, COSCO Barreda et al. [2024] improved few-shot multivariate TSC from the optimization perspective by combining sharpness-aware minimization with a prototypical loss. These methods demonstrate the importance of transferable priors and structured inductive biases, but they remain essentially discriminative pipelines and mostly operate within a single temporal representation space.

LLMs and foundation models for time series. Recent surveys show that foundation-model research for time series has rapidly expanded from adapting pretrained language models to building native temporal foundation models Jiang et al. [2024], Ye et al. [2024], Shi et al. [2025]. Representative adaptation-based approaches include TEST Sun et al. [2023], which aligns time-series embeddings with the LLM embedding space through text-prototype-aware contrastive learning; Time-LLM Jin et al. [2024], which reprograms time-series patches into a frozen LLM via text prototypes and prompt-as-prefix; and S^2IP-LLM Pan et al. [2024b], which aligns time-series embeddings with semantic anchors derived from pretrained word embeddings. LLM4TS Chang et al. [2025] further adopts a two-stage alignment-and-task fine-tuning strategy, while OpenTSLM Langer et al. [2025] moves toward native time-series language models with soft-prompt and cross-attention interfaces. A recent line of work has also begun to study few-shot classification with LLM-based architectures. The closest to our setting is LLMFew Chen et al. [2025], which uses a patch-wise temporal encoder, LoRA-tuned decoder, and a conventional classification head for few-shot multivariate TSC. Compared with this line, we focus on dual-view temporal evidence, explicit stage-wise alignment before few-shot adaptation, and generative label-token prediction rather than an external classifier. We are also motivated by recent ablation studies suggesting that naively attaching time-series inputs to pretrained LLMs is insufficient unless the temporal interface is meaningfully designed and aligned Tan et al. [2024].

Vision-based representations for time series. Beyond sequence-space modeling, a growing body of work represents time series as images so that waveform morphology, texture, periodicity, and holistic structural patterns can be captured by vision backbones. Recent surveys identify this as an increasingly important branch of time-series foundation modeling Ni et al. [2025]. In the context of multimodal reasoning, Liu et al. Liu et al. [2025] show that visualization can substantially improve LLM reasoning over time series. For classification, TiViT Roschmann et al. [2025] demonstrates that pretrained vision transformers applied to imaged time series can yield highly competitive representations and are complementary to time-series foundation models. These results suggest that visualized signals expose discriminative cues that are not fully captured by sequence-only modeling. However, existing visual approaches are typically used as alternatives to temporal encoders or as interfaces for reasoning, rather than being fused at the token level with a temporal branch for few-shot generative classification.

Generative classification and constrained decoding. Casting classification as generation has been widely explored in NLP, most notably in the text-to-text paradigm of T5 Raffel et al. [2020]. Rather than relying on an external discriminative head, this paradigm formulates prediction as conditional token generation, making the decision process more naturally aligned with the pretrained objective of language models. In parallel, constrained decoding has been studied as a general mechanism for restricting generation to valid outputs under lexical or structural constraints Hokamp and Liu [2017], Koo et al. [2024]. Inspired by these ideas, we formulate few-shot time series classification as generation of class-specific label tokens conditioned on textual instructions and time-series soft prompts. Instead of attaching a separate classifier, our model performs prediction in the native autoregressive manner of a frozen causal LLM, while restricting decoding to the candidate label space of the current episode further reduces irrelevant output freedom and improves decision stability under low-data uncertainty.

In summary, our method lies at the intersection of these four lines of work: few-shot TSC, LLM-based time-series adaptation, vision-based time-series representation learning, and generative classification with constrained decoding. It differs from prior work by jointly combining dual-view token-level encoding, stage-wise time-series-to-LLM alignment, and episode-wise constrained label-token generation in a unified framework for few-shot TSC.

3 Method

3.1 Problem Setup and Unified Time-Series-to-Language Formulation

We study few-shot time series classification with a frozen causal large language model (LLM). Given a time series $x \in \mathbb{R}^T$, we construct each training example as a unified sequence-to-sequence instance consisting of a textual prefix p^{pre}, an optional textual description p^{ts}, the time series itself, a textual suffix p^{post}, and a target answer y. The answer can be either a natural-language response (e.g., for question answering or captioning) or a dataset-specific class token (e.g., for classification).

Our key design principle is to treat time series as a native conditioning modality for language modeling, rather than serializing raw values into text. Concretely, the model converts the input series into a sequence of continuous soft tokens, projects them into the hidden space of the LLM, and inserts them into the token stream together with the textual prompt. The resulting mixed sequence is processed by the frozen LLM under the standard autoregressive objective.

Let $\mathrm{Tok}(\cdot)$ denote the tokenizer, $\mathrm{Emb}(\cdot)$ the input embedding layer of the LLM, and $P(\cdot)$ the projector from time-series features to the LLM hidden space. If Z_x denotes the time-series soft prompt extracted from x, the final input sequence to the LLM is

$$H = \left[\mathrm{Emb}(\mathrm{Tok}(p^{\text{pre}})); \mathrm{Emb}(\mathrm{Tok}(p^{\text{ts}})); P(Z_x); \mathrm{Emb}(\mathrm{Tok}(p^{\text{post}}))\right]. \tag{1}$$

During training, the answer tokens are appended to the end of this mixed sequence, and the causal language modeling loss is computed only on answer positions. This formulation provides a single interface for all stages of training and all downstream tasks. In particular, few-shot classification is not treated as a separate discriminative problem with an additional classifier head, but as a constrained answer generation problem within the same autoregressive framework. Figure 1 provides an overview of the proposed pipeline.

3.2 Dual-View Native Time-Series Prompting

As illustrated in Figure 1, our method consists of three tightly coupled components: dual-view native time-series prompting, curriculum bridging from time series to language, and constrained generative label prediction for few-shot classification. A central challenge in few-shot time series learning is that class-discriminative cues are not concentrated in a single representational view. Some cues arise from local temporal relations, long-range dependencies, phase shifts, and trend evolution, while others are more naturally expressed as global waveform shapes, texture-like patterns, or cross-local morphological structures. Relying on a single encoding pathway therefore risks discarding useful inductive bias precisely when supervision is scarce.

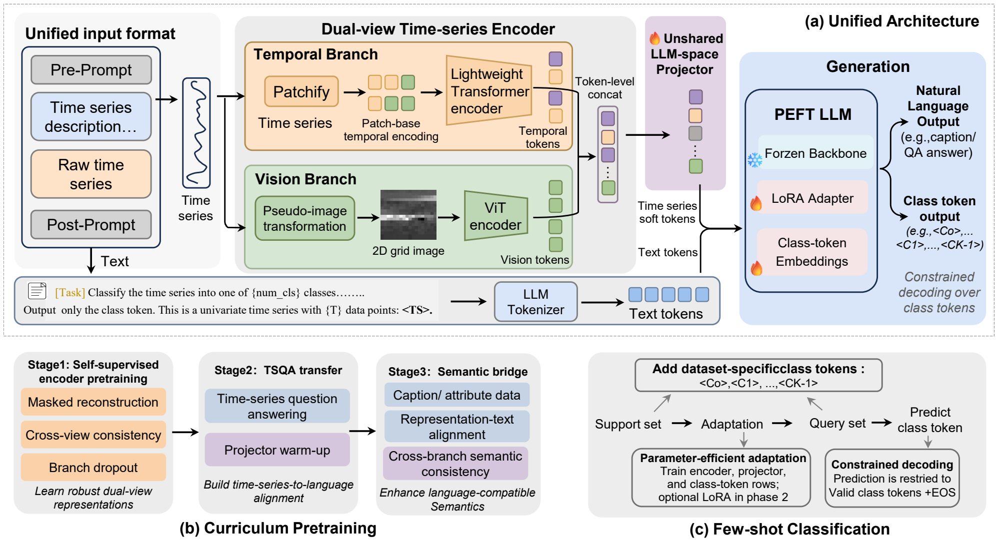

Figure 1: Overview of the proposed framework. Given a raw time series, we construct dual-view native time-series prompts using a temporal branch and a visual branch. The fused soft tokens are projected into the hidden space of a frozen LLM and combined with textual prompts for autoregressive generation. Training proceeds in three stages: dual-view self-supervised representation learning, generative transfer via time-series question answering, and semantic bridging with captioning and attribute supervision. For downstream few-shot classification, we introduce dataset-specific class tokens and perform constrained decoding in the label subspace.

To address this issue, we introduce a *dual-view native time-series prompting* mechanism that generates soft time-series tokens from two complementary branches under a unified interface.

Temporal branch. The temporal branch preserves the sequential nature of the signal. We divide the input series into overlapping patches and encode them with a Transformer-style time-series backbone, producing a sequence of temporally contextualized tokens,

$$Z^t = E_t(x), \tag{2}$$

where E_t denotes the temporal encoder. This branch is designed to capture local patch interactions as well as longer-range temporal dependencies.

Visual branch. The visual branch captures morphology-level patterns that may be difficult to express through a purely sequential pathway. We first transform the 1D series into a pseudo-image $I(x)$, and then feed it into a pretrained vision transformer to obtain visual tokens,

$$Z^v = E_v(I(x)), \tag{3}$$

where E_v denotes the visual encoder. This branch emphasizes global waveform geometry, texture-like regularities, and structural patterns across distant parts of the sequence.

After branch-specific projection into a shared encoder space, we fuse the two token streams at the token level to form the final soft time-series prompt,

$$Z_x = F(Z^t, Z^v). \tag{4}$$

In our main instantiation, $F(\cdot)$ is implemented by direct token concatenation, which preserves branch-specific information without prematurely collapsing the two views into a single vector. We further maintain pooled representations $\bar{z}^t$, $\bar{z}^v$, and $\bar{z}^f$ for branch-level and fused-level auxiliary objectives.

This design differs from standard "encoder + projector + LLM" pipelines in an important way. We do not map time series to language through a single pathway. Instead, we construct native time-series prompts from dual-view inductive biases, so that both temporal dependency patterns and morphology-level structures are preserved before language conditioning. It also differs from pure image-based approaches, where the visual view replaces time-series modeling. In our framework, the visual pathway complements rather than substitutes temporal encoding.

3.3 Curriculum Bridging from Time Series to Language

Simply attaching a time-series encoder to a frozen LLM is often unstable, especially in the few-shot regime. The difficulty is not only representational, but also architectural and semantic: the model must first learn robust dual-view features, then learn how these features should condition language generation, and finally organize the internal representation into a more language-compatible semantic space. We therefore adopt a three-stage curriculum that progressively bridges time-series representation and language modeling.

3.3.1 Stage I: Dual-view Self-Supervised Representation Learning

The first stage trains the dual-branch encoder before interfacing it with the LLM. The goal is to obtain stable and complementary representations from both branches while preventing the fused representation from collapsing onto a single dominant view.

We use a combination of masked reconstruction and multi-view self-supervision. Specifically, the temporal branch is trained with a masked reconstruction objective L_{rec}, which encourages recovery of masked temporal content from partially observed inputs. In addition, we apply branch-specific and fused self-supervised regularization across augmented views of the same series, yielding losses L_{ssl}^{t}, L_{ssl}^{v}, and L_{ssl}^{f} for the temporal, visual, and fused representations, respectively. The overall objective of Stage I is

$$L_{\text{I}} = \lambda_{\text{rec}} L_{\text{rec}} + \lambda_t L_{\text{ssl}}^{t} + \lambda_v L_{\text{ssl}}^{v} + \lambda_f L_{\text{ssl}}^{f}. \tag{5}$$

To reduce the risk that training overfits to a single branch, we further introduce *branch dropout*: during pretraining, each forward pass randomly uses the temporal branch only, the visual branch only, or both branches together. This simple strategy encourages each branch to remain individually informative and leads to a more balanced fused representation.

3.3.2 Stage II: Generative Transfer via Time-Series Question Answering

The second stage establishes the functional path from time-series soft prompts to language outputs. We initialize the dual-branch encoder from Stage I, attach the projector and the frozen LLM, and train the model on time-series question answering data. Given the mixed prompt sequence defined in Section 3.1, the model is optimized with the standard causal language modeling objective,

$$L_{\text{II}} = L_{\text{LM}}^{\text{TSQA}}. \tag{6}$$

This stage is not merely another pretraining phase. It is the first stage that explicitly trains the model to generate text conditioned on time-series soft prompts, thereby converting a time-series encoder into a time-series-conditioned language model. To mitigate optimization instability when moving from the encoder feature space to the LLM hidden space, we first warm up the projector while keeping the encoder fixed, and then jointly optimize the encoder and projector. This warm-start procedure stabilizes cross-space interfacing and reduces alignment shock at the beginning of language-conditioned training.

3.3.3 Stage III: Semantic Bridging with Captioning and Attribute Supervision

Although Stage II teaches the model to answer language queries from time-series prompts, the learned internal representation may still be overly task-specific. For downstream few-shot classification, we want the representation to be more explicitly organized around semantically expressible properties of time series.

We therefore perform a third stage of training on a mixture of time-series question answering, captioning, and automatically constructed attribute-oriented supervision. The synthetic attribute data describes interpretable time-series properties such as trend, periodicity, spikes, abrupt changes, volatility, or stationarity in textual form. These tasks encourage the model to connect latent time-series structure with language-compatible semantics.

In addition to the standard language modeling loss, Stage III introduces two auxiliary objectives. First, we apply a representation-text alignment loss L_{align} that pulls time-series representations closer to their matched textual semantics in a shared alignment space. Second, we use a branch-consistency loss L_{cons} that encourages the temporal and visual branches to produce semantically consistent representations for the same input. The Stage III objective is

$$L_{\text{III}} = L_{\text{LM}}^{\text{sem}} + \lambda_{\text{align}} L_{\text{align}} + \lambda_{\text{cons}} L_{\text{cons}}. \tag{7}$$

The three stages are thus not redundant training tricks. They progressively address three distinct obstacles in time-series-to-language transfer: representation learning, generative interfacing, and semantic bridging.

3.4 Few-Shot Classification as Constrained Generative Label Selection

Most existing methods that combine time-series features with LLMs still revert to a conventional classifier head for final decision making. While effective in some settings, this design breaks the generative interface of the frozen LLM and introduces a separate discriminative output space that must be learned from very limited supervision.

Instead, we reformulate few-shot classification as a constrained generative label prediction problem. For each dataset D with K classes, we dynamically introduce a set of class tokens

$$\mathcal{C}_D = \{\langle c_0 \rangle, \langle c_1 \rangle, \ldots, \langle c_{K-1} \rangle\}. \tag{8}$$

The input prompt ends with a task-specific classification suffix (e.g., "Class:"), and the target answer is the corresponding class token. Therefore, downstream classification remains a next-token prediction problem under the same autoregressive objective used throughout training.

To keep adaptation compact and stable in the few-shot regime, we do not open the full vocabulary for task-specific learning. Instead, only the embedding vectors and output-layer rows corresponding to the newly introduced class tokens are made trainable, while the rest of the vocabulary remains fixed. This creates a small dataset-specific label subspace in which task adaptation is concentrated.

At inference time, we further apply *constrained decoding* by restricting the candidate output set to $\mathcal{C}_D \cup \{\texttt{eos}\}$. The prediction is therefore

$$\hat{y} = \arg \max_{c \in \mathcal{C}_D \cup \{\texttt{eos}\}} p(c \mid H). \tag{9}$$

This procedure preserves the generative interface of the LLM while eliminating the unnecessary freedom of open-vocabulary generation. As a result, few-shot adaptation becomes both more parameter-efficient and more decision-stable.

3.5 Downstream Adaptation

For downstream few-shot adaptation, we use a two-phase training schedule. In the first phase, we warm up the task-critical components, namely the time-series encoder, the projector, and the dataset-specific class-token parameters. In the second phase, we optionally enable lightweight language-side adaptation (e.g., LoRA) and jointly tune the trainable components. This staged adaptation avoids introducing excessive optimization freedom at the beginning of few-shot learning, where the number of labeled examples is extremely limited.

Overall, our method is built around a simple but crucial claim: few-shot time series classification with frozen LLMs becomes effective only when representation completeness, cross-modal transfer, and label-space decision are co-designed as a single pipeline. Dual-view native prompting allows the model to see more complete evidence, curriculum bridging allows the LLM to consume time-series information more reliably, and constrained generative label selection allows the final decision to remain stable under scarce supervision.

4 Experiments

4.1 Experimental Setup

We evaluate DuoTSP on the UCRArchive 2018 benchmark under a fixed full-label-space few-shot classification protocol. For each dataset, we keep its native class set instead of sampling episode-specific class subsets, draw $K \in \{1, 2, 5, 10\}$ support examples per class from the training split, and

Table 1: Few-shot classification accuracy (%) on 128 shared UCR datasets. We report the mean over 5 runs. Best is bold and second-best is underlined.

Model	1-shot	2-shot	5-shot	10-shot	Avg	AvgRank
ResNet	47.79	53.98	61.27	66.24	57.32	4.89
TapNet	45.40	50.79	59.47	64.93	55.15	5.45
COSCO-ResNet	<u>47.88</u>	<u>54.95</u>	<u>63.55</u>	<u>67.18</u>	<u>58.39</u>	<u>4.70</u>
DLinear	44.21	48.01	52.30	55.43	49.99	7.24
PatchTST	42.39	48.43	56.94	62.72	52.62	6.54
TimesNet	45.16	49.95	57.86	62.79	53.94	5.90
Autoformer	36.78	39.45	43.20	46.04	41.37	9.47
Crossformer	44.31	49.45	57.20	61.94	53.23	6.04
FEDformer	37.71	41.71	47.30	51.50	44.56	8.68
Informer	46.42	51.14	58.35	63.13	54.76	5.59
GPT4TS	35.61	38.30	42.94	46.88	40.93	9.26
Ours	**48.31**	**55.55**	**64.14**	**69.16**	**59.29**	**4.24**

evaluate on the corresponding test split. All compared methods are reported on the 128 datasets that are jointly covered by every baseline in the final benchmark. Due to the main-paper page budget, we defer the complete per-dataset results and implementation details to Appendix A and Appendix B.

We compare DuoTSP against three groups of baselines: classical few-shot or supervised TSC architectures (ResNet, TapNet, and COSCO-ResNet), Transformer-style or decomposition-based time-series backbones (PatchTST, DLinear, TimesNet, Autoformer, Crossformer, FEDformer, and Informer), and a foundation-style baseline (GPT4TS). Following the migrated batch results used for the paper, all methods are evaluated over five runs and we report mean accuracy in the main paper; standard deviations are provided in the appendix tables.

Our main instantiation of DuoTSP uses `meta-llama/Llama-3.2-1B` as the frozen causal LLM and `facebook/dinov2-base` as the visual backbone. The temporal branch uses a patch length of 16, stride 8, and hidden size 128. The visual branch uses the `tivit_sqrt_overlap` pseudo-image construction. During few-shot adaptation, we adopt a two-phase schedule with 5 warm-up epochs followed by 55 joint optimization epochs, together with LoRA rank 16 and LoRA alpha 32. Additional hyperparameters are summarized in Appendix B.

4.2 Main Results

Table 1 reports the benchmark-level mean accuracy and average rank across the 128 shared UCR datasets. DuoTSP achieves the best average accuracy at every shot setting, reaching 48.31, 55.55, 64.14, and 69.16 for 1-, 2-, 5-, and 10-shot evaluation, respectively. The strongest non-DuoTSP baseline is COSCO-ResNet, which obtains 47.88, 54.95, 63.55, and 67.18. This corresponds to absolute gains of +0.43, +0.60, +0.59, and +1.98 points for DuoTSP, with the largest margin appearing at 10-shot. DuoTSP also attains the lowest average rank (4.24), improving over COSCO-ResNet (4.70) and ResNet (4.89), which indicates stronger overall robustness across the benchmark rather than gains on a small number of favorable datasets.

The head-to-head Win/Tie/Loss comparison in Table 2 further clarifies where the gains come from. DuoTSP wins on a clear majority of datasets against PatchTST, TimesNet, and GPT4TS across all shot settings. The comparison with COSCO-ResNet and ResNet is much closer on a per-dataset basis, but DuoTSP still yields the best aggregate accuracy and the lowest overall rank. This pattern is consistent with our claim that dual-view prompting, curriculum bridging, and constrained generative prediction improve benchmark-level stability rather than depending on isolated outlier datasets.

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Table 2: Win/Tie/Loss of Ours against selected baselines on 128 shared UCR datasets.

Baseline	1-shot	2-shot	5-shot	10-shot
COSCO-ResNet	68 / 0 / 60	68 / 1 / 59	58 / 0 / 70	64 / 0 / 64
PatchTST	97 / 0 / 31	97 / 0 / 31	99 / 0 / 29	94 / 2 / 32
TimesNet	88 / 0 / 40	92 / 0 / 36	86 / 0 / 42	93 / 1 / 34
GPT4TS	106 / 2 / 20	110 / 0 / 18	114 / 1 / 13	121 / 1 / 6
ResNet	64 / 1 / 63	66 / 1 / 61	67 / 0 / 61	69 / 0 / 59
TapNet	75 / 0 / 53	75 / 3 / 50	76 / 1 / 51	72 / 3 / 53

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A Full UCR Benchmark Results

This appendix reports the complete per-dataset few-shot benchmark results for all 128 shared UCR datasets. Each cell is shown as mean ± standard deviation over five runs, and the summary rows at the end of each table report average accuracy, average rank, and the number of datasets attaining the best mean accuracy (counting ties).

Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
ACSF1	**53.80 ± 4.97**	42.00 ± 4.95	48.40 ± 7.30	24.00 ± 4.24	35.60 ± 3.21	27.67 ± 7.02	24.33 ± 9.71	28.33 ± 2.52	27.80 ± 7.09	31.67 ± 9.81	18.60 ± 4.51	44.00 ± 5.20
Adiac	33.81 ± 1.23	12.84 ± 3.09	33.81 ± 1.23	8.85 ± 2.36	11.97 ± 1.69	21.40 ± 4.24	34.53 ± 1.77	8.10 ± 1.92	40.82 ± 3.61	24.21 ± 3.24	3.02 ± 1.05	**45.17 ± 5.28**
AllGestureWiimoteX	25.71 ± 4.01	**26.86 ± 2.52**	24.43 ± 2.55	15.57 ± 2.95	21.86 ± 4.59	20.71 ± 2.87	11.33 ± 1.19	16.76 ± 2.52	13.43 ± 1.74	18.90 ± 1.08	21.86 ± 4.59	25.11 ± 4.65
AllGestureWiimoteY	**26.37 ± 2.89**	26.17 ± 2.88	22.71 ± 4.62	17.49 ± 2.02	25.17 ± 5.50	21.81 ± 2.78	12.29 ± 1.27	20.29 ± 1.98	13.06 ± 1.34	21.33 ± 2.38	25.17 ± 5.50	25.54 ± 4.42
AllGestureWiimoteZ	**27.11 ± 1.95**	22.66 ± 4.13	25.49 ± 2.57	18.26 ± 2.01	19.71 ± 2.87	19.24 ± 1.57	13.29 ± 0.43	18.76 ± 2.79	14.14 ± 1.72	19.71 ± 1.79	19.71 ± 2.87	22.26 ± 3.96
ArrowHead	47.66 ± 8.94	46.63 ± 9.45	48.00 ± 7.74	52.00 ± 6.38	49.71 ± 1.40	47.62 ± 9.46	38.29 ± 4.00	50.86 ± 1.71	35.89 ± 2.60	50.86 ± 6.05	41.49 ± 12.34	**52.57 ± 3.43**
BME	49.60 ± 2.85	50.80 ± 2.13	49.73 ± 3.42	56.67 ± 3.43	62.93 ± 5.35	55.33 ± 7.86	51.56 ± 3.15	61.33 ± 4.06	55.87 ± 7.71	58.89 ± 6.19	35.73 ± 3.22	**66.40 ± 3.88**
Beef	36.00 ± 6.41	31.33 ± 5.06	27.33 ± 7.96	38.67 ± 9.89	32.00 ± 7.67	40.00 ± 8.82	34.44 ± 6.94	**41.11 ± 1.92**	30.67 ± 6.83	40.00 ± 6.67	31.33 ± 9.31	34.00 ± 2.79
BeetleFly	55.00 ± 24.24	60.00 ± 25.00	61.00 ± 24.85	67.00 ± 4.47	61.00 ± 9.62	60.00 ± 13.23	55.00 ± 5.00	61.67 ± 11.55	48.00 ± 2.74	60.00 ± 13.23	**70.00 ± 11.18**	**70.00 ± 11.73**
BirdChicken	62.00 ± 8.37	**71.00 ± 9.62**	70.00 ± 17.68	56.00 ± 4.18	56.00 ± 4.18	51.67 ± 2.89	51.67 ± 12.58	53.33 ± 5.77	55.00 ± 5.00	51.67 ± 7.64	59.00 ± 9.62	68.00 ± 14.83
CBF	81.67 ± 6.47	**87.62 ± 2.73**	71.71 ± 7.91	54.02 ± 3.44	66.38 ± 7.40	55.85 ± 2.16	33.63 ± 1.20	58.00 ± 5.61	36.89 ± 2.37	54.78 ± 1.68	35.20 ± 3.75	72.48 ± 4.85
Car	37.00 ± 4.62	39.00 ± 4.80	41.67 ± 9.28	38.67 ± 11.63	27.33 ± 9.90	42.22 ± 5.09	24.44 ± 3.47	43.89 ± 5.85	27.00 ± 4.15	40.00 ± 7.26	30.00 ± 9.65	**55.67 ± 8.94**
Chinatown	66.53 ± 19.14	63.73 ± 26.98	62.51 ± 27.20	67.41 ± 23.86	67.35 ± 11.32	77.07 ± 16.93	81.83 ± 2.19	77.94 ± 9.20	77.61 ± 8.88	75.41 ± 7.92	56.68 ± 20.52	**93.59 ± 3.92**
ChlorineConcentration	34.61 ± 2.26	35.41 ± 4.36	34.04 ± 2.01	33.26 ± 2.96	35.54 ± 3.20	33.64 ± 3.96	32.92 ± 4.26	33.42 ± 5.09	33.78 ± 2.74	34.11 ± 3.50	**39.93 ± 15.64**	32.61 ± 3.30
CinCECGTorso	37.20 ± 9.73	36.03 ± 6.56	36.51 ± 8.82	40.06 ± 3.86	35.72 ± 7.09	44.86 ± 6.03	35.36 ± 5.77	**50.53 ± 4.66**	34.97 ± 4.68	45.68 ± 5.50	42.49 ± 6.76	48.74 ± 6.47
Coffee	78.57 ± 9.78	**92.86 ± 8.38**	84.29 ± 13.02	72.86 ± 14.42	74.29 ± 18.80	63.10 ± 10.91	65.48 ± 7.43	71.43 ± 16.37	76.43 ± 4.07	64.29 ± 7.14	51.43 ± 4.07	75.71 ± 13.69
Computers	57.52 ± 7.64	**60.40 ± 6.13**	59.12 ± 4.47	47.12 ± 5.87	52.64 ± 10.26	50.93 ± 1.29	48.80 ± 0.69	51.87 ± 2.57	48.96 ± 2.79	52.27 ± 3.23	45.20 ± 5.79	56.40 ± 5.77
CricketX	26.15 ± 3.29	26.10 ± 3.67	20.56 ± 4.67	17.28 ± 1.96	18.15 ± 4.50	20.34 ± 0.39	8.89 ± 0.82	17.52 ± 1.26	11.38 ± 2.35	20.68 ± 0.74	7.28 ± 1.05	**26.26 ± 5.29**
CricketY	**26.97 ± 3.03**	24.46 ± 4.25	22.72 ± 1.74	18.56 ± 4.13	17.90 ± 3.61	22.82 ± 1.56	10.34 ± 1.89	20.94 ± 3.72	11.23 ± 0.80	21.20 ± 3.98	9.54 ± 2.33	23.33 ± 5.19
CricketZ	**32.51 ± 2.32**	28.72 ± 2.15	26.56 ± 4.19	19.64 ± 2.73	22.26 ± 3.91	20.34 ± 4.11	11.03 ± 1.54	20.00 ± 2.31	12.26 ± 1.25	20.34 ± 2.57	7.64 ± 1.05	29.28 ± 4.82
Crop	25.10 ± 4.94	28.14 ± 2.90	25.10 ± 4.94	10.23 ± 2.66	17.27 ± 1.43	27.73 ± 2.89	24.10 ± 4.30	26.19 ± 2.96	27.27 ± 3.58	30.31 ± 5.00	3.80 ± 0.58	**36.48 ± 3.38**
DiatomSizeReduction	79.67 ± 5.51	55.36 ± 19.14	84.12 ± 4.82	**89.22 ± 8.91**	49.61 ± 24.86	75.71 ± 7.74	55.01 ± 11.77	56.43 ± 20.40	50.07 ± 12.66	76.47 ± 11.70	41.44 ± 13.25	85.82 ± 8.55
DistalPhalanxOutlineAgeGroup	43.02 ± 17.16	43.02 ± 18.34	43.60 ± 14.92	42.73 ± 19.01	46.76 ± 16.49	45.80 ± 19.81	46.28 ± 21.27	**56.59 ± 9.36**	50.94 ± 19.66	46.28 ± 20.78	35.68 ± 14.61	42.01 ± 16.44
DistalPhalanxOutlineCorrect	62.10 ± 1.32	57.32 ± 9.22	**63.04 ± 2.44**	57.54 ± 11.13	53.55 ± 11.63	53.99 ± 14.78	53.26 ± 12.27	53.62 ± 11.97	55.14 ± 10.52	53.26 ± 12.97	56.74 ± 7.30	60.58 ± 2.98
DistalPhalanxTW	51.22 ± 5.15	44.03 ± 11.70	53.38 ± 4.02	43.74 ± 5.17	51.22 ± 6.58	45.32 ± 9.89	42.69 ± 4.40	38.13 ± 7.51	48.20 ± 5.84	47.48 ± 5.62	18.56 ± 14.51	**54.53 ± 1.72**
DodgerLoopDay	32.50 ± 7.34	25.00 ± 4.05	29.75 ± 7.68	**41.50 ± 2.05**	26.00 ± 6.58	40.42 ± 4.73	20.42 ± 1.91	32.92 ± 3.15	21.50 ± 4.45	40.42 ± 7.94	26.00 ± 6.58	34.50 ± 4.38
DodgerLoopGame	52.17 ± 6.30	55.07 ± 17.38	49.13 ± 8.90	**60.87 ± 12.35**	59.86 ± 22.05	51.45 ± 9.96	51.21 ± 5.14	56.52 ± 10.82	50.43 ± 2.44	51.45 ± 9.59	59.86 ± 22.05	56.96 ± 11.69
DodgerLoopWeekend	84.49 ± 8.37	74.78 ± 12.27	83.19 ± 10.24	93.62 ± 3.81	**96.23 ± 1.19**	95.65 ± 1.26	63.77 ± 4.41	94.44 ± 0.42	68.41 ± 4.57	93.72 ± 5.86	**96.23 ± 1.19**	85.36 ± 4.30
ECG200	63.00 ± 12.61	**72.00 ± 6.75**	66.20 ± 8.32	71.80 ± 4.44	61.60 ± 12.82	61.33 ± 10.97	65.00 ± 11.79	63.00 ± 20.66	64.20 ± 14.29	68.67 ± 7.09	51.40 ± 12.93	60.60 ± 6.91
ECG5000	54.22 ± 21.00	50.05 ± 20.59	61.53 ± 13.66	62.80 ± 19.19	53.35 ± 20.14	**69.07 ± 12.52**	54.44 ± 9.79	57.20 ± 16.09	62.35 ± 12.99	68.84 ± 13.99	19.70 ± 25.30	67.18 ± 10.57
ECGFiveDays	58.65 ± 7.69	57.24 ± 6.51	54.80 ± 13.00	57.63 ± 2.63	**62.21 ± 8.35**	57.99 ± 4.84	54.86 ± 3.17	55.87 ± 1.56	54.08 ± 6.73	58.46 ± 5.08	51.54 ± 6.38	61.77 ± 6.61
EOGHorizontalSignal	20.11 ± 2.01	22.38 ± 1.86	22.60 ± 2.77	22.43 ± 4.01	21.22 ± 3.15	20.63 ± 4.63	15.29 ± 2.63	23.57 ± 5.45	10.39 ± 2.26	22.01 ± 4.37	27.24 ± 2.99	**29.06 ± 4.16**
EOGVerticalSignal	11.27 ± 3.67	14.09 ± 3.23	13.26 ± 3.09	24.20 ± 3.27	23.76 ± 3.73	23.39 ± 2.79	11.23 ± 0.70	24.86 ± 4.17	11.27 ± 3.83	23.11 ± 4.07	19.06 ± 6.08	**27.35 ± 1.90**
Earthquakes	50.07 ± 8.55	51.08 ± 6.33	50.94 ± 6.46	49.35 ± 5.05	45.61 ± 20.94	48.68 ± 19.16	49.40 ± 1.10	49.40 ± 14.03	47.05 ± 5.90	47.00 ± 24.65	46.04 ± 13.71	**55.54 ± 11.57**
ElectricDevices	28.83 ± 9.01	32.82 ± 12.31	28.35 ± 12.24	17.59 ± 1.67	27.21 ± 7.38	21.93 ± 4.54	17.11 ± 1.87	15.12 ± 2.81	17.04 ± 2.05	15.61 ± 7.10	17.06 ± 3.10	**33.14 ± 10.96**
EthanolLevel	24.44 ± 1.30	25.40 ± 0.85	25.28 ± 1.57	24.92 ± 1.50	23.80 ± 1.65	26.13 ± 0.58	26.53 ± 1.55	26.67 ± 1.29	25.32 ± 1.50	25.80 ± 1.39	24.76 ± 1.40	**26.68 ± 1.86**
FaceAll	**42.17 ± 4.79**	32.36 ± 9.47	**42.17 ± 4.79**	28.66 ± 9.09	19.36 ± 3.07	29.11 ± 4.19	33.12 ± 5.46	25.94 ± 1.75	34.05 ± 3.40	32.92 ± 2.10	8.64 ± 3.46	32.84 ± 6.86
FaceFour	68.64 ± 10.56	47.50 ± 11.12	56.36 ± 4.30	64.32 ± 4.73	54.09 ± 8.34	70.08 ± 4.30	52.65 ± 3.99	67.05 ± 7.10	51.36 ± 4.28	**73.11 ± 3.99**	42.95 ± 6.60	54.77 ± 8.25
FacesUCR	**54.83 ± 5.21**	37.63 ± 5.89	43.65 ± 3.88	35.69 ± 7.47	20.81 ± 1.95	36.24 ± 3.26	41.98 ± 2.95	40.05 ± 2.80	41.87 ± 3.09	42.63 ± 2.58	16.92 ± 7.54	50.37 ± 2.98
FiftyWords	13.27 ± 1.89	12.79 ± 3.45	13.27 ± 1.89	32.70 ± 4.33	19.60 ± 4.90	35.90 ± 3.31	3.44 ± 1.59	**38.97 ± 6.44**	11.30 ± 2.86	36.70 ± 4.41	3.74 ± 2.40	36.48 ± 2.66
Fish	**49.03 ± 5.96**	21.26 ± 3.60	45.71 ± 1.71	36.00 ± 3.98	23.20 ± 3.80	32.57 ± 4.57	16.95 ± 2.93	30.29 ± 2.62	21.71 ± 2.77	31.81 ± 4.76	20.80 ± 5.10	34.74 ± 4.91
FordA	59.23 ± 9.68	57.68 ± 10.00	**59.83 ± 8.76**	50.73 ± 1.01	56.70 ± 8.04	50.30 ± 0.33	49.57 ± 1.42	49.72 ± 0.63	49.68 ± 1.11	49.92 ± 0.80	50.61 ± 1.68	50.71 ± 0.98
FordB	52.54 ± 6.44	52.64 ± 6.88	52.40 ± 4.53	50.12 ± 1.83	**53.60 ± 4.09**	49.96 ± 0.94	50.53 ± 0.72	50.21 ± 0.80	50.81 ± 0.65	49.88 ± 0.98	50.05 ± 1.16	49.75 ± 2.56
FreezerRegularTrain	61.33 ± 21.55	58.22 ± 20.21	56.76 ± 19.80	61.31 ± 21.84	57.07 ± 18.69	59.91 ± 27.91	48.78 ± 1.68	53.81 ± 23.62	49.55 ± 8.37	60.27 ± 28.68	61.45 ± 8.09	**61.90 ± 22.25**
FreezerSmallTrain	67.85 ± 12.31	66.48 ± 12.45	67.42 ± 12.38	67.47 ± 12.29	63.19 ± 10.91	**73.32 ± 3.65**	55.33 ± 0.97	69.60 ± 5.96	57.00 ± 2.79	73.03 ± 4.44	65.92 ± 7.33	66.53 ± 10.64
Fungi	93.23 ± 4.00	35.81 ± 4.97	64.62 ± 2.56	68.39 ± 5.78	67.20 ± 19.03	**95.70 ± 0.93**	73.66 ± 2.69	86.74 ± 1.89	75.16 ± 1.03	**95.70 ± 0.93**	14.09 ± 7.48	79.35 ± 5.30
GestureMidAirD1	17.08 ± 4.26	19.38 ± 4.43	17.08 ± 4.26	34.31 ± 3.87	19.08 ± 6.45	31.54 ± 2.04	17.44 ± 0.89	34.62 ± 4.07	21.85 ± 4.63	**34.87 ± 2.47**	19.08 ± 6.45	32.00 ± 3.94
GestureMidAirD2	13.23 ± 5.14	17.08 ± 3.90	13.23 ± 5.14	29.08 ± 4.19	20.46 ± 5.77	29.74 ± 3.20	11.79 ± 2.22	29.23 ± 5.38	17.38 ± 5.95	**36.41 ± 0.89**	20.46 ± 5.77	33.85 ± 3.03
GestureMidAirD3	8.62 ± 1.84	9.08 ± 2.52	8.62 ± 1.84	20.00 ± 3.17	10.92 ± 1.26	**24.87 ± 4.24**	9.49 ± 1.60	23.08 ± 0.77	14.77 ± 4.47	22.56 ± 2.91	10.92 ± 1.26	18.46 ± 2.61

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Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
GesturePebbleZ1	49.07 ± 5.03	38.37 ± 6.03	**53.26 ± 6.97**	46.16 ± 13.08	23.60 ± 6.95	40.50 ± 15.60	20.93 ± 3.54	42.44 ± 7.05	18.95 ± 2.08	43.99 ± 8.75	23.60 ± 6.95	35.81 ± 4.48
GesturePebbleZ2	55.19 ± 11.38	42.91 ± 15.92	**57.47 ± 8.89**	48.35 ± 13.33	21.01 ± 3.48	47.68 ± 13.93	18.35 ± 4.94	43.88 ± 16.17	16.20 ± 5.17	55.06 ± 14.60	21.01 ± 3.48	33.29 ± 5.85
GunPoint	55.47 ± 4.79	58.13 ± 9.13	55.87 ± 5.32	53.20 ± 9.11	56.40 ± 7.58	**64.00 ± 8.08**	58.89 ± 10.59	63.33 ± 4.81	58.93 ± 8.45	62.89 ± 3.01	52.40 ± 11.44	58.00 ± 7.79
GunPointAgeSpan	49.94 ± 8.02	48.67 ± 3.53	49.11 ± 3.99	60.82 ± 22.09	58.73 ± 19.72	59.70 ± 21.56	52.11 ± 7.60	59.07 ± 20.81	52.41 ± 6.32	**61.92 ± 19.97**	48.10 ± 2.61	60.51 ± 10.32
GunPointMaleVersusFemale	63.99 ± 11.97	63.29 ± 11.59	64.30 ± 12.20	54.56 ± 4.47	56.65 ± 3.27	55.91 ± 6.97	50.95 ± 7.17	55.59 ± 9.92	50.19 ± 6.98	56.75 ± 8.76	58.04 ± 11.38	**81.01 ± 10.12**
GunPointOldVersusYoung	**92.38 ± 10.44**	89.97 ± 13.74	90.35 ± 13.23	53.90 ± 25.42	53.33 ± 25.91	58.31 ± 30.15	54.81 ± 4.33	57.46 ± 29.14	54.79 ± 8.98	58.10 ± 28.87	78.41 ± 20.62	57.97 ± 10.78
Ham	50.10 ± 9.32	51.24 ± 7.01	47.81 ± 11.28	55.24 ± 11.13	49.33 ± 2.81	48.89 ± 5.25	53.65 ± 3.06	51.43 ± 7.44	**55.81 ± 6.79**	50.16 ± 4.89	51.81 ± 10.55	52.76 ± 9.30
HandOutlines	46.81 ± 15.76	58.86 ± 12.82	55.84 ± 9.47	70.59 ± 7.20	75.68 ± 8.02	71.17 ± 7.55	51.08 ± 2.34	**76.76 ± 10.41**	52.81 ± 15.40	74.86 ± 11.22	61.19 ± 14.73	61.73 ± 6.31
Haptics	23.96 ± 5.22	22.66 ± 2.12	22.86 ± 4.94	**26.49 ± 6.85**	22.66 ± 4.38	25.11 ± 9.19	19.16 ± 1.81	23.16 ± 7.32	23.44 ± 3.46	25.43 ± 9.51	25.65 ± 5.26	25.65 ± 4.99
Herring	49.69 ± 6.39	49.69 ± 10.50	50.94 ± 11.51	48.12 ± 8.44	49.69 ± 7.44	47.40 ± 10.40	**51.04 ± 2.39**	49.48 ± 9.55	48.75 ± 8.44	48.44 ± 9.50	48.12 ± 10.39	50.00 ± 10.00
HouseTwenty	**80.84 ± 7.67**	74.62 ± 5.52	68.40 ± 16.13	50.25 ± 3.68	57.65 ± 8.25	68.63 ± 3.50	49.30 ± 7.15	61.62 ± 2.57	46.05 ± 6.81	66.67 ± 1.28	53.61 ± 8.52	62.18 ± 8.04
InlineSkate	16.40 ± 1.60	15.64 ± 0.76	18.00 ± 0.73	15.89 ± 3.32	14.69 ± 2.92	18.06 ± 2.11	15.15 ± 0.73	18.00 ± 2.36	15.02 ± 3.34	**18.67 ± 3.56**	14.87 ± 3.22	17.13 ± 2.24
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	43.53 ± 11.83	50.44 ± 13.06	48.19 ± 5.92	42.03 ± 6.13	45.25 ± 3.22	46.27 ± 8.59	49.26 ± 6.69	87.15 ± 20.64	55.34 ± 14.47
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	47.95 ± 6.69	55.10 ± 5.49	51.14 ± 8.16	39.36 ± 1.75	52.88 ± 6.50	37.67 ± 14.87	52.48 ± 11.47	85.78 ± 19.79	50.76 ± 7.86
InsectWingbeatSound	16.18 ± 3.00	14.23 ± 3.87	14.35 ± 2.82	39.88 ± 4.03	22.79 ± 2.65	43.42 ± 0.26	17.56 ± 4.92	43.16 ± 3.28	21.14 ± 2.99	**45.61 ± 5.13**	12.52 ± 3.37	25.27 ± 2.20
ItalyPowerDemand	61.83 ± 17.15	62.29 ± 17.84	64.74 ± 16.58	55.32 ± 7.10	60.35 ± 16.10	69.81 ± 17.24	68.35 ± 9.65	**78.33 ± 24.61**	68.98 ± 15.67	71.62 ± 21.13	47.31 ± 2.37	58.52 ± 15.27
LargeKitchenAppliances	48.48 ± 12.88	52.85 ± 11.74	46.72 ± 15.63	33.39 ± 3.38	**54.88 ± 8.28**	40.09 ± 11.55	34.93 ± 5.70	32.53 ± 3.23	31.47 ± 3.69	41.69 ± 12.56	33.81 ± 2.87	45.44 ± 5.60
Lightning2	50.49 ± 6.80	55.08 ± 10.15	51.80 ± 4.27	56.72 ± 11.39	51.80 ± 12.52	58.47 ± 11.16	48.63 ± 3.41	57.38 ± 11.36	51.48 ± 5.26	59.02 ± 8.67	55.08 ± 13.25	**59.67 ± 8.87**
Lightning7	44.93 ± 4.48	44.38 ± 4.60	40.82 ± 7.47	43.01 ± 4.18	42.19 ± 7.34	53.42 ± 10.70	15.07 ± 6.28	45.66 ± 6.76	18.63 ± 4.40	**56.16 ± 8.33**	24.11 ± 7.72	41.64 ± 7.60
Mallat	64.03 ± 14.52	50.42 ± 9.84	64.49 ± 9.43	80.37 ± 4.76	56.96 ± 10.09	**82.32 ± 3.73**	54.67 ± 2.15	75.96 ± 4.82	64.55 ± 4.53	75.72 ± 2.60	37.30 ± 5.46	68.24 ± 7.54
Meat	42.00 ± 9.96	44.00 ± 10.18	58.33 ± 10.99	**77.67 ± 17.26**	52.67 ± 11.94	46.11 ± 9.18	38.33 ± 8.66	33.33 ± 0.00	36.00 ± 2.53	66.11 ± 20.97	33.67 ± 0.75	55.67 ± 9.62
MedicalImages	27.92 ± 5.84	28.45 ± 4.75	24.95 ± 5.95	26.34 ± 8.67	24.16 ± 4.26	22.32 ± 5.35	26.84 ± 12.59	23.86 ± 6.12	**28.53 ± 8.23**	23.16 ± 6.10	4.21 ± 2.27	27.66 ± 4.41
MelbournePedestrian	**51.29 ± 5.58**	44.27 ± 4.81	47.47 ± 6.50	9.85 ± 0.23	31.99 ± 4.95	21.74 ± 7.37	46.44 ± 8.26	40.62 ± 4.65	43.40 ± 7.11	43.73 ± 5.33	31.99 ± 4.95	39.16 ± 5.22
MiddlePhalanxOutlineAgeGroup	38.83 ± 7.74	37.53 ± 10.57	38.05 ± 7.83	34.03 ± 9.03	35.97 ± 8.21	37.45 ± 11.78	29.87 ± 4.55	38.53 ± 15.64	33.90 ± 7.76	32.90 ± 7.86	**41.69 ± 18.94**	39.48 ± 7.40
MiddlePhalanxOutlineCorrect	53.68 ± 13.63	52.10 ± 14.21	54.43 ± 13.33	53.75 ± 17.89	56.49 ± 9.58	**58.42 ± 17.26**	55.56 ± 15.16	54.07 ± 23.81	54.50 ± 10.22	57.50 ± 21.09	52.85 ± 12.08	51.68 ± 5.00
MiddlePhalanxTW	34.03 ± 16.27	44.68 ± 7.76	32.47 ± 17.97	30.52 ± 10.75	32.34 ± 10.78	29.00 ± 15.96	38.96 ± 3.37	**46.10 ± 3.62**	35.45 ± 13.57	42.21 ± 5.07	15.19 ± 11.64	37.14 ± 16.84
MixedShapesRegularTrain	45.36 ± 5.29	40.59 ± 8.80	43.58 ± 10.39	47.60 ± 10.74	39.59 ± 5.62	43.05 ± 15.67	17.31 ± 1.15	45.53 ± 11.76	25.71 ± 3.77	42.76 ± 16.81	45.03 ± 12.47	**56.06 ± 5.34**
MixedShapesSmallTrain	52.78 ± 8.02	39.55 ± 6.53	48.84 ± 9.40	53.90 ± 7.66	45.57 ± 7.91	52.81 ± 6.78	17.32 ± 0.61	53.84 ± 4.25	24.17 ± 3.24	53.31 ± 5.69	48.73 ± 8.44	**63.30 ± 3.34**
MoteStrain	62.99 ± 13.96	60.67 ± 12.58	64.35 ± 11.70	63.40 ± 15.50	63.27 ± 13.76	68.45 ± 11.94	56.26 ± 12.55	63.82 ± 16.98	61.65 ± 9.57	**68.66 ± 11.89**	57.99 ± 9.98	54.36 ± 12.77
NonInvasiveFetalECGThorax1	19.05 ± 4.42	8.04 ± 2.52	19.05 ± 4.42	38.23 ± 1.81	12.07 ± 7.41	36.76 ± 3.30	12.57 ± 2.38	28.62 ± 3.97	14.44 ± 2.07	**42.58 ± 4.62**	5.58 ± 2.48	39.16 ± 2.28
NonInvasiveFetalECGThorax2	20.69 ± 5.30	9.71 ± 2.65	20.69 ± 5.30	47.48 ± 2.72	14.81 ± 5.33	46.67 ± 0.18	20.97 ± 3.22	34.59 ± 7.78	23.69 ± 1.73	**49.89 ± 4.96**	8.30 ± 1.72	47.48 ± 2.06
OSULeaf	**45.62 ± 7.45**	30.41 ± 8.35	38.35 ± 12.00	24.05 ± 2.14	22.15 ± 5.63	27.82 ± 3.13	17.77 ± 1.80	26.03 ± 1.89	20.00 ± 5.09	26.17 ± 1.56	17.60 ± 2.44	29.83 ± 5.82
OliveOil	28.00 ± 23.64	26.67 ± 11.06	47.33 ± 13.21	**65.33 ± 15.20**	36.00 ± 22.66	32.22 ± 16.44	44.44 ± 10.72	28.89 ± 11.71	32.00 ± 9.60	61.11 ± 15.40	27.33 ± 11.88	28.00 ± 14.26
PLAID	15.46 ± 6.16	13.97 ± 4.45	16.95 ± 4.17	17.77 ± 3.34	21.79 ± 0.97	21.54 ± 5.77	13.78 ± 3.08	19.93 ± 5.41	14.41 ± 2.63	20.30 ± 4.75	21.79 ± 0.97	**32.10 ± 5.72**
PhalangesOutlinesCorrect	**54.90 ± 6.98**	51.28 ± 10.93	54.41 ± 7.38	53.40 ± 8.33	52.54 ± 10.38	48.87 ± 4.48	51.01 ± 10.34	53.38 ± 8.94	52.73 ± 9.70	52.49 ± 9.30	54.83 ± 6.32	51.93 ± 9.54
Phoneme	**10.43 ± 1.42**	8.46 ± 1.00	**10.43 ± 1.42**	4.21 ± 0.54	5.61 ± 2.38	6.14 ± 1.81	2.39 ± 1.01	5.87 ± 1.54	3.20 ± 2.06	4.98 ± 3.20	3.42 ± 0.41	6.87 ± 1.15
PickupGestureWiimoteZ	35.60 ± 7.80	44.80 ± 4.82	42.00 ± 5.10	40.80 ± 5.40	36.00 ± 3.16	46.67 ± 9.87	17.33 ± 5.03	39.33 ± 6.43	20.40 ± 4.98	**48.67 ± 8.33**	36.00 ± 3.16	47.20 ± 7.29
PigAirwayPressure	23.94 ± 3.76	9.13 ± 2.66	**38.65 ± 2.92**	3.65 ± 1.47	3.75 ± 1.33	5.29 ± 0.83	4.01 ± 0.73	4.01 ± 1.11	3.27 ± 0.92	4.81 ± 0.96	6.15 ± 1.64	5.48 ± 2.19
PigArtPressure	40.58 ± 5.14	10.10 ± 2.36	**60.10 ± 2.28**	9.90 ± 2.42	2.21 ± 0.80	10.58 ± 2.10	8.01 ± 2.00	10.58 ± 1.73	4.81 ± 0.76	9.94 ± 2.27	15.58 ± 3.05	15.38 ± 4.07
PigCVP	22.50 ± 3.47	14.33 ± 2.79	**50.29 ± 2.90**	5.00 ± 1.25	3.27 ± 0.71	5.93 ± 0.73	3.53 ± 1.39	6.89 ± 0.56	2.02 ± 0.22	4.33 ± 0.83	8.94 ± 1.94	11.44 ± 2.08
Plane	**98.86 ± 2.56**	91.62 ± 6.26	95.43 ± 3.65	87.24 ± 8.67	85.14 ± 9.49	82.86 ± 4.15	78.73 ± 5.25	88.57 ± 9.09	86.48 ± 4.39	88.57 ± 6.67	26.10 ± 13.02	93.90 ± 6.92
PowerCons	76.89 ± 9.68	**80.22 ± 9.41**	78.11 ± 13.88	60.22 ± 11.82	61.00 ± 7.74	62.96 ± 3.70	49.44 ± 7.47	58.33 ± 2.78	51.33 ± 4.56	62.78 ± 6.41	55.22 ± 21.33	60.89 ± 11.09
ProximalPhalanxOutlineAgeGroup	48.49 ± 34.51	48.20 ± 31.13	48.68 ± 30.82	39.61 ± 19.82	39.12 ± 18.60	43.74 ± 32.89	49.59 ± 25.02	**60.81 ± 40.85**	38.15 ± 22.87	46.50 ± 27.14	32.68 ± 16.57	48.98 ± 23.27
ProximalPhalanxOutlineCorrect	54.43 ± 13.05	53.40 ± 15.85	51.20 ± 16.57	56.22 ± 5.47	53.26 ± 10.68	44.67 ± 13.09	46.96 ± 12.63	**58.76 ± 6.63**	52.78 ± 12.27	43.99 ± 6.30	57.59 ± 13.02	52.23 ± 12.03
ProximalPhalanxTW	54.63 ± 10.31	57.56 ± 9.33	58.15 ± 6.64	54.34 ± 11.16	49.76 ± 11.67	**60.81 ± 11.71**	55.93 ± 9.23	38.37 ± 0.75	55.80 ± 7.24	57.07 ± 10.70	17.37 ± 16.23	55.22 ± 6.63
RefrigerationDevices	36.05 ± 6.49	35.68 ± 5.16	37.44 ± 5.37	34.99 ± 2.77	33.97 ± 4.76	**41.16 ± 5.10**	34.58 ± 1.47	34.93 ± 0.92	34.72 ± 0.99	38.84 ± 5.04	32.75 ± 2.24	29.76 ± 11.02
Rock	29.20 ± 8.07	34.40 ± 7.92	28.40 ± 6.54	41.60 ± 16.52	26.40 ± 8.53	38.67 ± 25.32	**57.33 ± 5.03**	44.67 ± 18.90	36.80 ± 8.07	40.00 ± 24.25	36.40 ± 14.93	39.20 ± 13.75
ScreenType	33.92 ± 4.73	34.19 ± 7.39	**34.29 ± 6.61**	33.87 ± 2.72	33.87 ± 4.61	32.44 ± 5.27	32.18 ± 3.99	28.00 ± 3.40	30.99 ± 3.01	33.60 ± 4.19	30.61 ± 2.66	32.53 ± 4.84
SemgHandGenderCh2	50.87 ± 2.35	55.37 ± 5.38	53.17 ± 4.35	50.17 ± 11.45	**61.00 ± 15.57**	42.78 ± 13.47	52.78 ± 1.84	39.89 ± 8.47	44.67 ± 12.14	35.61 ± 1.06	55.73 ± 4.74	51.90 ± 14.92
SemgHandMovementCh2	28.89 ± 4.22	**29.87 ± 4.86**	29.60 ± 5.24	25.29 ± 1.13	24.89 ± 3.57	20.37 ± 0.13	21.70 ± 2.31	22.52 ± 2.76	18.71 ± 1.91	20.07 ± 3.46	27.02 ± 5.10	28.13 ± 4.95

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Table 3: Per-dataset 1-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
SemgHandSubjectCh2	28.89 ± 5.77	29.07 ± 4.60	28.00 ± 2.65	35.60 ± 4.19	29.91 ± 10.77	26.37 ± 1.89	23.11 ± 2.89	26.67 ± 4.87	21.56 ± 1.94	26.96 ± 3.68	27.91 ± 3.11	**37.42 ± 8.82**
ShakeGestureWiimoteZ	63.20 ± 7.69	58.00 ± 11.83	**67.60 ± 11.61**	40.40 ± 11.26	43.20 ± 9.55	45.33 ± 3.06	16.00 ± 7.21	40.00 ± 3.46	24.40 ± 4.98	46.00 ± 6.00	43.20 ± 9.55	60.40 ± 7.92
ShapeletSim	66.33 ± 12.25	**87.00 ± 16.91**	73.89 ± 23.85	50.11 ± 2.27	75.22 ± 8.30	53.89 ± 1.67	52.96 ± 1.40	48.89 ± 5.64	50.44 ± 1.82	51.11 ± 0.96	47.78 ± 3.07	62.44 ± 7.44
ShapesAll	23.63 ± 4.60	15.60 ± 1.98	23.63 ± 4.60	45.70 ± 2.59	19.27 ± 4.58	44.72 ± 2.58	6.00 ± 0.58	43.94 ± 2.14	8.07 ± 3.90	44.06 ± 1.11	8.13 ± 2.75	**50.20 ± 3.03**
SmallKitchenAppliances	40.37 ± 15.51	**46.24 ± 15.42**	44.32 ± 15.50	42.19 ± 3.16	38.13 ± 22.42	32.98 ± 5.23	32.80 ± 3.67	35.82 ± 4.27	35.41 ± 3.13	36.89 ± 7.74	38.88 ± 7.80	42.88 ± 15.27
SmoothSubspace	**74.93 ± 5.59**	74.40 ± 5.13	74.80 ± 8.48	31.73 ± 3.96	50.00 ± 10.52	46.67 ± 1.76	53.33 ± 9.45	50.22 ± 10.96	62.27 ± 6.56	60.67 ± 10.35	56.67 ± 5.19	68.53 ± 8.95
SonyAIBORobotSurface1	77.57 ± 10.30	**80.80 ± 8.99**	80.37 ± 5.89	59.80 ± 6.73	64.53 ± 18.83	52.86 ± 10.38	56.96 ± 5.82	55.35 ± 9.37	56.07 ± 4.43	57.68 ± 9.32	55.61 ± 9.09	57.20 ± 8.62
SonyAIBORobotSurface2	80.29 ± 6.43	**84.09 ± 5.18**	80.90 ± 6.99	67.39 ± 19.08	73.03 ± 13.78	66.18 ± 11.61	60.37 ± 27.05	61.35 ± 27.10	69.49 ± 15.13	68.07 ± 16.06	57.31 ± 15.03	66.09 ± 10.78
StarLightCurves	63.75 ± 3.45	62.30 ± 9.88	57.59 ± 9.08	57.39 ± 13.40	57.55 ± 15.28	**67.54 ± 17.11**	37.78 ± 3.53	57.92 ± 15.55	27.99 ± 0.00	67.36 ± 17.40	64.81 ± 14.61	63.03 ± 14.23
Strawberry	52.27 ± 4.26	51.84 ± 3.86	54.43 ± 4.33	54.86 ± 4.34	53.62 ± 3.45	58.29 ± 0.95	54.86 ± 13.21	**60.18 ± 0.68**	56.11 ± 9.26	57.93 ± 0.83	45.89 ± 11.15	51.46 ± 5.21
SwedishLeaf	**65.95 ± 6.01**	46.08 ± 6.36	**65.95 ± 6.01**	27.42 ± 3.44	40.83 ± 4.40	36.69 ± 1.99	27.20 ± 2.42	24.53 ± 4.08	31.65 ± 2.48	38.19 ± 0.81	8.10 ± 1.23	57.98 ± 2.72
Symbols	**85.77 ± 4.83**	70.17 ± 4.91	76.68 ± 7.53	79.60 ± 7.65	72.50 ± 6.12	81.54 ± 3.92	37.15 ± 0.97	82.18 ± 0.42	46.13 ± 9.06	80.57 ± 0.95	51.54 ± 6.72	80.94 ± 7.54
SyntheticControl	72.53 ± 6.61	76.40 ± 5.24	68.67 ± 10.59	38.47 ± 6.28	80.33 ± 9.12	47.78 ± 5.87	44.89 ± 3.20	48.22 ± 0.38	42.13 ± 2.87	50.00 ± 4.48	24.53 ± 7.68	**84.20 ± 6.54**
ToeSegmentation1	**69.82 ± 7.39**	69.65 ± 4.27	65.09 ± 8.05	51.67 ± 1.47	55.79 ± 6.85	49.27 ± 1.10	49.71 ± 3.29	46.64 ± 4.73	52.46 ± 2.82	48.39 ± 3.23	49.47 ± 2.47	58.16 ± 3.70
ToeSegmentation2	68.92 ± 8.63	70.31 ± 3.38	70.92 ± 10.76	50.31 ± 5.37	65.54 ± 9.84	50.26 ± 2.91	35.90 ± 10.81	50.51 ± 4.24	33.23 ± 16.47	52.05 ± 6.54	41.54 ± 14.95	**75.23 ± 12.13**
Trace	61.00 ± 4.64	65.20 ± 9.09	65.80 ± 6.61	49.60 ± 3.65	74.40 ± 6.15	51.33 ± 5.51	42.33 ± 4.51	47.67 ± 0.58	44.80 ± 7.69	51.33 ± 4.93	37.80 ± 6.42	**99.20 ± 1.79**
TwoLeadECG	**69.48 ± 12.74**	64.97 ± 10.26	66.51 ± 13.23	54.03 ± 2.33	57.82 ± 5.04	59.32 ± 3.05	53.32 ± 7.30	52.62 ± 3.32	54.47 ± 5.41	60.32 ± 1.06	50.73 ± 1.02	64.88 ± 11.28
TwoPatterns	48.77 ± 2.41	**61.51 ± 6.85**	46.17 ± 11.65	31.86 ± 3.24	36.86 ± 6.10	29.76 ± 2.56	25.52 ± 1.93	28.23 ± 0.74	26.06 ± 1.27	29.87 ± 1.68	26.32 ± 2.53	45.74 ± 2.64
UMD	43.06 ± 4.02	**54.58 ± 9.53**	45.56 ± 6.14	46.81 ± 7.70	50.14 ± 6.28	46.30 ± 5.02	45.60 ± 3.21	47.69 ± 4.19	48.61 ± 2.55	47.45 ± 6.56	31.39 ± 4.35	45.47 ± 8.73
UWaveGestureLibraryAll	29.36 ± 4.20	24.46 ± 5.19	28.34 ± 3.82	66.77 ± 5.89	41.63 ± 5.40	69.19 ± 5.01	25.78 ± 1.68	67.23 ± 2.91	26.67 ± 3.53	**69.21 ± 6.24**	50.70 ± 5.42	33.37 ± 0.26
UWaveGestureLibraryX	26.45 ± 4.19	26.60 ± 3.44	24.71 ± 1.83	47.27 ± 6.01	35.36 ± 5.04	**49.30 ± 7.17**	22.37 ± 0.14	47.22 ± 8.02	24.34 ± 5.30	49.17 ± 7.61	17.28 ± 8.20	30.33 ± 0.23
UWaveGestureLibraryY	25.37 ± 4.63	26.13 ± 4.45	24.62 ± 4.93	43.95 ± 4.51	38.35 ± 7.26	47.05 ± 4.02	24.58 ± 1.89	46.37 ± 4.93	26.81 ± 2.91	**47.18 ± 5.20**	20.37 ± 4.71	40.36 ± 0.25
UWaveGestureLibraryZ	24.77 ± 4.74	27.03 ± 3.18	24.45 ± 3.15	42.90 ± 1.96	36.37 ± 3.72	44.61 ± 1.78	22.47 ± 2.82	44.70 ± 2.64	25.10 ± 2.80	44.75 ± 1.63	22.77 ± 3.47	**47.44 ± 2.61**
Wafer	50.39 ± 4.56	44.94 ± 10.31	49.91 ± 13.57	**67.87 ± 4.81**	51.56 ± 17.59	63.90 ± 3.73	64.15 ± 3.76	63.99 ± 9.60	63.82 ± 9.82	65.48 ± 1.03	43.02 ± 26.21	46.27 ± 7.11
Wine	**57.41 ± 14.70**	54.07 ± 4.61	54.81 ± 11.31	52.96 ± 8.34	50.74 ± 6.23	48.77 ± 3.85	50.00 ± 0.00	50.00 ± 0.00	52.22 ± 4.97	37.65 ± 19.80	51.48 ± 3.31	51.11 ± 7.48
WordSynonyms	14.92 ± 3.03	13.70 ± 5.79	14.92 ± 3.03	19.78 ± 2.70	16.96 ± 4.46	19.75 ± 1.90	6.79 ± 1.46	**21.16 ± 1.50**	10.06 ± 3.54	20.53 ± 2.41	4.98 ± 1.34	20.91 ± 2.73
Worms	**36.10 ± 9.87**	33.25 ± 6.66	35.06 ± 11.43	17.14 ± 4.04	23.90 ± 9.39	19.91 ± 6.54	23.38 ± 4.50	18.61 ± 5.86	17.66 ± 7.03	19.05 ± 5.25	17.66 ± 2.36	24.68 ± 8.95
WormsTwoClass	44.79 ± 3.48	49.09 ± 6.96	52.99 ± 6.71	49.09 ± 3.72	50.39 ± 3.23	49.35 ± 3.44	48.92 ± 10.50	52.38 ± 1.98	53.51 ± 4.81	**53.68 ± 3.75**	52.99 ± 2.82	47.53 ± 10.69
Yoga	**55.60 ± 4.57**	50.68 ± 5.18	53.54 ± 4.28	49.99 ± 1.44	49.75 ± 4.01	49.48 ± 0.91	51.11 ± 3.62	49.62 ± 0.72	51.12 ± 3.30	48.62 ± 0.52	49.35 ± 2.85	50.42 ± 4.51
Avg. Acc.	47.79	45.40	47.88	44.21	42.39	45.16	36.78	44.31	37.71	46.42	35.61	**48.31**
Avg. Rank	5.11	5.80	5.19	6.45	6.92	6.04	9.11	6.17	8.51	5.30	8.82	**4.58**
#Best	25	16	14	7	5	11	2	11	2	15	4	**26**

Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
ACSF1	**61.60 ± 7.86**	47.20 ± 6.30	59.20 ± 5.26	25.80 ± 4.38	42.00 ± 3.16	40.33 ± 2.31	30.67 ± 2.52	44.00 ± 6.24	35.80 ± 3.19	40.67 ± 5.69	24.80 ± 4.66	54.33 ± 4.16
Adiac	35.45 ± 2.60	14.27 ± 3.74	35.45 ± 2.60	18.87 ± 2.86	17.80 ± 5.08	27.79 ± 2.14	44.67 ± 0.39	16.62 ± 1.77	55.24 ± 4.67	32.91 ± 3.49	3.48 ± 1.27	**56.01 ± 1.25**
AllGestureWiimoteX	**32.43 ± 2.94**	27.69 ± 3.48	32.09 ± 3.58	19.46 ± 2.47	23.77 ± 3.63	24.43 ± 4.71	11.86 ± 1.00	22.67 ± 2.29	14.09 ± 1.28	21.10 ± 5.46	23.77 ± 3.63	26.51 ± 4.36
AllGestureWiimoteY	**35.11 ± 5.43**	28.57 ± 4.31	28.91 ± 5.81	20.54 ± 2.69	33.31 ± 2.65	24.33 ± 2.50	12.00 ± 1.38	21.90 ± 1.33	16.14 ± 1.67	22.57 ± 1.00	33.31 ± 2.65	29.43 ± 1.69
AllGestureWiimoteZ	**29.80 ± 2.53**	26.94 ± 2.84	29.31 ± 4.56	16.77 ± 2.02	22.97 ± 5.35	19.86 ± 4.12	12.76 ± 0.36	19.57 ± 2.48	14.97 ± 0.98	20.29 ± 3.48	22.97 ± 5.35	27.91 ± 5.21
ArrowHead	50.17 ± 5.13	44.80 ± 3.82	45.14 ± 5.25	**57.03 ± 4.92**	49.14 ± 13.02	53.14 ± 6.02	34.10 ± 6.27	50.67 ± 2.01	42.29 ± 4.92	55.43 ± 6.44	43.77 ± 8.76	**57.03 ± 2.81**
BME	48.40 ± 4.91	55.33 ± 7.02	54.40 ± 8.12	50.80 ± 5.11	67.33 ± 7.90	48.89 ± 2.14	64.00 ± 5.21	58.89 ± 3.91	67.33 ± 7.44	54.89 ± 9.20	50.80 ± 8.05	**79.47 ± 11.40**
Beef	42.00 ± 2.98	36.67 ± 8.50	36.70 ± 9.60	49.33 ± 7.23	39.33 ± 6.41	48.89 ± 5.09	40.00 ± 3.33	34.44 ± 5.09	42.67 ± 4.94	**51.11 ± 7.70**	32.67 ± 7.23	43.33 ± 5.27
BeetleFly	50.00 ± 9.35	67.00 ± 18.23	55.00 ± 12.75	52.00 ± 11.51	57.00 ± 16.43	45.00 ± 10.00	48.33 ± 12.58	38.33 ± 7.64	45.00 ± 9.35	56.67 ± 7.64	57.00 ± 4.47	**75.00 ± 18.03**
BirdChicken	71.00 ± 6.52	72.00 ± 4.47	**76.00 ± 7.42**	52.00 ± 14.40	65.00 ± 10.61	45.00 ± 8.66	51.67 ± 7.64	46.67 ± 5.77	53.00 ± 2.74	50.00 ± 13.23	49.00 ± 2.24	69.00 ± 22.19
CBF	85.76 ± 5.83	**90.62 ± 7.16**	82.18 ± 10.67	64.78 ± 6.67	82.69 ± 10.89	68.07 ± 6.00	34.89 ± 2.41	73.78 ± 6.88	36.78 ± 1.75	70.00 ± 6.65	42.84 ± 7.09	85.91 ± 4.23
Car	42.00 ± 15.43	41.67 ± 13.49	52.67 ± 7.87	55.33 ± 10.63	43.00 ± 11.45	54.44 ± 6.74	32.78 ± 0.96	52.78 ± 10.84	37.00 ± 7.76	55.56 ± 4.19	31.00 ± 10.71	**63.33 ± 7.45**
Chinatown	95.74 ± 1.26	**96.73 ± 1.19**	**96.73 ± 0.81**	54.52 ± 24.75	91.08 ± 5.39	85.33 ± 8.50	86.10 ± 2.71	84.55 ± 3.36	80.70 ± 5.41	82.80 ± 4.37	45.01 ± 24.65	91.84 ± 4.85
ChlorineConcentration	34.61 ± 4.95	35.24 ± 5.79	35.19 ± 5.79	34.54 ± 2.93	34.28 ± 2.22	31.22 ± 6.21	33.24 ± 2.95	33.84 ± 2.69	33.91 ± 2.07	33.69 ± 2.59	**39.08 ± 9.47**	32.47 ± 3.03
CinCECGTorso	46.52 ± 12.75	48.35 ± 9.58	51.06 ± 10.06	39.77 ± 4.46	54.91 ± 12.73	48.14 ± 2.63	36.18 ± 3.15	53.94 ± 2.51	50.30 ± 12.26	48.89 ± 2.81	49.00 ± 5.40	**62.95 ± 6.66**
Coffee	**94.29 ± 7.41**	88.57 ± 12.73	**94.29 ± 3.19**	92.14 ± 3.91	90.71 ± 10.89	73.81 ± 8.99	63.10 ± 7.43	85.71 ± 3.57	84.29 ± 10.29	77.38 ± 4.12	65.71 ± 15.07	88.57 ± 4.66
Computers	52.24 ± 10.15	54.40 ± 10.81	52.72 ± 12.14	48.08 ± 7.70	53.28 ± 11.25	48.00 ± 10.51	48.67 ± 4.60	48.00 ± 11.45	50.40 ± 1.72	46.67 ± 11.22	50.16 ± 7.53	**57.73 ± 17.44**
CricketX	**36.51 ± 3.27**	34.10 ± 3.98	33.03 ± 3.86	20.62 ± 2.07	30.56 ± 3.87	27.35 ± 2.06	8.12 ± 1.41	26.15 ± 1.85	12.15 ± 1.24	27.44 ± 2.96	11.54 ± 1.86	35.98 ± 3.13
CricketY	**37.54 ± 5.48**	30.62 ± 2.69	33.28 ± 3.86	22.56 ± 4.44	24.21 ± 2.69	26.50 ± 7.27	9.06 ± 0.39	25.90 ± 6.29	12.67 ± 0.94	27.52 ± 6.80	10.00 ± 3.24	30.43 ± 5.23
CricketZ	**40.41 ± 4.15**	35.49 ± 2.72	33.69 ± 5.21	20.36 ± 1.13	24.92 ± 3.09	26.75 ± 2.72	8.12 ± 1.50	24.44 ± 0.78	12.97 ± 2.47	26.50 ± 2.74	11.18 ± 2.42	33.33 ± 3.02
Crop	31.31 ± 3.87	37.82 ± 2.09	31.31 ± 3.87	14.67 ± 0.98	24.66 ± 2.54	36.33 ± 2.63	35.06 ± 3.15	37.89 ± 1.28	35.83 ± 2.39	38.66 ± 2.99	4.63 ± 0.75	**43.94 ± 2.11**
DiatomSizeReduction	78.43 ± 9.98	84.44 ± 4.38	89.74 ± 8.40	83.73 ± 5.41	63.01 ± 13.25	84.42 ± 6.89	59.91 ± 2.78	84.97 ± 5.38	59.28 ± 18.44	82.35 ± 7.47	35.10 ± 12.79	**90.39 ± 5.62**
DistalPhalanxOutlineAgeGroup	66.91 ± 6.08	**67.77 ± 3.31**	65.76 ± 6.53	55.54 ± 12.17	58.42 ± 11.15	57.07 ± 19.57	50.84 ± 14.56	56.12 ± 13.32	56.26 ± 12.92	57.07 ± 14.47	38.99 ± 8.80	**67.77 ± 3.39**
DistalPhalanxOutlineCorrect	52.32 ± 12.15	51.59 ± 11.49	52.54 ± 12.44	49.57 ± 11.83	51.45 ± 15.36	**57.49 ± 13.09**	56.76 ± 11.56	55.80 ± 11.93	51.38 ± 12.34	56.64 ± 12.97	50.65 ± 9.31	55.80 ± 14.95
DistalPhalanxTW	49.50 ± 8.83	**55.54 ± 4.21**	54.10 ± 4.92	39.28 ± 8.61	53.96 ± 7.56	41.97 ± 11.56	42.45 ± 10.74	39.81 ± 18.49	47.05 ± 8.79	45.80 ± 16.17	22.45 ± 8.85	51.37 ± 8.97
DodgerLoopDay	28.75 ± 2.34	23.75 ± 2.50	28.00 ± 5.70	**44.00 ± 4.09**	38.50 ± 7.42	39.58 ± 4.02	23.75 ± 2.50	42.50 ± 7.60	19.75 ± 4.09	42.50 ± 3.31	38.50 ± 7.42	35.50 ± 8.78
DodgerLoopGame	54.93 ± 6.45	63.04 ± 8.02	58.41 ± 11.09	53.62 ± 8.51	65.65 ± 18.08	67.87 ± 7.26	48.55 ± 4.53	**71.50 ± 0.84**	50.14 ± 3.81	68.36 ± 3.42	65.65 ± 18.08	61.01 ± 13.03
DodgerLoopWeekend	92.75 ± 2.40	87.68 ± 5.59	92.03 ± 3.69	94.64 ± 2.74	96.09 ± 1.41	96.38 ± 1.26	64.98 ± 4.71	**97.34 ± 0.42**	65.22 ± 8.45	96.14 ± 0.84	96.09 ± 1.41	84.64 ± 2.73
ECG200	69.40 ± 7.77	75.20 ± 7.05	71.80 ± 5.54	76.80 ± 3.56	71.80 ± 4.38	**78.33 ± 1.53**	74.33 ± 5.69	75.67 ± 3.21	73.60 ± 3.65	75.33 ± 0.58	52.20 ± 18.20	73.00 ± 3.46
ECG5000	61.90 ± 13.09	69.05 ± 11.38	61.86 ± 10.65	**74.23 ± 10.96**	58.61 ± 14.82	72.42 ± 4.03	65.61 ± 11.76	69.40 ± 4.84	67.93 ± 11.62	66.36 ± 9.81	30.14 ± 19.40	67.68 ± 9.76
ECGFiveDays	68.78 ± 9.46	56.79 ± 4.91	71.73 ± 8.76	57.12 ± 2.23	67.29 ± 3.58	66.55 ± 6.20	74.64 ± 13.99	**75.18 ± 2.37**	68.76 ± 10.24	68.52 ± 6.54	57.79 ± 7.24	69.12 ± 9.82
EOGHorizontalSignal	20.66 ± 1.70	23.43 ± 5.48	21.38 ± 4.60	25.86 ± 4.54	28.07 ± 2.90	30.66 ± 2.91	12.06 ± 1.96	31.68 ± 2.61	9.56 ± 1.50	29.01 ± 2.49	29.83 ± 3.23	**37.35 ± 4.07**
EOGVerticalSignal	13.37 ± 1.32	16.57 ± 3.25	14.86 ± 2.77	26.80 ± 1.98	29.56 ± 4.11	26.98 ± 4.28	11.88 ± 1.54	26.34 ± 4.16	10.83 ± 1.45	26.61 ± 6.09	20.88 ± 2.94	**31.82 ± 3.47**
Earthquakes	52.95 ± 7.17	57.84 ± 11.11	49.35 ± 10.38	53.24 ± 4.93	**63.88 ± 6.42**	63.31 ± 10.45	49.16 ± 1.10	54.92 ± 17.17	56.55 ± 8.34	60.43 ± 21.88	57.12 ± 15.34	55.25 ± 14.33
ElectricDevices	37.63 ± 9.44	41.80 ± 2.08	34.70 ± 5.95	21.27 ± 5.73	37.69 ± 10.00	28.63 ± 3.21	22.56 ± 2.12	20.63 ± 2.87	22.76 ± 4.51	29.88 ± 3.14	14.36 ± 4.97	**47.14 ± 3.99**
EthanolLevel	25.32 ± 0.64	25.52 ± 1.56	25.12 ± 2.09	25.76 ± 1.71	26.96 ± 0.82	26.87 ± 2.61	**29.20 ± 1.91**	25.67 ± 3.18	25.24 ± 0.98	26.47 ± 2.08	25.12 ± 0.81	28.00 ± 3.11
FaceAll	**55.98 ± 6.19**	45.22 ± 8.18	**55.98 ± 6.19**	40.27 ± 5.76	24.12 ± 4.85	38.56 ± 3.11	38.84 ± 3.81	36.71 ± 2.55	38.89 ± 3.64	41.40 ± 3.79	11.35 ± 5.01	39.09 ± 1.28
FaceFour	77.73 ± 4.99	52.05 ± 10.37	81.82 ± 7.14	74.09 ± 5.98	59.32 ± 5.76	79.55 ± 9.02	57.58 ± 7.31	80.68 ± 3.41	62.50 ± 7.14	**84.85 ± 2.37**	40.45 ± 14.16	68.55 ± 4.91
FacesUCR	**67.20 ± 6.42**	51.90 ± 6.85	62.04 ± 8.47	41.67 ± 3.36	25.42 ± 6.93	44.55 ± 2.52	45.95 ± 4.27	49.07 ± 5.45	48.56 ± 3.79	49.27 ± 6.47	10.96 ± 5.26	55.08 ± 2.32
FiftyWords	19.03 ± 1.51	13.49 ± 1.95	19.03 ± 1.51	40.09 ± 3.22	28.35 ± 3.05	44.69 ± 2.97	5.05 ± 0.44	46.74 ± 2.43	16.53 ± 1.75	44.18 ± 3.70	15.16 ± 3.75	**47.30 ± 2.89**
Fish	**50.97 ± 10.77**	26.97 ± 9.88	50.51 ± 16.65	50.17 ± 6.36	31.54 ± 5.18	45.90 ± 8.88	20.00 ± 4.12	39.05 ± 5.64	22.97 ± 3.62	45.71 ± 10.47	22.86 ± 2.83	49.91 ± 10.25
FordA	**64.98 ± 8.59**	62.27 ± 9.82	62.82 ± 9.33	49.92 ± 1.84	60.48 ± 7.29	50.30 ± 1.31	50.66 ± 0.90	50.15 ± 0.79	50.27 ± 1.01	50.58 ± 1.43	50.76 ± 2.11	53.35 ± 2.84
FordB	**61.19 ± 3.67**	58.10 ± 1.40	58.67 ± 2.63	49.09 ± 1.35	56.89 ± 2.99	49.84 ± 0.63	50.99 ± 2.05	50.25 ± 1.29	49.95 ± 1.77	50.58 ± 0.68	50.77 ± 0.87	50.42 ± 2.72
FreezerRegularTrain	68.01 ± 9.57	67.73 ± 22.81	61.99 ± 17.00	67.56 ± 20.68	71.73 ± 5.42	59.79 ± 26.51	53.13 ± 0.84	60.39 ± 25.70	58.85 ± 2.14	60.74 ± 25.80	72.39 ± 6.35	**79.14 ± 7.94**
FreezerSmallTrain	66.47 ± 12.08	65.70 ± 14.41	66.00 ± 12.69	66.60 ± 13.71	64.67 ± 11.73	**71.47 ± 6.03**	56.34 ± 1.10	68.49 ± 7.01	56.88 ± 4.22	71.17 ± 6.80	65.57 ± 16.06	64.30 ± 13.28
Fungi	97.10 ± 2.45	34.84 ± 7.53	66.24 ± 2.06	68.17 ± 5.25	68.17 ± 13.75	**97.13 ± 0.82**	75.99 ± 1.64	82.62 ± 2.42	74.73 ± 0.85	96.95 ± 0.31	8.06 ± 3.93	81.97 ± 3.24
GestureMidAirD1	24.62 ± 1.44	29.08 ± 3.94	20.62 ± 1.44	38.31 ± 3.66	29.38 ± 2.74	37.69 ± 3.35	17.95 ± 2.70	36.67 ± 1.78	35.69 ± 2.53	36.92 ± 2.77	29.38 ± 2.74	**44.92 ± 6.31**
GestureMidAirD2	16.46 ± 2.22	21.85 ± 4.16	16.46 ± 2.22	35.69 ± 3.47	33.69 ± 3.66	42.05 ± 0.44	12.05 ± 1.60	44.36 ± 2.47	30.92 ± 4.26	**45.64 ± 3.20**	33.69 ± 3.66	43.85 ± 4.03
GestureMidAirD3	10.15 ± 1.99	11.85 ± 3.91	10.15 ± 1.99	25.08 ± 3.59	18.00 ± 2.64	26.41 ± 1.18	14.62 ± 4.07	**26.92 ± 1.54**	22.00 ± 3.63	**26.92 ± 2.77**	18.00 ± 2.64	21.08 ± 1.85

Continued on next page

Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
GesturePebbleZ1	62.44 ± 5.24	41.40 ± 5.40	**64.88 ± 2.42**	53.26 ± 4.18	29.53 ± 10.46	58.91 ± 3.41	16.86 ± 3.54	57.17 ± 0.89	24.19 ± 4.00	61.82 ± 3.41	29.53 ± 10.46	37.91 ± 6.35
GesturePebbleZ2	**66.58 ± 5.17**	44.05 ± 14.70	61.70 ± 6.13	51.90 ± 9.19	35.32 ± 9.94	55.06 ± 7.46	15.82 ± 0.63	46.84 ± 8.79	20.76 ± 3.15	56.75 ± 4.12	35.32 ± 9.94	41.39 ± 4.24
GunPoint	67.07 ± 8.15	65.20 ± 5.95	66.00 ± 5.89	60.13 ± 7.74	65.47 ± 7.68	55.33 ± 6.57	64.44 ± 5.18	58.44 ± 9.46	68.13 ± 5.80	57.11 ± 7.95	54.00 ± 6.53	**79.33 ± 16.25**
GunPointAgeSpan	65.38 ± 2.37	63.73 ± 4.38	**68.61 ± 1.37**	67.41 ± 13.19	61.08 ± 13.41	61.08 ± 8.21	52.22 ± 8.74	55.49 ± 10.60	52.41 ± 8.13	58.76 ± 12.22	47.41 ± 14.13	64.49 ± 7.90
GunPointMaleVersusFemale	79.30 ± 6.81	84.62 ± 8.74	81.65 ± 8.32	57.15 ± 7.06	65.95 ± 9.57	60.65 ± 4.86	50.11 ± 5.57	60.76 ± 10.27	52.59 ± 5.35	65.82 ± 3.12	65.00 ± 5.79	**87.66 ± 5.66**
GunPointOldVersusYoung	**91.94 ± 11.58**	89.97 ± 13.74	91.37 ± 12.08	72.89 ± 4.20	73.14 ± 5.24	73.54 ± 7.07	56.51 ± 2.29	72.49 ± 6.74	59.30 ± 3.02	75.87 ± 9.07	78.03 ± 17.26	70.92 ± 5.05
Ham	45.71 ± 9.69	50.86 ± 8.18	48.00 ± 10.27	54.86 ± 11.40	53.33 ± 7.28	54.92 ± 1.98	**58.73 ± 5.25**	50.48 ± 13.83	55.24 ± 4.62	57.78 ± 9.77	52.57 ± 10.93	50.86 ± 8.07
HandOutlines	57.19 ± 12.29	63.30 ± 2.05	60.65 ± 3.52	69.19 ± 14.14	63.57 ± 17.16	67.21 ± 16.82	52.16 ± 12.39	68.11 ± 20.52	41.57 ± 12.57	67.48 ± 15.69	**69.89 ± 13.52**	64.59 ± 10.14
Haptics	29.16 ± 3.65	24.35 ± 4.42	25.32 ± 2.75	28.31 ± 7.72	28.25 ± 7.08	**29.33 ± 7.26**	23.81 ± 0.68	29.22 ± 2.66	20.97 ± 3.07	29.00 ± 7.31	26.95 ± 8.77	26.62 ± 4.64
Herring	52.81 ± 11.18	**56.88 ± 7.13**	52.81 ± 11.23	51.25 ± 5.89	52.19 ± 12.13	47.92 ± 6.51	46.35 ± 3.61	45.83 ± 3.25	50.00 ± 4.69	48.96 ± 4.51	45.00 ± 9.84	55.00 ± 4.34
HouseTwenty	**76.30 ± 9.04**	69.24 ± 11.83	67.39 ± 13.31	60.00 ± 7.10	63.03 ± 13.56	68.91 ± 7.47	47.34 ± 6.31	59.94 ± 1.75	52.10 ± 8.21	71.15 ± 7.81	60.34 ± 11.20	71.93 ± 10.30
InlineSkate	18.58 ± 2.99	19.56 ± 1.81	**19.93 ± 2.98**	18.29 ± 2.68	17.05 ± 3.64	16.85 ± 1.00	15.76 ± 0.93	15.82 ± 1.62	16.29 ± 1.31	17.03 ± 0.46	16.36 ± 2.46	18.98 ± 2.54
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.97 ± 7.58	58.39 ± 6.76	58.37 ± 4.44	37.08 ± 10.70	54.89 ± 2.35	37.99 ± 13.89	59.71 ± 4.91	81.69 ± 17.66	68.76 ± 9.60
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	51.33 ± 4.40	58.47 ± 6.25	57.56 ± 4.71	41.37 ± 7.24	58.23 ± 4.08	42.65 ± 10.05	58.10 ± 4.04	80.40 ± 17.17	67.79 ± 2.45
InsectWingbeatSound	20.38 ± 3.08	16.62 ± 4.40	18.70 ± 3.47	47.87 ± 2.61	33.18 ± 5.98	50.44 ± 4.17	20.69 ± 3.83	49.41 ± 1.90	27.56 ± 2.55	**51.03 ± 4.83**	16.68 ± 7.51	31.49 ± 3.72
ItalyPowerDemand	68.88 ± 12.96	75.63 ± 13.47	73.61 ± 12.06	50.03 ± 0.16	79.18 ± 15.47	54.49 ± 7.77	75.87 ± 16.00	76.35 ± 22.01	**83.15 ± 15.39**	73.57 ± 21.06	50.07 ± 0.34	65.03 ± 13.70
LargeKitchenAppliances	45.65 ± 8.51	**54.19 ± 10.41**	50.93 ± 13.42	33.60 ± 7.89	44.85 ± 7.66	37.96 ± 12.60	32.00 ± 3.23	34.40 ± 7.77	32.96 ± 7.01	36.62 ± 11.05	32.00 ± 6.67	42.56 ± 5.00
Lightning2	57.05 ± 8.63	53.77 ± 7.90	59.34 ± 7.64	54.10 ± 10.50	51.80 ± 7.11	57.38 ± 4.34	49.73 ± 9.88	51.91 ± 4.13	52.13 ± 11.90	55.19 ± 6.21	50.16 ± 8.80	**63.28 ± 8.25**
Lightning7	45.48 ± 6.31	43.01 ± 5.94	42.74 ± 6.38	38.90 ± 8.91	47.95 ± 8.16	**59.36 ± 11.07**	19.63 ± 3.45	50.23 ± 12.43	27.40 ± 4.94	56.16 ± 10.34	20.82 ± 11.60	46.03 ± 6.17
Mallat	73.35 ± 15.31	47.53 ± 14.01	80.84 ± 4.13	86.79 ± 1.54	65.16 ± 19.15	**91.78 ± 0.99**	63.31 ± 2.66	78.79 ± 4.89	70.90 ± 5.59	87.65 ± 2.43	49.21 ± 6.11	80.67 ± 4.06
Meat	54.33 ± 7.42	42.00 ± 7.76	73.00 ± 16.39	**89.67 ± 7.85**	52.00 ± 12.16	53.33 ± 6.01	43.33 ± 8.33	41.11 ± 7.52	51.33 ± 18.61	57.22 ± 7.88	36.33 ± 5.82	57.67 ± 17.30
MedicalImages	33.45 ± 8.28	25.74 ± 5.92	33.61 ± 5.62	24.45 ± 6.71	27.45 ± 4.51	28.46 ± 3.05	27.85 ± 6.61	28.55 ± 2.80	**34.53 ± 1.25**	30.00 ± 1.71	29.82 ± 16.29	31.63 ± 5.59
MelbournePedestrian	**63.62 ± 4.54**	57.50 ± 8.49	63.00 ± 2.07	14.05 ± 5.17	42.42 ± 4.22	40.29 ± 3.90	51.77 ± 5.29	50.24 ± 2.74	54.32 ± 5.40	46.37 ± 2.70	42.42 ± 4.22	49.78 ± 4.68
MiddlePhalanxOutlineAgeGroup	47.01 ± 9.52	44.55 ± 12.64	45.45 ± 9.45	**50.39 ± 7.58**	44.29 ± 10.12	37.45 ± 10.11	46.75 ± 13.26	45.24 ± 16.76	48.57 ± 12.10	48.92 ± 14.26	22.86 ± 4.12	42.99 ± 12.95
MiddlePhalanxOutlineCorrect	47.01 ± 7.12	51.13 ± 7.83	49.42 ± 8.07	53.75 ± 12.47	44.54 ± 7.47	52.92 ± 7.99	47.54 ± 10.52	**54.07 ± 11.30**	48.59 ± 8.66	53.26 ± 15.07	50.72 ± 11.13	48.87 ± 3.56
MiddlePhalanxTW	37.92 ± 11.48	44.68 ± 3.99	42.73 ± 9.27	38.96 ± 14.59	39.22 ± 13.51	45.45 ± 5.95	43.07 ± 3.27	36.36 ± 13.61	**46.36 ± 2.32**	45.24 ± 3.58	16.36 ± 7.24	44.81 ± 7.78
MixedShapesRegularTrain	55.18 ± 4.21	33.76 ± 13.30	55.82 ± 8.68	58.38 ± 6.27	54.48 ± 12.34	62.60 ± 3.23	19.42 ± 0.39	63.00 ± 3.88	22.36 ± 6.85	63.64 ± 2.51	54.05 ± 4.44	**67.98 ± 4.06**
MixedShapesSmallTrain	45.73 ± 9.51	37.76 ± 12.10	51.77 ± 7.21	61.69 ± 3.51	54.28 ± 10.55	63.00 ± 2.46	19.05 ± 0.77	64.04 ± 1.32	24.01 ± 6.26	62.64 ± 3.39	57.53 ± 4.77	**68.35 ± 3.26**
MoteStrain	81.95 ± 4.07	75.34 ± 9.74	78.95 ± 3.25	70.97 ± 10.57	75.59 ± 8.13	78.41 ± 9.87	68.53 ± 3.31	82.48 ± 1.96	68.63 ± 10.07	**83.65 ± 4.68**	55.75 ± 11.79	58.56 ± 15.12
NonInvasiveFetalECGThorax1	25.55 ± 2.02	14.44 ± 3.90	25.55 ± 2.02	**55.96 ± 3.30**	18.44 ± 9.38	50.77 ± 4.84	18.37 ± 2.16	51.65 ± 2.21	21.42 ± 2.01	53.89 ± 1.99	13.55 ± 0.51	50.58 ± 2.13
NonInvasiveFetalECGThorax2	27.07 ± 5.56	22.30 ± 6.32	27.07 ± 5.56	**62.09 ± 2.37**	28.27 ± 8.71	55.69 ± 4.28	27.23 ± 2.03	56.05 ± 0.94	30.73 ± 3.23	58.63 ± 2.42	18.29 ± 1.75	59.72 ± 2.76
OSULeaf	55.70 ± 9.28	38.35 ± 6.31	**56.03 ± 5.89**	25.79 ± 2.77	25.29 ± 5.19	29.89 ± 3.34	18.18 ± 1.43	28.10 ± 2.07	21.90 ± 3.33	28.51 ± 4.37	23.31 ± 3.85	36.28 ± 6.97
OliveOil	64.67 ± 10.70	32.67 ± 5.48	71.33 ± 10.70	**80.67 ± 3.65**	17.33 ± 10.90	32.22 ± 11.71	38.89 ± 7.70	25.56 ± 12.62	24.00 ± 10.90	58.89 ± 22.69	22.00 ± 13.04	23.33 ± 9.72
PLAID	19.29 ± 3.99	16.65 ± 2.54	20.89 ± 4.30	22.98 ± 3.80	35.53 ± 7.89	27.56 ± 4.82	17.44 ± 2.97	25.88 ± 4.40	19.81 ± 3.24	28.68 ± 3.91	35.53 ± 7.89	**40.19 ± 5.06**
PhalangesOutlinesCorrect	52.91 ± 9.24	53.03 ± 10.79	52.61 ± 9.16	56.64 ± 11.15	54.13 ± 11.04	**60.64 ± 3.81**	58.70 ± 2.16	60.37 ± 4.20	52.26 ± 8.88	60.06 ± 3.62	49.04 ± 14.25	50.84 ± 2.93
Phoneme	**12.22 ± 1.88**	10.75 ± 2.12	11.22 ± 1.88	5.41 ± 0.40	7.31 ± 0.69	7.77 ± 0.84	4.13 ± 1.74	7.00 ± 0.45	6.19 ± 2.34	5.29 ± 0.61	5.20 ± 0.75	10.02 ± 0.59
PickupGestureWiimoteZ	43.20 ± 11.37	53.20 ± 9.55	**62.40 ± 1.67**	43.20 ± 6.72	40.40 ± 8.29	52.67 ± 10.07	21.33 ± 1.15	53.33 ± 4.16	27.60 ± 3.58	51.33 ± 6.43	40.40 ± 8.29	58.80 ± 3.03
PigAirwayPressure	28.37 ± 3.10	13.08 ± 1.90	**49.52 ± 0.76**	5.00 ± 0.26	3.65 ± 0.87	4.49 ± 0.28	3.21 ± 1.21	7.05 ± 0.73	4.62 ± 1.05	6.09 ± 0.73	5.10 ± 0.80	9.81 ± 0.43
PigArtPressure	50.58 ± 2.62	23.65 ± 3.56	**69.42 ± 3.86**	7.88 ± 0.26	5.29 ± 1.63	10.10 ± 0.48	8.01 ± 1.39	9.78 ± 1.11	7.12 ± 1.24	8.33 ± 0.56	19.13 ± 1.04	30.67 ± 1.38
PigCVP	31.54 ± 0.94	18.08 ± 2.27	**61.92 ± 1.78**	7.02 ± 0.43	3.94 ± 1.10	6.73 ± 0.00	4.01 ± 0.56	7.85 ± 1.00	4.13 ± 0.80	5.29 ± 0.83	10.10 ± 1.40	19.33 ± 0.99
Plane	**100.00 ± 0.00**	94.29 ± 3.75	99.05 ± 1.17	86.10 ± 6.69	91.62 ± 3.39	86.98 ± 7.02	84.44 ± 7.02	81.90 ± 1.90	87.24 ± 4.98	86.98 ± 6.34	21.71 ± 8.89	99.24 ± 0.80
PowerCons	82.11 ± 6.61	**83.78 ± 3.63**	82.33 ± 7.91	72.11 ± 2.98	70.78 ± 14.65	69.81 ± 4.10	52.22 ± 1.47	70.93 ± 7.46	50.67 ± 4.38	70.00 ± 8.73	65.89 ± 16.73	66.78 ± 15.52
ProximalPhalanxOutlineAgeGroup	**68.10 ± 13.07**	64.29 ± 28.01	64.78 ± 22.07	58.44 ± 28.63	68.00 ± 16.32	53.17 ± 31.21	56.26 ± 31.37	56.91 ± 37.90	59.22 ± 23.54	51.38 ± 35.67	33.95 ± 22.63	66.24 ± 17.89
ProximalPhalanxOutlineCorrect	**65.43 ± 5.55**	60.07 ± 16.64	63.85 ± 6.55	63.37 ± 8.02	62.96 ± 6.50	54.75 ± 14.29	54.30 ± 16.96	60.94 ± 6.65	59.86 ± 17.09	55.21 ± 16.86	44.88 ± 18.17	59.93 ± 12.82
ProximalPhalanxTW	50.83 ± 4.70	56.00 ± 9.92	**67.12 ± 8.12**	49.76 ± 11.43	53.37 ± 7.43	58.70 ± 5.35	53.98 ± 16.76	42.28 ± 24.98	56.00 ± 8.06	48.29 ± 19.01	16.98 ± 12.37	57.76 ± 7.59
RefrigerationDevices	**41.39 ± 9.49**	40.37 ± 10.57	41.23 ± 9.97	33.92 ± 0.58	35.31 ± 3.72	36.71 ± 1.61	33.87 ± 4.03	35.64 ± 1.78	33.81 ± 1.50	33.60 ± 3.03	33.07 ± 2.39	33.39 ± 7.13
Rock	32.00 ± 5.48	38.40 ± 8.65	37.80 ± 13.31	67.60 ± 10.53	44.80 ± 18.25	58.67 ± 13.61	**70.00 ± 11.14**	66.67 ± 8.08	47.60 ± 12.28	60.00 ± 14.00	44.40 ± 14.24	64.00 ± 7.07
ScreenType	36.91 ± 4.95	**40.21 ± 6.72**	38.40 ± 6.31	35.25 ± 3.07	36.75 ± 2.52	39.56 ± 4.47	33.78 ± 3.74	38.67 ± 2.63	33.71 ± 2.21	39.11 ± 4.89	33.81 ± 3.79	35.63 ± 4.40
SemgHandGenderCh2	55.13 ± 5.29	58.23 ± 11.97	54.27 ± 6.20	53.27 ± 13.91	**62.87 ± 8.65**	49.22 ± 10.20	55.56 ± 1.55	62.44 ± 3.53	43.87 ± 10.45	56.22 ± 13.04	53.70 ± 4.90	59.73 ± 10.87
SemgHandMovementCh2	33.51 ± 7.12	31.29 ± 5.18	**36.00 ± 6.56**	29.78 ± 3.16	23.56 ± 5.26	24.44 ± 2.52	18.89 ± 0.59	24.44 ± 1.39	18.71 ± 1.13	24.07 ± 5.35	27.91 ± 3.59	31.16 ± 4.70

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Table 4: Per-dataset 2-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
SemgHandSubjectCh2	32.00 ± 3.88	30.71 ± 4.13	33.56 ± 5.04	**45.91 ± 5.66**	40.67 ± 4.39	29.56 ± 4.02	23.56 ± 2.47	32.52 ± 7.62	24.09 ± 3.07	30.81 ± 0.84	29.47 ± 4.28	43.20 ± 6.72
ShakeGestureWiimoteZ	76.00 ± 6.63	76.80 ± 7.43	**81.60 ± 0.89**	47.20 ± 7.95	58.40 ± 5.18	54.00 ± 6.00	26.00 ± 4.00	54.67 ± 4.62	32.40 ± 2.61	56.00 ± 5.29	58.40 ± 5.18	**81.60 ± 2.61**
ShapeletSim	78.89 ± 14.16	**98.33 ± 2.08**	92.11 ± 5.26	50.11 ± 2.87	80.22 ± 16.23	47.59 ± 3.16	50.74 ± 3.06	48.15 ± 1.79	49.22 ± 2.93	46.48 ± 3.06	49.67 ± 1.50	67.44 ± 10.92
ShapesAll	36.80 ± 3.60	28.17 ± 3.75	36.80 ± 3.60	50.73 ± 1.71	32.23 ± 5.06	55.78 ± 1.50	6.11 ± 0.63	53.17 ± 1.00	18.33 ± 5.95	53.33 ± 2.62	27.63 ± 3.10	**63.03 ± 2.22**
SmallKitchenAppliances	48.85 ± 7.91	50.40 ± 8.87	50.19 ± 7.71	41.49 ± 3.33	47.15 ± 6.04	49.60 ± 6.54	35.20 ± 2.93	40.09 ± 2.14	42.03 ± 4.49	44.09 ± 8.36	34.61 ± 4.49	**55.68 ± 8.74**
SmoothSubspace	78.93 ± 10.17	77.87 ± 5.55	80.13 ± 5.80	33.33 ± 0.00	51.20 ± 5.65	58.22 ± 4.73	68.89 ± 7.31	59.11 ± 4.23	70.67 ± 4.88	81.33 ± 6.36	48.27 ± 9.05	**85.07 ± 5.20**
SonyAIBORobotSurface1	90.42 ± 4.34	**94.08 ± 1.77**	89.38 ± 5.35	64.83 ± 12.29	70.05 ± 3.77	62.90 ± 13.91	61.90 ± 7.89	66.61 ± 10.89	59.10 ± 5.18	62.73 ± 9.45	52.65 ± 8.42	71.85 ± 8.28
SonyAIBORobotSurface2	**88.00 ± 2.82**	84.87 ± 6.58	83.67 ± 9.41	74.59 ± 3.28	78.70 ± 2.68	69.60 ± 7.04	77.26 ± 1.79	78.49 ± 0.86	78.30 ± 1.49	79.29 ± 1.17	56.81 ± 17.39	68.21 ± 5.28
StarLightCurves	**76.49 ± 11.97**	66.10 ± 9.91	66.22 ± 10.70	67.92 ± 13.52	71.57 ± 12.97	63.90 ± 14.70	35.37 ± 11.93	65.31 ± 14.87	34.69 ± 14.98	63.78 ± 14.15	68.45 ± 12.08	74.47 ± 15.08
Strawberry	62.81 ± 6.83	57.14 ± 6.89	60.22 ± 9.39	51.84 ± 8.28	56.59 ± 2.41	53.33 ± 7.10	**66.58 ± 2.89**	52.97 ± 8.18	59.62 ± 11.05	54.59 ± 4.47	51.03 ± 11.13	59.41 ± 6.38
SwedishLeaf	**72.45 ± 3.37**	54.30 ± 6.89	**72.45 ± 3.37**	34.02 ± 3.00	43.26 ± 7.26	48.91 ± 2.97	33.28 ± 3.94	46.24 ± 1.25	38.85 ± 4.16	47.52 ± 2.78	7.94 ± 0.74	64.99 ± 3.83
Symbols	**90.49 ± 1.42**	75.88 ± 7.72	88.48 ± 4.41	81.23 ± 0.99	76.18 ± 6.87	81.37 ± 1.84	37.42 ± 3.12	82.04 ± 0.58	58.95 ± 6.86	80.00 ± 1.74	51.50 ± 7.93	82.69 ± 3.43
SyntheticControl	79.67 ± 4.05	85.07 ± 4.63	82.60 ± 6.01	41.40 ± 6.37	83.80 ± 5.24	55.00 ± 3.38	44.11 ± 1.39	59.89 ± 3.15	47.93 ± 4.58	62.89 ± 3.15	20.80 ± 4.95	**87.93 ± 3.95**
ToeSegmentation1	**68.42 ± 7.02**	60.79 ± 8.14	61.58 ± 7.54	53.95 ± 2.19	58.60 ± 10.97	51.90 ± 4.30	53.80 ± 2.21	53.95 ± 0.44	52.37 ± 4.44	51.61 ± 3.08	50.70 ± 2.53	60.26 ± 8.54
ToeSegmentation2	76.15 ± 9.44	**79.69 ± 9.24**	72.15 ± 7.68	48.92 ± 5.64	68.62 ± 12.68	66.41 ± 15.71	51.54 ± 5.38	63.08 ± 15.78	47.08 ± 8.33	63.59 ± 15.79	54.31 ± 14.96	76.15 ± 12.44
Trace	76.00 ± 3.00	76.40 ± 3.05	73.80 ± 2.95	52.40 ± 4.04	76.60 ± 4.39	47.33 ± 3.21	51.00 ± 4.00	49.00 ± 3.00	55.60 ± 4.62	49.00 ± 5.00	46.00 ± 6.78	**98.40 ± 3.05**
TwoLeadECG	**82.53 ± 6.47**	79.19 ± 7.45	70.63 ± 4.92	55.68 ± 3.84	59.95 ± 3.74	62.36 ± 5.86	56.60 ± 1.27	55.90 ± 1.72	55.49 ± 2.80	60.14 ± 7.03	48.48 ± 5.21	74.63 ± 8.09
TwoPatterns	52.71 ± 4.44	71.75 ± 5.17	65.30 ± 11.45	32.62 ± 1.84	44.61 ± 9.45	31.66 ± 1.59	24.88 ± 1.30	28.57 ± 1.14	25.14 ± 0.61	29.88 ± 2.34	27.30 ± 2.15	**78.23 ± 0.41**
UMD	54.86 ± 7.95	71.67 ± 9.36	68.75 ± 16.74	54.17 ± 4.55	62.50 ± 12.00	52.31 ± 6.22	47.69 ± 9.93	60.19 ± 1.45	53.06 ± 4.83	56.25 ± 6.05	47.50 ± 4.16	**78.86 ± 14.11**
UWaveGestureLibraryAll	32.50 ± 3.38	28.30 ± 5.01	31.51 ± 4.14	**69.44 ± 6.55**	43.65 ± 6.39	66.54 ± 6.96	26.72 ± 0.50	66.58 ± 7.48	30.82 ± 2.10	66.61 ± 7.59	54.22 ± 7.86	42.54 ± 0.28
UWaveGestureLibraryX	31.94 ± 2.72	28.97 ± 3.44	30.99 ± 3.77	50.42 ± 3.39	42.01 ± 2.19	**51.69 ± 1.72**	21.35 ± 1.20	51.05 ± 1.76	28.72 ± 2.91	50.93 ± 1.76	22.53 ± 5.80	42.54 ± 0.28
UWaveGestureLibraryY	28.00 ± 4.17	26.42 ± 2.97	27.96 ± 5.26	43.36 ± 3.42	39.43 ± 5.20	42.54 ± 7.86	19.88 ± 2.66	**43.61 ± 6.37**	28.02 ± 2.78	41.75 ± 7.40	28.22 ± 7.42	42.54 ± 0.28
UWaveGestureLibraryZ	31.19 ± 2.60	30.19 ± 2.10	28.42 ± 4.12	44.60 ± 3.60	41.59 ± 3.78	42.17 ± 5.87	23.93 ± 2.40	43.23 ± 7.30	29.59 ± 1.71	40.53 ± 6.44	23.38 ± 7.02	**59.88 ± 2.03**
Wafer	47.08 ± 18.17	57.95 ± 22.06	52.16 ± 16.40	56.77 ± 13.38	49.59 ± 17.01	67.22 ± 1.00	55.02 ± 6.93	66.32 ± 1.28	56.21 ± 9.73	**67.64 ± 1.29**	54.99 ± 20.43	55.01 ± 15.62
Wine	**60.74 ± 8.43**	50.74 ± 7.92	60.48 ± 12.52	53.70 ± 8.88	47.41 ± 11.54	53.70 ± 3.21	53.09 ± 5.35	50.00 ± 0.00	50.74 ± 1.66	54.94 ± 26.28	50.74 ± 1.66	48.89 ± 5.34
WordSynonyms	20.63 ± 4.03	16.05 ± 1.67	20.63 ± 4.03	21.35 ± 4.96	20.06 ± 3.25	29.05 ± 0.77	6.11 ± 1.90	29.73 ± 1.73	13.92 ± 2.48	29.57 ± 1.41	6.43 ± 4.20	**34.92 ± 2.64**
Worms	**54.03 ± 11.78**	44.94 ± 11.42	52.73 ± 7.32	22.60 ± 2.99	28.57 ± 6.43	19.48 ± 1.30	17.32 ± 5.41	17.75 ± 5.25	15.58 ± 4.00	20.35 ± 3.75	18.70 ± 5.00	38.70 ± 4.63
WormsTwoClass	55.32 ± 9.03	52.47 ± 8.50	**55.84 ± 3.31**	48.57 ± 2.69	49.09 ± 4.63	49.78 ± 2.70	51.52 ± 8.84	47.19 ± 3.97	47.27 ± 4.37	46.32 ± 1.50	48.83 ± 7.82	54.03 ± 7.72
Yoga	**52.99 ± 4.56**	50.68 ± 2.33	50.48 ± 3.16	50.86 ± 0.91	51.49 ± 3.75	50.12 ± 2.91	51.48 ± 2.93	50.70 ± 2.90	51.46 ± 3.58	50.58 ± 2.12	51.69 ± 4.15	50.64 ± 4.90
Avg. Acc.	53.98	50.79	54.95	48.01	48.43	49.95	39.45	49.45	41.71	51.14	38.30	**55.55**
Avg. Rank	4.84	5.63	4.83	6.78	6.42	6.13	9.36	6.29	8.63	5.64	9.17	**4.28**
#Best	**33**	13	19	10	2	9	4	6	3	7	2	32

Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
ACSF1	**76.00 ± 2.92**	64.60 ± 5.18	72.80 ± 1.64	36.20 ± 3.49	50.40 ± 2.61	60.00 ± 2.00	40.00 ± 1.00	62.67 ± 2.52	46.80 ± 3.70	58.67 ± 1.53	30.60 ± 4.83	68.33 ± 2.89
Adiac	56.21 ± 4.18	42.81 ± 4.46	56.21 ± 4.18	35.29 ± 1.32	34.58 ± 2.33	43.82 ± 1.54	46.97 ± 3.07	38.62 ± 1.68	64.50 ± 5.47	43.99 ± 3.86	7.11 ± 1.15	**69.51 ± 2.42**
AllGestureWiimoteX	**40.46 ± 4.50**	34.14 ± 3.58	38.71 ± 5.08	23.63 ± 2.19	35.40 ± 4.31	32.24 ± 3.43	13.57 ± 1.57	29.95 ± 4.86	18.17 ± 3.28	29.38 ± 3.89	35.40 ± 4.31	39.40 ± 2.86
AllGestureWiimoteY	43.20 ± 1.70	40.60 ± 1.51	36.20 ± 3.42	27.89 ± 2.07	40.97 ± 3.06	36.76 ± 2.07	15.38 ± 0.81	34.43 ± 2.02	19.49 ± 2.37	35.43 ± 1.55	40.97 ± 3.06	**48.11 ± 2.89**
AllGestureWiimoteZ	**41.46 ± 4.33**	35.71 ± 5.12	36.46 ± 5.30	20.20 ± 1.22	27.11 ± 2.51	26.57 ± 0.25	13.86 ± 1.71	26.67 ± 2.65	17.37 ± 2.23	26.86 ± 0.99	27.11 ± 2.51	36.14 ± 0.97
ArrowHead	66.40 ± 5.44	61.03 ± 5.46	64.23 ± 7.62	59.43 ± 5.27	58.86 ± 12.66	60.19 ± 7.46	42.67 ± 0.87	60.76 ± 1.19	48.11 ± 2.87	63.43 ± 8.42	54.06 ± 9.79	**73.60 ± 3.71**
BME	55.60 ± 7.80	56.93 ± 3.90	68.00 ± 9.90	58.53 ± 7.20	89.60 ± 10.65	58.89 ± 5.55	73.33 ± 8.11	76.89 ± 7.34	80.00 ± 6.70	73.33 ± 5.93	53.07 ± 5.63	**97.73 ± 1.53**
Beef	54.67 ± 4.47	47.33 ± 2.79	55.33 ± 2.98	52.67 ± 7.60	43.33 ± 4.71	64.44 ± 3.85	50.00 ± 3.33	52.22 ± 8.39	62.00 ± 8.03	**84.44 ± 1.92**	35.33 ± 4.47	51.33 ± 6.91
BeetleFly	72.00 ± 7.58	60.00 ± 11.73	79.00 ± 10.84	71.00 ± 14.32	57.00 ± 8.37	70.00 ± 8.66	51.67 ± 2.89	66.67 ± 16.07	49.00 ± 6.52	70.00 ± 10.00	59.00 ± 18.51	**91.67 ± 2.89**
BirdChicken	69.00 ± 12.94	77.00 ± 9.75	**90.00 ± 3.54**	57.00 ± 5.70	61.00 ± 16.73	63.33 ± 7.64	36.67 ± 2.89	65.00 ± 10.00	54.00 ± 8.94	68.33 ± 15.28	54.00 ± 4.18	85.00 ± 11.73
CBF	95.87 ± 2.59	95.58 ± 3.67	**95.89 ± 3.27**	61.47 ± 7.78	95.67 ± 4.48	74.52 ± 5.66	39.04 ± 0.80	80.11 ± 4.97	42.29 ± 1.09	80.59 ± 4.09	41.20 ± 4.62	93.76 ± 3.90
Car	59.33 ± 9.02	51.00 ± 7.96	65.67 ± 5.73	52.33 ± 6.41	45.67 ± 2.79	68.89 ± 6.31	31.67 ± 5.00	48.89 ± 4.19	41.00 ± 12.56	57.78 ± 5.85	36.00 ± 4.50	**69.67 ± 3.98**
Chinatown	96.15 ± 2.16	96.97 ± 0.49	**97.43 ± 0.38**	50.96 ± 33.10	96.03 ± 2.24	84.94 ± 15.12	91.06 ± 1.94	93.00 ± 5.85	89.56 ± 2.00	87.95 ± 8.75	73.00 ± 0.76	93.46 ± 10.56
ChlorineConcentration	35.97 ± 2.09	35.63 ± 3.63	36.15 ± 2.53	35.57 ± 3.02	35.80 ± 3.16	35.02 ± 2.94	36.15 ± 1.24	35.49 ± 4.56	37.02 ± 1.05	35.96 ± 1.50	38.60 ± 5.65	**39.58 ± 3.20**
CinCECGTorso	49.41 ± 5.43	57.77 ± 5.88	56.17 ± 3.91	43.81 ± 1.84	69.26 ± 4.67	63.41 ± 2.73	37.80 ± 1.42	64.95 ± 2.40	64.83 ± 11.83	68.41 ± 5.18	54.58 ± 3.46	**82.56 ± 7.22**
Coffee	86.43 ± 14.81	97.86 ± 1.96	**100.00 ± 0.00**	92.86 ± 3.57	97.86 ± 1.96	86.90 ± 2.06	76.19 ± 2.06	82.14 ± 9.45	92.86 ± 10.41	85.71 ± 0.00	50.00 ± 6.19	94.29 ± 1.96
Computers	58.80 ± 7.81	61.60 ± 5.76	60.96 ± 6.58	46.64 ± 4.37	54.48 ± 7.54	49.87 ± 5.33	50.53 ± 1.29	48.00 ± 8.43	49.04 ± 4.09	48.67 ± 4.31	46.56 ± 5.23	**62.67 ± 4.74**
CricketX	45.90 ± 4.41	43.08 ± 3.97	43.08 ± 2.15	25.23 ± 1.80	36.10 ± 5.19	34.87 ± 4.89	11.79 ± 1.43	30.77 ± 0.51	16.92 ± 1.32	33.68 ± 3.44	11.33 ± 1.41	**49.06 ± 3.21**
CricketY	42.67 ± 4.49	43.08 ± 3.40	43.28 ± 6.01	30.41 ± 3.78	36.92 ± 6.92	37.86 ± 2.42	9.06 ± 1.21	36.58 ± 2.06	15.23 ± 1.85	37.26 ± 2.44	12.05 ± 2.18	**49.66 ± 1.41**
CricketZ	51.59 ± 3.02	49.64 ± 3.11	47.79 ± 6.26	26.31 ± 2.67	36.31 ± 4.35	37.18 ± 1.18	10.77 ± 0.51	34.87 ± 1.43	15.44 ± 1.54	37.18 ± 0.77	10.92 ± 1.59	**53.68 ± 2.06**
Crop	49.93 ± 1.45	46.93 ± 2.27	49.93 ± 1.45	31.91 ± 2.21	38.13 ± 2.76	46.92 ± 2.69	44.47 ± 0.97	47.82 ± 2.55	47.76 ± 1.51	48.69 ± 2.69	8.15 ± 3.22	**52.18 ± 2.24**
DiatomSizeReduction	85.75 ± 6.98	70.26 ± 16.84	**95.95 ± 1.26**	61.57 ± 2.74	65.49 ± 10.17	87.69 ± 0.50	68.63 ± 2.55	66.34 ± 8.52	73.53 ± 9.37	87.25 ± 0.00	41.83 ± 13.83	93.66 ± 1.97
DistalPhalanxOutlineAgeGroup	65.76 ± 4.54	68.63 ± 3.96	67.19 ± 5.75	58.99 ± 14.81	63.31 ± 4.46	64.75 ± 3.74	67.63 ± 2.16	69.30 ± 3.24	68.20 ± 2.35	68.59 ± 2.31	53.67 ± 12.57	**69.35 ± 1.20**
DistalPhalanxOutlineCorrect	66.81 ± 3.92	61.96 ± 7.85	**68.62 ± 6.57**	56.23 ± 10.81	60.07 ± 6.07	62.44 ± 3.65	62.68 ± 3.32	58.94 ± 3.37	63.41 ± 7.40	59.30 ± 3.08	55.87 ± 8.58	63.77 ± 11.63
DistalPhalanxTW	58.99 ± 2.69	57.12 ± 1.40	58.56 ± 3.40	49.64 ± 7.01	**60.58 ± 2.61**	57.31 ± 0.42	51.32 ± 3.62	59.47 ± 4.40	54.10 ± 4.24	56.59 ± 5.05	31.94 ± 15.91	56.40 ± 4.87
DodgerLoopDay	37.75 ± 3.35	40.50 ± 4.11	37.00 ± 3.26	46.75 ± 4.01	44.25 ± 2.09	**52.50 ± 5.45**	22.08 ± 0.72	45.42 ± 2.60	24.25 ± 3.49	50.83 ± 4.73	44.25 ± 2.09	45.75 ± 3.26
DodgerLoopGame	66.52 ± 4.02	82.32 ± 3.38	72.17 ± 3.64	76.09 ± 5.15	**84.49 ± 1.89**	75.36 ± 5.02	50.00 ± 2.61	78.99 ± 7.63	53.33 ± 2.09	77.29 ± 4.82	**84.49 ± 1.89**	72.61 ± 10.25
DodgerLoopWeekend	95.51 ± 1.65	92.46 ± 3.46	94.78 ± 1.88	97.39 ± 0.40	97.25 ± 0.61	96.62 ± 0.84	66.43 ± 4.25	**97.83 ± 0.00**	72.90 ± 1.82	96.86 ± 0.42	97.25 ± 0.61	93.62 ± 2.37
ECG200	76.40 ± 3.58	77.80 ± 6.26	76.20 ± 4.60	75.00 ± 1.58	62.80 ± 16.36	77.67 ± 3.06	76.67 ± 3.79	78.00 ± 2.00	77.60 ± 3.21	**79.33 ± 4.93**	69.60 ± 4.51	67.40 ± 8.91
ECG5000	78.85 ± 9.03	75.51 ± 9.29	75.56 ± 6.88	**84.88 ± 3.51**	71.69 ± 7.99	82.01 ± 5.12	76.33 ± 7.90	83.48 ± 5.73	74.58 ± 8.80	82.23 ± 5.84	18.64 ± 16.84	73.89 ± 8.21
ECGFiveDays	71.54 ± 13.30	70.27 ± 8.41	70.22 ± 9.52	69.38 ± 6.83	77.54 ± 3.65	72.94 ± 9.66	**87.65 ± 8.42**	75.22 ± 2.60	83.30 ± 11.51	75.61 ± 4.26	60.49 ± 6.59	70.96 ± 3.91
EOGHorizontalSignal	24.97 ± 1.88	32.71 ± 1.62	27.35 ± 3.43	29.56 ± 3.95	37.73 ± 6.23	36.92 ± 1.62	10.96 ± 1.15	38.40 ± 2.19	9.94 ± 1.90	35.17 ± 4.76	31.99 ± 1.89	**46.35 ± 2.27**
EOGVerticalSignal	18.18 ± 3.62	22.10 ± 3.22	18.78 ± 4.22	29.01 ± 2.45	**38.23 ± 4.43**	31.77 ± 5.00	8.47 ± 1.80	31.95 ± 3.43	10.39 ± 1.59	31.22 ± 4.29	22.82 ± 2.72	36.74 ± 2.24
Earthquakes	50.94 ± 12.41	50.22 ± 8.85	52.23 ± 11.61	**56.83 ± 2.92**	52.09 ± 5.05	56.35 ± 11.87	51.32 ± 9.11	56.12 ± 3.14	53.38 ± 4.56	54.68 ± 7.61	52.81 ± 9.99	54.68 ± 9.31
ElectricDevices	36.69 ± 5.52	36.63 ± 6.97	36.08 ± 5.74	23.65 ± 1.86	38.77 ± 8.33	38.73 ± 7.16	22.89 ± 3.05	25.09 ± 5.42	31.29 ± 6.08	38.61 ± 5.70	11.47 ± 4.02	**47.06 ± 4.18**
EthanolLevel	27.52 ± 1.92	26.36 ± 0.86	26.60 ± 1.66	27.88 ± 1.25	24.88 ± 0.18	28.40 ± 1.73	28.60 ± 2.78	26.53 ± 0.12	25.40 ± 0.75	**29.53 ± 0.64**	26.32 ± 1.07	25.92 ± 1.40
FaceAll	67.63 ± 4.66	**67.75 ± 7.28**	67.63 ± 4.66	53.96 ± 2.14	41.22 ± 3.29	55.76 ± 2.36	53.79 ± 5.19	55.44 ± 2.51	51.51 ± 5.08	54.30 ± 2.41	17.64 ± 1.73	54.54 ± 1.26
FaceFour	88.18 ± 2.85	61.82 ± 13.74	**89.77 ± 4.62**	78.86 ± 3.17	57.50 ± 8.82	86.36 ± 1.14	64.77 ± 2.27	87.88 ± 1.74	71.82 ± 1.24	85.61 ± 1.74	41.14 ± 6.88	75.68 ± 4.37
FacesUCR	**78.67 ± 2.69**	73.00 ± 4.70	76.25 ± 3.16	50.94 ± 2.11	43.00 ± 7.53	68.33 ± 3.31	58.46 ± 2.45	68.99 ± 1.59	60.62 ± 1.49	67.61 ± 2.78	27.20 ± 3.34	68.58 ± 2.57
FiftyWords	30.07 ± 3.14	27.91 ± 3.50	30.07 ± 3.14	47.03 ± 2.26	37.71 ± 3.50	52.23 ± 1.46	9.45 ± 3.91	56.19 ± 3.63	25.10 ± 2.19	49.38 ± 1.41	40.84 ± 2.35	**61.58 ± 2.55**
Fish	61.26 ± 15.63	48.69 ± 6.91	67.66 ± 6.55	56.80 ± 6.22	42.97 ± 5.32	**68.76 ± 3.49**	25.52 ± 2.93	56.57 ± 5.51	40.80 ± 3.78	67.05 ± 1.44	26.40 ± 8.24	64.57 ± 5.92
FordA	71.41 ± 10.22	**71.94 ± 9.09**	70.55 ± 10.29	50.33 ± 1.09	63.62 ± 6.82	50.28 ± 2.11	51.67 ± 1.61	49.67 ± 1.72	51.11 ± 2.04	49.82 ± 1.42	50.91 ± 1.25	54.00 ± 2.22
FordB	**61.26 ± 2.75**	58.96 ± 1.47	59.04 ± 3.16	50.00 ± 1.41	55.88 ± 4.39	50.86 ± 0.77	49.51 ± 0.81	48.81 ± 1.26	49.56 ± 0.80	50.21 ± 1.36	50.27 ± 2.02	52.32 ± 2.78
FreezerRegularTrain	71.93 ± 7.16	**86.24 ± 2.60**	79.17 ± 4.44	76.42 ± 0.69	77.88 ± 9.36	76.43 ± 0.19	57.27 ± 4.31	76.76 ± 0.69	62.60 ± 6.21	71.04 ± 6.76	59.41 ± 11.73	85.11 ± 3.83
FreezerSmallTrain	**77.10 ± 3.40**	75.64 ± 1.31	75.24 ± 0.61	75.24 ± 1.24	71.21 ± 6.86	75.59 ± 1.06	60.63 ± 1.62	76.02 ± 0.68	66.32 ± 5.59	73.06 ± 2.85	63.24 ± 10.73	75.05 ± 3.10
Fungi	96.34 ± 4.07	34.19 ± 4.54	66.45 ± 5.80	74.09 ± 4.22	74.73 ± 9.52	**97.13 ± 0.31**	75.63 ± 1.35	90.14 ± 2.24	75.27 ± 0.54	96.06 ± 0.82	10.75 ± 6.51	81.18 ± 3.16
GestureMidAirD1	37.54 ± 4.63	52.92 ± 3.41	37.54 ± 4.63	46.00 ± 0.64	43.38 ± 3.51	53.85 ± 0.77	21.03 ± 3.11	52.05 ± 3.11	40.46 ± 4.85	52.82 ± 1.94	43.38 ± 3.51	**60.31 ± 2.35**
GestureMidAirD2	28.15 ± 3.51	42.15 ± 7.00	28.15 ± 3.51	48.31 ± 1.58	46.46 ± 3.63	50.00 ± 1.54	12.82 ± 3.55	49.49 ± 1.60	33.08 ± 9.04	50.00 ± 0.77	46.46 ± 3.63	**54.62 ± 1.96**
GestureMidAirD3	14.00 ± 2.69	19.85 ± 2.33	14.00 ± 2.69	30.92 ± 2.20	26.92 ± 3.31	25.13 ± 1.18	14.10 ± 3.11	31.79 ± 1.60	26.31 ± 4.82	27.18 ± 1.78	26.92 ± 3.31	**32.15 ± 3.33**

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Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
GesturePebbleZ1	71.74 $\pm$ 3.41	64.53 $\pm$ 3.05	73.60 $\pm$ 2.12	65.35 $\pm$ 3.15	63.95 $\pm$ 8.94	65.89 $\pm$ 3.50	23.45 $\pm$ 1.21	73.64 $\pm$ 2.20	28.95 $\pm$ 4.06	**76.16 $\pm$ 6.07**	63.95 $\pm$ 8.94	63.84 $\pm$ 8.99
GesturePebbleZ2	67.34 $\pm$ 3.97	52.53 $\pm$ 17.40	**73.04 $\pm$ 4.07**	58.35 $\pm$ 6.18	48.10 $\pm$ 7.12	62.03 $\pm$ 7.30	19.41 $\pm$ 4.45	64.98 $\pm$ 9.05	24.81 $\pm$ 3.02	60.34 $\pm$ 8.06	48.10 $\pm$ 7.12	60.89 $\pm$ 8.06
GunPoint	82.53 $\pm$ 12.46	77.33 $\pm$ 12.19	79.87 $\pm$ 13.86	59.60 $\pm$ 4.87	73.07 $\pm$ 2.03	66.00 $\pm$ 6.36	55.11 $\pm$ 6.19	70.22 $\pm$ 2.52	65.87 $\pm$ 7.58	66.00 $\pm$ 2.91	62.40 $\pm$ 5.79	**90.13 $\pm$ 10.49**
GunPointAgeSpan	75.70 $\pm$ 7.16	69.81 $\pm$ 6.17	76.46 $\pm$ 8.05	75.51 $\pm$ 2.81	73.67 $\pm$ 12.74	76.79 $\pm$ 5.43	49.26 $\pm$ 6.34	75.74 $\pm$ 4.37	60.38 $\pm$ 3.16	69.20 $\pm$ 7.84	65.70 $\pm$ 10.14	**81.52 $\pm$ 9.45**
GunPointMaleVersusFemale	91.20 $\pm$ 8.64	**96.65 $\pm$ 4.17**	90.32 $\pm$ 9.07	61.14 $\pm$ 7.95	87.53 $\pm$ 12.91	74.37 $\pm$ 13.69	55.17 $\pm$ 0.48	75.00 $\pm$ 13.22	61.20 $\pm$ 4.73	77.95 $\pm$ 14.78	63.92 $\pm$ 13.57	94.62 $\pm$ 6.24
GunPointOldVersusYoung	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	75.87 $\pm$ 2.49	70.29 $\pm$ 8.33	70.79 $\pm$ 4.48	56.93 $\pm$ 3.97	73.23 $\pm$ 4.31	58.98 $\pm$ 5.43	70.48 $\pm$ 8.59	92.38 $\pm$ 8.67	80.44 $\pm$ 8.63
Ham	60.95 $\pm$ 6.21	60.00 $\pm$ 6.77	62.10 $\pm$ 4.59	62.86 $\pm$ 11.45	58.86 $\pm$ 7.48	60.00 $\pm$ 11.70	63.49 $\pm$ 0.55	56.51 $\pm$ 8.59	63.62 $\pm$ 3.26	**66.35 $\pm$ 16.28**	56.76 $\pm$ 9.56	60.57 $\pm$ 9.91
HandOutlines	46.86 $\pm$ 11.04	52.86 $\pm$ 13.07	67.24 $\pm$ 3.85	80.76 $\pm$ 4.40	74.49 $\pm$ 7.28	**83.06 $\pm$ 3.92**	47.84 $\pm$ 8.25	81.62 $\pm$ 4.95	35.95 $\pm$ 0.00	82.34 $\pm$ 4.91	75.57 $\pm$ 5.75	68.54 $\pm$ 6.21
Haptics	32.14 $\pm$ 2.79	28.25 $\pm$ 2.44	31.49 $\pm$ 1.92	35.39 $\pm$ 2.49	28.18 $\pm$ 3.44	36.58 $\pm$ 0.19	21.32 $\pm$ 3.65	**37.12 $\pm$ 1.14**	24.68 $\pm$ 3.46	**37.12 $\pm$ 0.94**	31.82 $\pm$ 7.39	26.95 $\pm$ 4.88
Herring	50.94 $\pm$ 5.70	55.00 $\pm$ 8.93	52.50 $\pm$ 7.46	51.88 $\pm$ 6.76	50.94 $\pm$ 6.68	48.96 $\pm$ 5.02	**57.29 $\pm$ 4.51**	51.56 $\pm$ 7.16	46.88 $\pm$ 13.58	49.48 $\pm$ 2.39	49.06 $\pm$ 7.13	45.00 $\pm$ 4.87
HouseTwenty	83.70 $\pm$ 6.31	81.18 $\pm$ 9.08	81.85 $\pm$ 9.95	58.99 $\pm$ 5.13	76.81 $\pm$ 7.55	73.67 $\pm$ 7.34	54.90 $\pm$ 2.57	66.95 $\pm$ 4.78	59.66 $\pm$ 6.85	72.83 $\pm$ 4.23	63.03 $\pm$ 4.28	**84.37 $\pm$ 7.51**
InlineSkate	20.15 $\pm$ 4.70	22.95 $\pm$ 2.93	19.75 $\pm$ 2.66	21.05 $\pm$ 2.12	18.98 $\pm$ 2.35	24.85 $\pm$ 1.07	16.00 $\pm$ 1.31	24.06 $\pm$ 0.93	16.51 $\pm$ 0.15	24.06 $\pm$ 0.90	19.20 $\pm$ 1.41	**28.07 $\pm$ 1.19**
InsectEPGRegularTrain	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	56.47 $\pm$ 1.83	78.96 $\pm$ 9.81	59.57 $\pm$ 2.67	38.96 $\pm$ 2.13	57.30 $\pm$ 1.41	53.17 $\pm$ 6.32	61.31 $\pm$ 3.42	89.88 $\pm$ 9.24	80.24 $\pm$ 3.80
InsectEPGSmallTrain	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	**100.00 $\pm$ 0.00**	53.98 $\pm$ 1.90	77.03 $\pm$ 8.38	59.44 $\pm$ 2.44	52.61 $\pm$ 2.78	60.51 $\pm$ 1.98	55.66 $\pm$ 1.65	62.12 $\pm$ 1.62	83.13 $\pm$ 0.00	83.05 $\pm$ 2.80
InsectWingbeatSound	25.39 $\pm$ 1.91	24.63 $\pm$ 1.27	26.10 $\pm$ 5.93	53.74 $\pm$ 0.54	42.18 $\pm$ 1.58	56.25 $\pm$ 3.65	25.84 $\pm$ 1.75	**58.96 $\pm$ 0.98**	38.75 $\pm$ 2.42	57.51 $\pm$ 1.69	20.02 $\pm$ 5.68	41.44 $\pm$ 2.25
ItalyPowerDemand	87.66 $\pm$ 4.88	89.29 $\pm$ 4.25	88.36 $\pm$ 4.24	62.66 $\pm$ 11.58	93.39 $\pm$ 3.05	58.50 $\pm$ 7.42	89.57 $\pm$ 5.31	93.78 $\pm$ 2.95	**95.72 $\pm$ 1.14**	85.46 $\pm$ 2.44	56.64 $\pm$ 14.68	84.26 $\pm$ 5.94
LargeKitchenAppliances	60.00 $\pm$ 4.27	58.51 $\pm$ 9.39	**62.67 $\pm$ 10.64**	37.55 $\pm$ 2.23	60.48 $\pm$ 4.88	44.09 $\pm$ 3.07	36.98 $\pm$ 2.46	40.44 $\pm$ 3.60	37.71 $\pm$ 3.74	44.09 $\pm$ 6.61	37.17 $\pm$ 1.99	50.13 $\pm$ 3.07
Lightning2	64.92 $\pm$ 6.82	62.30 $\pm$ 7.42	67.87 $\pm$ 5.51	60.33 $\pm$ 3.91	64.92 $\pm$ 4.86	**69.95 $\pm$ 2.50**	50.82 $\pm$ 5.68	68.85 $\pm$ 5.68	61.64 $\pm$ 7.91	68.31 $\pm$ 3.41	59.02 $\pm$ 6.86	69.51 $\pm$ 10.47
Lightning7	58.90 $\pm$ 7.12	54.25 $\pm$ 7.09	60.55 $\pm$ 3.80	51.23 $\pm$ 2.84	50.96 $\pm$ 8.07	67.58 $\pm$ 3.45	18.26 $\pm$ 3.95	61.64 $\pm$ 3.62	30.41 $\pm$ 0.61	**70.32 $\pm$ 2.85**	22.19 $\pm$ 2.63	66.30 $\pm$ 3.83
Mallat	71.03 $\pm$ 13.07	70.33 $\pm$ 5.20	93.72 $\pm$ 2.08	88.67 $\pm$ 2.40	77.56 $\pm$ 9.50	**94.66 $\pm$ 0.33**	65.67 $\pm$ 5.01	87.51 $\pm$ 2.78	79.72 $\pm$ 5.50	90.06 $\pm$ 1.99	56.21 $\pm$ 5.78	90.88 $\pm$ 1.15
Meat	64.33 $\pm$ 14.22	50.00 $\pm$ 9.93	81.00 $\pm$ 8.47	85.33 $\pm$ 10.70	49.33 $\pm$ 15.48	76.67 $\pm$ 10.41	47.78 $\pm$ 15.03	37.22 $\pm$ 5.36	50.00 $\pm$ 11.61	**88.89 $\pm$ 2.55**	34.00 $\pm$ 1.49	64.00 $\pm$ 11.82
MedicalImages	**45.61 $\pm$ 3.94**	43.95 $\pm$ 4.74	42.63 $\pm$ 4.22	27.26 $\pm$ 7.75	32.26 $\pm$ 2.93	37.41 $\pm$ 0.59	36.84 $\pm$ 3.23	38.03 $\pm$ 3.53	40.16 $\pm$ 6.60	38.20 $\pm$ 1.38	5.24 $\pm$ 3.29	42.03 $\pm$ 2.75
MelbournePedestrian	76.79 $\pm$ 2.31	74.96 $\pm$ 2.28	**77.84 $\pm$ 3.11**	11.65 $\pm$ 1.29	58.25 $\pm$ 3.82	62.66 $\pm$ 1.83	70.73 $\pm$ 4.35	71.76 $\pm$ 3.72	70.23 $\pm$ 2.95	67.90 $\pm$ 5.27	58.25 $\pm$ 3.82	65.81 $\pm$ 3.77
MiddlePhalanxOutlineAgeGroup	47.01 $\pm$ 8.99	54.03 $\pm$ 8.97	50.78 $\pm$ 7.01	52.08 $\pm$ 8.63	49.09 $\pm$ 6.85	50.22 $\pm$ 4.61	51.73 $\pm$ 7.75	53.90 $\pm$ 7.87	**55.06 $\pm$ 4.76**	54.11 $\pm$ 9.46	31.82 $\pm$ 16.75	49.74 $\pm$ 4.72
MiddlePhalanxOutlineCorrect	53.20 $\pm$ 7.83	52.71 $\pm$ 15.62	58.14 $\pm$ 12.45	46.94 $\pm$ 12.76	48.93 $\pm$ 10.44	49.60 $\pm$ 3.67	**58.65 $\pm$ 6.95**	55.44 $\pm$ 15.07	51.68 $\pm$ 10.94	49.60 $\pm$ 9.23	50.93 $\pm$ 8.48	48.45 $\pm$ 7.93
MiddlePhalanxTW	**45.84 $\pm$ 9.64**	**45.84 $\pm$ 3.45**	45.71 $\pm$ 4.74	**45.84 $\pm$ 4.10**	43.12 $\pm$ 4.67	44.37 $\pm$ 0.37	40.04 $\pm$ 2.62	43.72 $\pm$ 4.56	44.16 $\pm$ 4.95	44.59 $\pm$ 2.28	22.86 $\pm$ 6.13	43.64 $\pm$ 5.71
MixedShapesRegularTrain	63.93 $\pm$ 9.77	61.94 $\pm$ 4.99	68.21 $\pm$ 8.35	68.04 $\pm$ 1.34	68.63 $\pm$ 3.58	67.15 $\pm$ 2.55	20.78 $\pm$ 0.21	70.05 $\pm$ 0.70	30.67 $\pm$ 6.49	67.15 $\pm$ 1.60	64.11 $\pm$ 1.37	**76.68 $\pm$ 5.11**
MixedShapesSmallTrain	66.24 $\pm$ 12.23	57.21 $\pm$ 4.26	72.37 $\pm$ 7.91	66.37 $\pm$ 2.62	66.13 $\pm$ 3.66	68.30 $\pm$ 1.42	22.61 $\pm$ 0.88	67.60 $\pm$ 3.67	25.23 $\pm$ 4.04	66.78 $\pm$ 1.26	60.00 $\pm$ 5.54	**78.71 $\pm$ 1.70**
MoteStrain	84.74 $\pm$ 4.04	83.85 $\pm$ 4.63	85.10 $\pm$ 2.03	80.19 $\pm$ 8.46	77.96 $\pm$ 3.58	**85.73 $\pm$ 4.20**	72.68 $\pm$ 2.25	80.09 $\pm$ 3.68	77.11 $\pm$ 6.04	85.36 $\pm$ 1.87	60.30 $\pm$ 15.64	80.37 $\pm$ 4.50
NonInvasiveFetalECGThorax1	40.80 $\pm$ 1.72	31.63 $\pm$ 3.55	40.80 $\pm$ 1.72	**72.10 $\pm$ 1.96**	51.79 $\pm$ 9.05	70.09 $\pm$ 3.02	22.04 $\pm$ 0.40	70.72 $\pm$ 1.13	32.89 $\pm$ 2.08	69.19 $\pm$ 0.16	28.56 $\pm$ 4.30	67.42 $\pm$ 0.56
NonInvasiveFetalECGThorax2	41.15 $\pm$ 5.46	43.13 $\pm$ 8.61	41.15 $\pm$ 5.46	**78.55 $\pm$ 0.96**	60.71 $\pm$ 3.55	75.84 $\pm$ 1.88	37.61 $\pm$ 1.37	75.40 $\pm$ 1.50	50.66 $\pm$ 2.46	74.93 $\pm$ 2.49	37.30 $\pm$ 2.25	75.31 $\pm$ 0.82
OSULeaf	57.85 $\pm$ 5.97	50.74 $\pm$ 10.99	**72.23 $\pm$ 4.84**	32.40 $\pm$ 1.86	35.37 $\pm$ 3.13	38.84 $\pm$ 1.09	16.80 $\pm$ 1.04	32.64 $\pm$ 2.71	24.05 $\pm$ 2.60	36.91 $\pm$ 1.04	23.06 $\pm$ 4.07	49.92 $\pm$ 6.46
OliveOil	59.33 $\pm$ 13.21	28.67 $\pm$ 13.66	**74.00 $\pm$ 6.83**	31.33 $\pm$ 9.60	26.67 $\pm$ 10.00	42.22 $\pm$ 25.02	40.00 $\pm$ 10.00	32.22 $\pm$ 13.47	40.00 $\pm$ 21.73	32.22 $\pm$ 22.19	26.67 $\pm$ 10.00	28.67 $\pm$ 11.69
PLAID	24.02 $\pm$ 4.96	24.51 $\pm$ 3.44	22.87 $\pm$ 2.95	30.06 $\pm$ 4.34	47.93 $\pm$ 3.93	36.56 $\pm$ 2.69	26.88 $\pm$ 2.01	40.16 $\pm$ 2.24	27.26 $\pm$ 3.48	37.93 $\pm$ 4.09	47.93 $\pm$ 3.93	**52.96 $\pm$ 5.39**
PhalangesOutlinesCorrect	56.22 $\pm$ 7.89	55.22 $\pm$ 9.82	**56.85 $\pm$ 6.96**	52.75 $\pm$ 11.57	53.15 $\pm$ 7.53	51.83 $\pm$ 11.55	50.35 $\pm$ 9.47	51.20 $\pm$ 13.21	53.05 $\pm$ 9.33	50.85 $\pm$ 11.66	53.61 $\pm$ 9.97	53.61 $\pm$ 8.69
Phoneme	**19.05 $\pm$ 2.97**	18.15 $\pm$ 2.47	**19.05 $\pm$ 2.97**	6.61 $\pm$ 0.25	9.00 $\pm$ 1.08	10.58 $\pm$ 0.95	6.87 $\pm$ 0.83	9.00 $\pm$ 0.29	9.61 $\pm$ 1.88	9.30 $\pm$ 1.34	8.21 $\pm$ 0.75	15.63 $\pm$ 1.56
PickupGestureWiimoteZ	56.00 $\pm$ 10.20	66.80 $\pm$ 5.93	**73.60 $\pm$ 4.56**	52.40 $\pm$ 1.67	56.80 $\pm$ 9.55	57.33 $\pm$ 3.06	27.33 $\pm$ 1.15	58.00 $\pm$ 3.46	42.00 $\pm$ 2.00	58.67 $\pm$ 1.15	56.80 $\pm$ 9.55	68.00 $\pm$ 2.45
PigAirwayPressure	25.00 $\pm$ 2.61	13.37 $\pm$ 2.05	**48.17 $\pm$ 0.40**	5.10 $\pm$ 0.26	4.62 $\pm$ 0.55	4.81 $\pm$ 0.00	4.17 $\pm$ 2.22	6.73 $\pm$ 0.48	4.04 $\pm$ 0.87	5.93 $\pm$ 0.56	4.42 $\pm$ 0.86	9.81 $\pm$ 0.73
PigArtPressure	50.96 $\pm$ 5.81	20.77 $\pm$ 3.29	**68.17 $\pm$ 4.63**	8.46 $\pm$ 0.65	5.10 $\pm$ 1.82	8.97 $\pm$ 1.21	6.57 $\pm$ 2.00	8.97 $\pm$ 0.73	6.92 $\pm$ 1.65	9.29 $\pm$ 1.11	18.17 $\pm$ 2.24	33.08 $\pm$ 1.33
PigCVP	29.33 $\pm$ 2.90	18.75 $\pm$ 2.40	**62.98 $\pm$ 1.73**	6.83 $\pm$ 0.22	3.37 $\pm$ 1.32	7.53 $\pm$ 0.56	4.65 $\pm$ 1.11	7.53 $\pm$ 0.28	3.75 $\pm$ 1.24	4.81 $\pm$ 0.48	10.38 $\pm$ 1.21	19.52 $\pm$ 0.55
Plane	**100.00 $\pm$ 0.00**	98.10 $\pm$ 2.43	99.62 $\pm$ 0.52	88.95 $\pm$ 7.93	95.24 $\pm$ 3.16	97.78 $\pm$ 0.55	96.51 $\pm$ 1.98	95.87 $\pm$ 1.45	97.14 $\pm$ 1.51	95.87 $\pm$ 0.55	23.05 $\pm$ 11.32	99.24 $\pm$ 1.24
PowerCons	80.56 $\pm$ 3.36	**84.44 $\pm$ 5.02**	84.00 $\pm$ 6.30	70.11 $\pm$ 5.26	81.11 $\pm$ 10.74	75.37 $\pm$ 5.45	54.26 $\pm$ 3.90	80.93 $\pm$ 10.72	52.22 $\pm$ 4.39	78.52 $\pm$ 10.32	68.89 $\pm$ 15.97	76.11 $\pm$ 6.55
ProximalPhalanxOutlineAgeGroup	73.37 $\pm$ 8.19	81.76 $\pm$ 3.12	79.90 $\pm$ 3.95	65.56 $\pm$ 9.78	75.41 $\pm$ 5.61	79.02 $\pm$ 7.19	79.02 $\pm$ 8.46	**84.07 $\pm$ 1.71**	74.34 $\pm$ 7.73	80.00 $\pm$ 3.41	43.61 $\pm$ 12.69	76.88 $\pm$ 5.79
ProximalPhalanxOutlineCorrect	**69.35 $\pm$ 11.27**	69.00 $\pm$ 5.91	69.28 $\pm$ 9.35	64.47 $\pm$ 0.52	63.71 $\pm$ 8.48	65.18 $\pm$ 3.44	64.60 $\pm$ 1.57	63.69 $\pm$ 1.73	63.16 $\pm$ 5.61	66.67 $\pm$ 4.47	55.26 $\pm$ 17.89	65.84 $\pm$ 4.70
ProximalPhalanxTW	62.34 $\pm$ 7.16	66.44 $\pm$ 4.62	**69.37 $\pm$ 2.52**	53.37 $\pm$ 11.45	65.27 $\pm$ 5.74	58.05 $\pm$ 7.67	58.21 $\pm$ 7.39	55.45 $\pm$ 4.15	60.00 $\pm$ 4.28	55.28 $\pm$ 3.66	27.32 $\pm$ 18.90	66.24 $\pm$ 3.19
RefrigerationDevices	42.83 $\pm$ 1.65	43.20 $\pm$ 6.13	**44.00 $\pm$ 6.18**	33.33 $\pm$ 2.73	37.92 $\pm$ 6.42	38.93 $\pm$ 3.24	31.20 $\pm$ 3.23	32.98 $\pm$ 2.22	32.32 $\pm$ 2.29	38.04 $\pm$ 3.74	34.61 $\pm$ 1.83	40.96 $\pm$ 5.59
Rock	42.00 $\pm$ 5.29	36.80 $\pm$ 7.16	38.40 $\pm$ 4.34	80.40 $\pm$ 2.97	60.80 $\pm$ 5.93	80.67 $\pm$ 1.15	81.33 $\pm$ 3.06	**88.67 $\pm$ 2.31**	59.20 $\pm$ 3.63	85.33 $\pm$ 3.06	33.60 $\pm$ 5.90	76.00 $\pm$ 3.16
ScreenType	**42.56 $\pm$ 7.61**	41.33 $\pm$ 8.02	40.64 $\pm$ 8.17	35.41 $\pm$ 2.78	36.16 $\pm$ 4.77	35.56 $\pm$ 2.39	32.18 $\pm$ 2.24	36.09 $\pm$ 2.30	32.53 $\pm$ 1.41	35.02 $\pm$ 0.41	34.08 $\pm$ 2.12	34.51 $\pm$ 4.14
SemgHandGenderCh2	58.13 $\pm$ 2.49	66.47 $\pm$ 7.67	56.07 $\pm$ 2.82	65.13 $\pm$ 3.61	**67.13 $\pm$ 2.65**	50.28 $\pm$ 5.31	55.83 $\pm$ 13.67	55.06 $\pm$ 2.04	54.63 $\pm$ 4.30	47.94 $\pm$ 16.07	59.97 $\pm$ 7.08	66.80 $\pm$ 7.25
SemgHandMovementCh2	30.89 $\pm$ 3.99	33.91 $\pm$ 3.59	34.40 $\pm$ 6.06	32.84 $\pm$ 5.27	26.49 $\pm$ 1.89	27.70 $\pm$ 2.36	20.59 $\pm$ 1.85	29.48 $\pm$ 2.44	21.64 $\pm$ 2.79	23.41 $\pm$ 3.03	33.16 $\pm$ 2.38	**36.31 $\pm$ 5.92**

Continued on next page

Table 5: Per-dataset 5-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
SemgHandSubjectCh2	37.11 ± 4.58	44.09 ± 1.20	35.60 ± 3.13	50.76 ± 1.90	49.02 ± 0.83	46.67 ± 2.12	27.78 ± 0.59	51.19 ± 2.74	30.62 ± 1.93	41.48 ± 6.03	36.62 ± 4.30	**54.31 ± 3.37**
ShakeGestureWiimoteZ	80.40 ± 4.34	91.60 ± 2.61	89.20 ± 1.10	56.40 ± 1.67	72.80 ± 5.22	62.00 ± 0.00	35.33 ± 1.15	60.67 ± 3.06	44.00 ± 2.45	55.33 ± 2.31	72.80 ± 5.22	**92.00 ± 0.00**
ShapeletSim	82.78 ± 10.96	**99.67 ± 0.75**	92.56 ± 6.55	49.44 ± 2.69	88.22 ± 10.06	55.93 ± 2.31	49.81 ± 2.80	49.63 ± 4.21	50.67 ± 1.82	49.44 ± 3.33	49.67 ± 3.98	92.00 ± 3.88
ShapesAll	57.57 ± 3.68	48.33 ± 5.58	57.57 ± 3.68	58.13 ± 1.68	51.50 ± 3.69	64.67 ± 1.64	9.22 ± 0.79	65.83 ± 3.88	13.83 ± 3.47	59.22 ± 2.28	56.47 ± 1.46	**76.20 ± 1.83**
SmallKitchenAppliances	59.47 ± 8.51	57.39 ± 6.38	**62.35 ± 6.85**	39.04 ± 4.11	57.55 ± 2.52	57.51 ± 6.62	38.67 ± 2.54	39.56 ± 5.26	46.67 ± 3.53	56.62 ± 3.96	44.00 ± 7.15	51.63 ± 12.25
SmoothSubspace	86.80 ± 2.13	82.00 ± 2.83	**88.67 ± 2.36**	33.87 ± 1.19	57.07 ± 4.58	65.78 ± 2.04	69.33 ± 3.71	61.11 ± 0.77	77.07 ± 5.51	84.00 ± 4.81	37.07 ± 6.74	88.00 ± 2.87
SonyAIBORobotSurface1	93.94 ± 2.22	**95.71 ± 1.41**	92.85 ± 2.46	61.83 ± 10.72	84.73 ± 2.02	71.21 ± 3.16	67.39 ± 1.20	71.60 ± 6.11	64.96 ± 2.40	64.89 ± 3.03	54.54 ± 12.34	84.86 ± 2.67
SonyAIBORobotSurface2	**92.51 ± 3.32**	91.14 ± 0.89	91.54 ± 4.19	78.15 ± 2.59	85.83 ± 4.10	79.96 ± 1.73	83.11 ± 2.73	81.78 ± 2.73	80.76 ± 4.57	83.21 ± 2.88	61.15 ± 14.84	77.21 ± 3.67
StarLightCurves	81.52 ± 8.85	60.67 ± 16.70	76.44 ± 1.22	74.70 ± 4.25	76.72 ± 3.78	75.63 ± 4.10	28.48 ± 9.45	75.33 ± 3.39	27.99 ± 0.00	75.69 ± 5.23	75.95 ± 2.54	**81.86 ± 7.93**
Strawberry	63.51 ± 2.99	64.22 ± 4.90	60.22 ± 9.83	58.49 ± 2.90	63.03 ± 7.09	61.89 ± 4.92	**67.21 ± 4.90**	59.82 ± 1.80	61.03 ± 3.64	65.77 ± 7.40	52.92 ± 6.34	58.05 ± 7.04
SwedishLeaf	**79.81 ± 1.70**	76.00 ± 4.97	**79.81 ± 1.70**	47.97 ± 1.68	65.66 ± 4.54	62.61 ± 4.53	49.23 ± 4.26	67.15 ± 4.00	56.93 ± 3.43	62.29 ± 1.68	15.90 ± 6.80	79.78 ± 2.56
Symbols	88.78 ± 7.99	88.74 ± 7.74	**91.24 ± 0.96**	83.44 ± 1.29	84.92 ± 3.31	84.29 ± 1.01	36.11 ± 0.23	86.40 ± 1.63	58.19 ± 7.40	83.75 ± 0.90	27.56 ± 4.43	88.04 ± 1.30
SyntheticControl	87.80 ± 2.38	**96.33 ± 0.33**	92.60 ± 1.01	48.53 ± 3.68	92.73 ± 2.94	73.22 ± 3.10	52.11 ± 3.20	66.22 ± 3.66	56.27 ± 3.45	72.78 ± 3.66	20.93 ± 4.87	95.07 ± 1.62
ToeSegmentation1	**85.96 ± 7.05**	77.37 ± 5.14	81.58 ± 5.14	54.74 ± 3.58	68.16 ± 3.79	55.56 ± 2.16	53.95 ± 2.74	51.61 ± 0.91	53.25 ± 3.38	54.82 ± 1.32	48.77 ± 4.31	70.53 ± 10.39
ToeSegmentation2	74.77 ± 12.53	77.85 ± 7.33	74.77 ± 8.79	50.77 ± 2.88	68.46 ± 6.25	58.97 ± 1.60	50.26 ± 0.44	59.49 ± 6.72	37.54 ± 3.14	64.10 ± 2.70	52.31 ± 9.12	**78.00 ± 6.36**
Trace	88.60 ± 7.37	99.00 ± 1.00	93.00 ± 8.00	53.00 ± 4.42	76.40 ± 5.77	51.67 ± 4.93	53.33 ± 2.08	49.67 ± 3.79	59.80 ± 2.59	55.67 ± 8.50	48.80 ± 10.71	**100.00 ± 0.00**
TwoLeadECG	**97.98 ± 4.17**	95.47 ± 5.89	75.84 ± 5.58	58.75 ± 2.38	82.55 ± 14.28	62.16 ± 1.52	60.81 ± 0.10	56.34 ± 2.59	61.88 ± 1.65	68.51 ± 9.61	52.10 ± 2.73	96.68 ± 5.58
TwoPatterns	64.62 ± 3.64	**74.98 ± 5.04**	74.94 ± 1.43	38.09 ± 3.07	51.91 ± 7.48	40.83 ± 3.89	26.00 ± 0.92	39.83 ± 3.80	26.50 ± 1.83	37.54 ± 1.63	27.50 ± 4.13	74.22 ± 2.81
UMD	87.08 ± 10.10	91.53 ± 5.28	**96.67 ± 2.05**	56.53 ± 3.69	74.44 ± 17.00	61.57 ± 1.45	62.96 ± 5.82	68.52 ± 10.79	71.81 ± 2.17	70.14 ± 10.69	48.47 ± 7.32	95.78 ± 2.33
UWaveGestureLibraryAll	42.53 ± 1.15	39.33 ± 5.11	44.53 ± 1.84	77.02 ± 5.21	57.17 ± 2.24	**78.18 ± 3.39**	26.42 ± 1.62	77.47 ± 3.61	35.67 ± 1.53	76.96 ± 4.32	65.94 ± 6.48	48.36 ± 0.30
UWaveGestureLibraryX	39.64 ± 3.12	46.31 ± 4.46	46.23 ± 4.40	56.88 ± 2.64	50.95 ± 2.22	58.42 ± 2.37	19.27 ± 3.18	**58.58 ± 2.34**	31.22 ± 3.55	57.86 ± 1.47	26.48 ± 5.59	47.36 ± 0.30
UWaveGestureLibraryY	37.00 ± 1.20	38.44 ± 2.57	40.23 ± 3.63	**52.69 ± 2.24**	48.41 ± 3.12	51.42 ± 3.10	21.22 ± 1.11	51.72 ± 2.98	30.92 ± 5.43	50.81 ± 4.53	19.79 ± 3.07	51.36 ± 0.31
UWaveGestureLibraryZ	38.74 ± 2.41	44.95 ± 4.51	40.85 ± 4.13	48.35 ± 3.41	47.76 ± 4.10	50.34 ± 2.37	20.23 ± 1.68	49.48 ± 1.66	31.23 ± 4.16	48.05 ± 3.36	25.20 ± 6.78	**55.54 ± 2.61**
Wafer	76.83 ± 9.15	69.99 ± 14.88	71.79 ± 15.46	55.60 ± 16.77	73.34 ± 19.66	80.42 ± 16.85	76.25 ± 9.34	**83.86 ± 16.37**	75.80 ± 12.54	83.26 ± 17.85	64.01 ± 22.29	71.65 ± 9.58
Wine	57.41 ± 13.03	48.15 ± 3.93	**61.11 ± 5.40**	55.93 ± 3.56	53.33 ± 5.62	47.53 ± 8.35	54.94 ± 8.55	50.00 ± 0.00	46.67 ± 6.47	37.65 ± 19.80	50.00 ± 0.00	51.48 ± 2.03
WordSynonyms	25.08 ± 1.54	21.22 ± 1.56	25.08 ± 1.54	29.56 ± 2.06	30.38 ± 2.74	45.87 ± 4.63	7.99 ± 1.24	44.67 ± 4.83	18.81 ± 2.07	41.33 ± 4.67	22.26 ± 2.64	**47.05 ± 3.87**
Worms	57.14 ± 9.76	55.58 ± 5.91	**62.86 ± 4.55**	24.68 ± 4.68	40.26 ± 5.11	22.94 ± 0.75	19.91 ± 3.27	25.97 ± 0.00	16.62 ± 2.32	22.08 ± 1.30	22.08 ± 5.11	44.42 ± 6.26
WormsTwoClass	**68.05 ± 7.21**	59.74 ± 6.69	62.08 ± 7.97	48.83 ± 3.13	51.17 ± 8.09	52.38 ± 2.70	42.86 ± 3.44	51.08 ± 5.25	48.83 ± 3.26	52.81 ± 3.27	48.83 ± 5.92	54.81 ± 7.59
Yoga	**55.71 ± 3.37**	53.57 ± 4.72	55.19 ± 2.57	50.13 ± 2.22	53.89 ± 3.78	49.79 ± 3.21	52.94 ± 1.57	47.50 ± 3.55	51.67 ± 1.99	49.68 ± 4.28	48.87 ± 2.05	51.31 ± 4.34
Avg. Acc.	61.27	59.47	63.55	52.30	56.94	57.86	43.20	57.20	47.30	58.35	42.94	**64.14**
Avg. Rank	4.77	5.34	4.23	7.73	6.60	5.69	9.55	5.94	8.70	5.79	9.48	**4.19**
#Best	21	13	30	6	4	8	4	7	2	8	1	**36**

Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets. Each value is reported as mean ± standard deviation over 5 runs.

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
ACSF1	**85.40 ± 2.19**	76.80 ± 3.35	80.80 ± 0.84	38.60 ± 0.89	58.80 ± 2.59	64.67 ± 0.58	42.67 ± 1.53	67.00 ± 2.00	53.60 ± 2.51	67.00 ± 2.00	36.80 ± 3.27	81.33 ± 1.53
Adiac	62.46 ± 4.53	56.98 ± 3.84	62.46 ± 4.53	45.32 ± 1.06	47.72 ± 5.48	56.01 ± 2.77	48.68 ± 0.78	47.91 ± 6.68	71.71 ± 1.94	48.00 ± 3.58	12.84 ± 1.42	**76.42 ± 1.68**
AllGestureWiimoteX	49.37 ± 3.15	43.11 ± 2.57	39.26 ± 2.50	23.91 ± 1.60	43.54 ± 2.85	37.14 ± 0.52	15.00 ± 0.87	35.19 ± 1.33	25.80 ± 2.24	31.00 ± 1.57	43.54 ± 2.85	**52.31 ± 2.26**
AllGestureWiimoteY	54.89 ± 2.09	50.74 ± 4.62	40.17 ± 2.24	27.26 ± 1.11	48.03 ± 1.58	42.57 ± 3.19	16.95 ± 2.22	42.57 ± 3.81	30.37 ± 4.65	41.24 ± 2.84	48.03 ± 1.58	**56.34 ± 3.54**
AllGestureWiimoteZ	**49.34 ± 1.38**	46.03 ± 2.76	39.80 ± 2.97	22.43 ± 1.80	34.54 ± 4.20	31.29 ± 1.13	16.67 ± 1.15	29.67 ± 3.32	21.71 ± 1.69	32.19 ± 0.59	34.54 ± 4.20	44.43 ± 2.91
ArrowHead	72.23 ± 5.22	63.89 ± 7.06	65.49 ± 9.09	64.34 ± 0.65	62.86 ± 7.57	60.38 ± 1.19	48.19 ± 2.58	59.62 ± 2.88	57.83 ± 5.80	63.43 ± 3.48	45.83 ± 13.23	**81.49 ± 3.22**
BME	64.27 ± 9.29	69.33 ± 1.56	78.67 ± 4.64	64.53 ± 0.30	97.60 ± 3.29	71.11 ± 2.34	78.89 ± 1.02	88.44 ± 1.02	91.33 ± 2.31	86.67 ± 0.00	46.40 ± 7.46	**99.20 ± 0.30**
Beef	51.33 ± 6.91	44.00 ± 6.83	52.00 ± 2.98	61.33 ± 4.47	46.00 ± 6.41	70.00 ± 3.33	51.11 ± 5.09	54.44 ± 3.85	64.00 ± 7.23	**80.00 ± 0.00**	37.33 ± 4.35	68.00 ± 1.83
BeetleFly	71.00 ± 8.94	74.00 ± 6.52	75.00 ± 6.12	86.00 ± 4.18	86.00 ± 2.24	81.67 ± 2.89	48.33 ± 2.89	83.33 ± 2.89	49.00 ± 11.94	75.00 ± 5.00	70.00 ± 12.25	**96.67 ± 2.89**
BirdChicken	84.00 ± 12.45	92.00 ± 5.70	**94.00 ± 4.18**	68.00 ± 2.74	81.00 ± 9.62	71.67 ± 7.64	41.67 ± 7.64	68.33 ± 16.07	56.00 ± 5.48	56.67 ± 7.64	56.00 ± 4.18	89.00 ± 5.48
CBF	98.96 ± 0.23	98.27 ± 0.96	**99.24 ± 0.32**	67.49 ± 2.09	95.98 ± 4.64	90.07 ± 1.09	39.37 ± 1.51	91.44 ± 1.93	51.24 ± 7.49	90.11 ± 0.88	49.31 ± 9.05	97.71 ± 0.69
Car	57.67 ± 10.84	58.67 ± 3.21	65.33 ± 2.17	73.00 ± 2.98	62.67 ± 5.22	70.00 ± 2.89	27.22 ± 5.85	65.56 ± 0.96	37.33 ± 3.84	72.22 ± 1.92	50.33 ± 8.28	**76.33 ± 5.45**
Chinatown	98.19 ± 0.13	98.13 ± 0.16	**98.54 ± 0.21**	54.58 ± 24.80	97.73 ± 0.75	94.75 ± 2.10	95.63 ± 1.01	97.28 ± 0.61	96.62 ± 0.60	96.60 ± 1.10	63.56 ± 20.21	94.56 ± 3.09
ChlorineConcentration	**41.18 ± 3.66**	34.41 ± 8.63	37.86 ± 5.20	34.42 ± 4.40	35.51 ± 6.32	35.06 ± 1.39	38.85 ± 1.62	32.99 ± 4.01	39.05 ± 4.08	39.50 ± 4.07	33.01 ± 11.68	34.77 ± 6.93
CinCECGTorso	54.03 ± 3.03	65.94 ± 5.62	64.64 ± 5.11	45.36 ± 1.09	80.65 ± 7.24	81.59 ± 1.76	35.82 ± 2.41	84.42 ± 2.94	63.93 ± 1.27	85.19 ± 1.24	52.77 ± 4.47	**94.71 ± 4.65**
Coffee	97.14 ± 6.39	97.14 ± 1.60	**100.00 ± 0.00**	96.43 ± 4.37	97.86 ± 1.96	91.67 ± 2.06	86.90 ± 16.50	92.86 ± 0.00	98.57 ± 3.19	90.48 ± 2.06	56.43 ± 7.74	95.71 ± 2.99
Computers	66.48 ± 3.03	63.20 ± 6.81	**68.08 ± 4.62**	48.88 ± 2.52	56.40 ± 10.98	52.13 ± 9.60	52.13 ± 6.02	50.93 ± 12.73	49.68 ± 5.04	53.87 ± 12.94	51.52 ± 3.69	65.60 ± 6.66
CricketX	60.72 ± 2.35	55.18 ± 4.36	51.28 ± 3.40	30.15 ± 2.84	42.92 ± 1.99	46.50 ± 1.21	14.27 ± 1.71	45.38 ± 2.45	20.87 ± 2.94	45.38 ± 1.94	19.33 ± 2.95	**66.92 ± 2.71**
CricketY	56.92 ± 2.81	49.44 ± 1.94	47.69 ± 2.43	32.82 ± 4.21	42.15 ± 5.17	44.62 ± 0.44	9.32 ± 1.04	44.02 ± 2.09	17.90 ± 2.44	41.11 ± 0.53	23.95 ± 3.97	**57.95 ± 1.33**
CricketZ	63.18 ± 1.66	57.38 ± 2.58	48.82 ± 4.98	30.56 ± 3.52	43.18 ± 5.27	46.32 ± 1.71	12.91 ± 2.15	47.78 ± 1.41	18.26 ± 2.08	45.30 ± 1.50	21.95 ± 3.02	**65.30 ± 1.16**
Crop	57.92 ± 1.22	54.52 ± 1.67	57.92 ± 1.22	42.46 ± 2.10	44.96 ± 1.35	54.14 ± 2.07	52.84 ± 0.15	54.82 ± 1.79	54.74 ± 0.79	54.60 ± 1.45	25.02 ± 2.10	**57.97 ± 1.25**
DiatomSizeReduction	87.32 ± 3.50	71.05 ± 13.58	**96.80 ± 0.91**	44.77 ± 1.16	62.68 ± 4.64	87.04 ± 1.80	66.88 ± 2.86	51.96 ± 5.03	48.89 ± 18.63	87.47 ± 0.19	30.13 ± 0.15	92.61 ± 2.00
DistalPhalanxOutlineAgeGroup	68.20 ± 2.98	68.78 ± 4.24	70.50 ± 1.61	69.50 ± 1.66	69.50 ± 2.81	66.67 ± 1.10	65.71 ± 3.62	70.50 ± 1.25	68.78 ± 2.72	66.91 ± 1.25	48.06 ± 11.76	**70.65 ± 2.18**
DistalPhalanxOutlineCorrect	64.78 ± 12.49	65.07 ± 11.29	63.62 ± 12.87	50.00 ± 13.55	60.14 ± 9.92	58.94 ± 4.04	**67.15 ± 2.41**	55.43 ± 14.84	62.39 ± 9.46	61.71 ± 6.72	47.61 ± 8.80	60.51 ± 11.85
DistalPhalanxTW	55.83 ± 3.65	59.14 ± 2.12	**60.58 ± 3.58**	47.34 ± 6.42	57.55 ± 6.08	53.96 ± 3.14	53.00 ± 2.91	54.92 ± 0.83	52.52 ± 4.35	54.92 ± 2.91	32.37 ± 14.50	54.10 ± 3.11
DodgerLoopDay	49.25 ± 3.71	47.25 ± 4.28	39.75 ± 3.11	41.50 ± 2.24	47.00 ± 4.11	**49.58 ± 5.64**	28.75 ± 2.50	45.42 ± 1.91	29.00 ± 2.40	48.75 ± 1.25	47.00 ± 4.11	48.75 ± 1.77
DodgerLoopGame	68.70 ± 6.09	83.33 ± 5.28	75.07 ± 2.64	79.28 ± 0.97	**88.84 ± 1.67**	76.81 ± 1.26	48.79 ± 3.42	81.40 ± 1.51	53.48 ± 2.83	78.99 ± 1.45	**88.84 ± 1.67**	83.62 ± 0.65
DodgerLoopWeekend	94.06 ± 1.39	94.20 ± 2.11	96.81 ± 0.83	**97.83 ± 0.00**	96.81 ± 0.65	96.86 ± 0.42	72.71 ± 1.82	97.58 ± 0.42	76.38 ± 1.82	97.10 ± 0.00	96.81 ± 0.65	**97.83 ± 0.00**
ECG200	80.00 ± 4.00	**83.00 ± 2.00**	80.40 ± 4.28	77.60 ± 4.34	79.80 ± 6.53	79.33 ± 3.21	78.00 ± 3.61	81.00 ± 5.00	82.00 ± 2.92	82.00 ± 3.61	63.00 ± 15.26	77.40 ± 4.88
ECG5000	83.56 ± 5.09	78.23 ± 2.20	80.72 ± 4.37	**88.37 ± 1.22**	76.31 ± 5.83	79.59 ± 2.75	73.64 ± 2.21	84.75 ± 1.16	79.28 ± 5.11	82.19 ± 1.79	24.56 ± 15.15	73.78 ± 6.18
ECGFiveDays	95.40 ± 1.88	88.99 ± 4.44	**97.56 ± 1.20**	74.87 ± 4.61	94.61 ± 2.00	83.82 ± 7.05	96.44 ± 1.42	91.64 ± 1.90	97.51 ± 1.12	86.18 ± 8.29	57.47 ± 4.38	75.28 ± 1.71
EOGHorizontalSignal	30.61 ± 3.99	43.15 ± 5.84	30.94 ± 3.48	29.89 ± 1.60	38.78 ± 6.14	39.59 ± 1.67	8.01 ± 1.27	41.62 ± 1.42	10.61 ± 3.44	40.06 ± 3.72	39.56 ± 4.17	**49.17 ± 1.62**
EOGVerticalSignal	21.71 ± 2.53	28.23 ± 2.47	19.83 ± 3.08	31.55 ± 2.19	40.06 ± 1.98	36.00 ± 1.12	8.47 ± 0.84	37.75 ± 1.67	11.99 ± 1.33	37.02 ± 1.99	30.55 ± 3.17	**42.21 ± 2.26**
Earthquakes	53.81 ± 8.62	53.96 ± 9.50	51.37 ± 3.47	52.52 ± 6.49	54.82 ± 2.85	60.91 ± 4.63	49.16 ± 8.34	59.71 ± 6.86	54.68 ± 5.26	**61.87 ± 3.60**	47.91 ± 6.66	54.68 ± 7.63
ElectricDevices	40.12 ± 4.42	38.65 ± 3.96	36.00 ± 4.64	28.04 ± 1.63	39.08 ± 8.26	48.59 ± 6.21	30.07 ± 3.26	33.74 ± 3.43	37.44 ± 1.49	48.14 ± 2.51	16.08 ± 5.57	**49.42 ± 5.02**
EthanolLevel	26.56 ± 1.96	25.76 ± 1.05	26.16 ± 0.71	29.28 ± 1.78	25.92 ± 1.70	28.07 ± 2.55	25.20 ± 3.34	26.80 ± 0.69	25.60 ± 1.13	**29.87 ± 2.47**	25.56 ± 1.05	28.24 ± 0.50
FaceAll	77.73 ± 3.48	**80.30 ± 1.67**	77.73 ± 3.48	67.09 ± 2.40	54.49 ± 5.41	67.46 ± 1.82	59.68 ± 1.83	67.32 ± 2.82	60.69 ± 2.63	64.06 ± 2.32	32.89 ± 2.71	63.01 ± 1.34
FaceFour	87.95 ± 3.07	59.32 ± 7.51	**92.95 ± 2.03**	74.77 ± 0.95	66.36 ± 10.77	85.98 ± 1.31	65.91 ± 2.27	88.26 ± 0.66	69.55 ± 1.69	83.71 ± 1.31	34.55 ± 3.82	79.09 ± 1.90
FacesUCR	**87.80 ± 1.60**	84.26 ± 2.06	82.41 ± 2.48	63.80 ± 1.79	54.40 ± 3.05	80.54 ± 1.15	70.11 ± 1.05	83.69 ± 1.13	72.08 ± 1.16	79.51 ± 1.53	34.67 ± 3.28	79.08 ± 1.41
FiftyWords	45.36 ± 2.39	36.70 ± 1.89	45.36 ± 2.39	53.67 ± 1.47	50.46 ± 2.83	64.32 ± 1.87	16.85 ± 0.67	67.99 ± 1.84	37.05 ± 3.62	57.36 ± 1.54	56.40 ± 1.98	**71.34 ± 2.10**
Fish	**83.31 ± 3.78**	56.00 ± 4.28	71.54 ± 6.28	65.60 ± 3.49	62.74 ± 2.15	68.95 ± 4.29	24.38 ± 3.80	71.05 ± 1.44	57.14 ± 3.45	75.05 ± 2.31	38.17 ± 5.05	78.29 ± 3.92
FordA	78.26 ± 5.14	75.61 ± 0.71	**78.70 ± 3.26**	49.09 ± 0.93	67.88 ± 3.90	51.99 ± 2.52	52.22 ± 0.95	48.81 ± 1.31	53.05 ± 1.09	48.21 ± 1.67	50.06 ± 2.65	57.15 ± 1.85
FordB	**63.83 ± 5.74**	62.17 ± 3.33	61.70 ± 3.19	48.54 ± 0.72	59.06 ± 1.85	46.75 ± 0.61	49.79 ± 1.45	47.24 ± 1.24	50.47 ± 2.25	47.33 ± 0.14	49.33 ± 1.67	55.60 ± 1.91
FreezerRegularTrain	86.27 ± 5.59	**89.20 ± 7.73**	79.00 ± 5.98	76.33 ± 0.75	80.48 ± 6.98	76.36 ± 0.25	61.75 ± 7.11	76.22 ± 0.12	68.27 ± 6.23	74.90 ± 1.40	66.57 ± 12.21	**89.20 ± 5.65**
FreezerSmallTrain	78.20 ± 4.17	80.08 ± 7.30	74.46 ± 2.32	76.11 ± 0.45	68.74 ± 3.00	73.18 ± 1.45	65.15 ± 4.43	71.92 ± 2.00	68.25 ± 2.23	70.97 ± 2.25	65.53 ± 10.69	**82.86 ± 3.49**
Fungi	93.98 ± 3.37	38.06 ± 6.98	67.85 ± 1.59	67.74 ± 4.64	69.57 ± 7.88	**95.88 ± 1.64**	72.94 ± 1.89	91.22 ± 1.73	74.73 ± 0.76	**95.88 ± 0.31**	12.47 ± 9.15	79.57 ± 1.20
GestureMidAirD1	49.38 ± 3.66	60.92 ± 1.48	49.38 ± 3.66	49.69 ± 1.40	48.31 ± 2.52	54.36 ± 1.94	26.15 ± 2.31	58.21 ± 2.22	44.15 ± 6.68	57.44 ± 1.60	48.31 ± 2.52	**67.08 ± 1.00**
GestureMidAirD2	38.00 ± 5.29	47.23 ± 3.34	38.00 ± 5.29	50.46 ± 0.69	50.46 ± 2.75	53.08 ± 2.66	11.28 ± 2.47	55.13 ± 1.18	34.00 ± 4.05	51.79 ± 1.18	50.46 ± 2.75	**55.69 ± 1.85**
GestureMidAirD3	20.15 ± 2.40	25.23 ± 1.38	20.15 ± 2.40	32.92 ± 0.34	29.23 ± 3.17	28.46 ± 0.77	15.13 ± 2.70	33.08 ± 2.04	29.38 ± 4.79	31.28 ± 0.44	29.23 ± 3.17	**33.69 ± 2.63**

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Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
GesturePebbleZ1	76.86 ± 5.40	75.47 ± 8.13	77.67 ± 2.20	66.63 ± 2.78	68.84 ± 4.61	75.00 ± 1.16	22.29 ± 3.87	77.33 ± 2.10	38.95 ± 10.62	**81.40 ± 2.53**	68.84 ± 4.61	75.12 ± 5.19
GesturePebbleZ2	73.16 ± 4.51	63.42 ± 5.15	73.16 ± 2.82	60.38 ± 3.46	67.59 ± 4.18	69.62 ± 2.90	19.20 ± 0.73	74.26 ± 2.56	28.86 ± 9.85	73.00 ± 1.59	67.59 ± 4.18	**75.57 ± 4.64**
GunPoint	85.73 ± 9.32	82.13 ± 8.02	91.73 ± 11.16	68.27 ± 5.09	77.60 ± 4.39	71.78 ± 3.79	64.00 ± 6.57	72.00 ± 1.15	77.73 ± 11.90	76.67 ± 6.67	62.00 ± 16.53	**97.47 ± 2.23**
GunPointAgeSpan	89.56 ± 7.42	72.47 ± 9.28	74.87 ± 11.59	83.54 ± 2.13	82.22 ± 4.04	85.44 ± 1.67	52.22 ± 3.12	85.13 ± 2.39	71.08 ± 9.43	83.54 ± 3.02	53.61 ± 10.43	**92.72 ± 5.80**
GunPointMaleVersusFemale	92.78 ± 8.49	93.48 ± 5.65	92.22 ± 9.65	67.09 ± 4.97	95.57 ± 2.50	91.14 ± 4.91	60.97 ± 6.06	93.57 ± 3.96	74.05 ± 7.11	94.41 ± 2.22	67.22 ± 11.33	**98.54 ± 0.94**
GunPointOldVersusYoung	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	76.19 ± 1.96	79.17 ± 3.15	78.20 ± 2.30	57.46 ± 1.38	77.25 ± 5.77	61.90 ± 5.09	79.26 ± 2.47	85.78 ± 19.21	86.79 ± 4.71
Ham	60.76 ± 5.19	66.48 ± 2.64	65.52 ± 4.83	**71.24 ± 5.65**	62.86 ± 4.86	66.03 ± 7.40	63.49 ± 10.09	68.89 ± 6.69	61.71 ± 8.23	68.89 ± 7.02	61.52 ± 3.79	62.86 ± 3.37
HandOutlines	50.76 ± 12.61	50.92 ± 12.38	57.46 ± 2.17	81.57 ± 4.57	**82.65 ± 3.66**	81.62 ± 2.31	38.65 ± 4.09	78.47 ± 5.74	35.95 ± 0.00	81.62 ± 1.95	76.97 ± 8.74	71.51 ± 5.35
Haptics	29.68 ± 4.36	29.16 ± 4.62	33.31 ± 5.65	37.73 ± 3.74	31.95 ± 2.95	37.66 ± 2.89	23.81 ± 5.22	37.66 ± 2.77	25.45 ± 4.31	38.20 ± 2.76	35.32 ± 3.26	**39.22 ± 4.29**
Herring	55.00 ± 9.20	55.62 ± 5.91	55.00 ± 7.28	50.31 ± 6.57	**59.69 ± 3.39**	52.60 ± 3.93	47.92 ± 2.39	53.65 ± 8.02	49.06 ± 4.50	50.52 ± 1.80	57.19 ± 6.68	55.31 ± 5.01
HouseTwenty	**90.92 ± 1.10**	88.91 ± 1.82	89.58 ± 3.18	65.55 ± 5.48	81.51 ± 3.71	81.23 ± 2.43	60.50 ± 3.36	74.79 ± 1.46	64.20 ± 2.95	81.79 ± 4.63	70.76 ± 6.04	89.24 ± 2.81
InlineSkate	30.04 ± 2.77	24.04 ± 1.78	21.27 ± 3.03	25.16 ± 1.55	21.67 ± 3.27	28.42 ± 1.51	16.91 ± 0.91	28.73 ± 0.83	18.22 ± 1.24	29.09 ± 2.20	22.62 ± 1.07	**33.49 ± 2.77**
InsectEPGRegularTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	54.62 ± 4.82	79.12 ± 3.30	61.31 ± 3.90	42.70 ± 1.23	60.37 ± 1.29	55.18 ± 4.83	64.39 ± 5.02	96.63 ± 7.54	86.59 ± 4.06
InsectEPGSmallTrain	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	60.00 ± 1.05	81.69 ± 6.46	69.08 ± 1.20	55.82 ± 1.06	69.88 ± 0.80	58.71 ± 4.87	66.40 ± 1.81	83.13 ± 0.00	85.06 ± 2.08
InsectWingbeatSound	34.13 ± 0.95	34.52 ± 3.87	30.58 ± 3.83	60.48 ± 0.59	47.36 ± 1.34	60.99 ± 1.05	39.46 ± 3.03	**61.67 ± 1.18**	49.66 ± 3.26	61.20 ± 0.34	46.64 ± 2.91	51.36 ± 2.05
ItalyPowerDemand	92.21 ± 1.02	92.56 ± 1.30	92.75 ± 1.07	53.43 ± 4.80	93.97 ± 0.64	87.66 ± 8.42	94.69 ± 1.03	95.59 ± 0.11	**95.82 ± 0.32**	94.33 ± 0.49	49.23 ± 14.84	92.19 ± 2.62
LargeKitchenAppliances	64.05 ± 4.75	66.13 ± 5.93	66.08 ± 5.44	40.85 ± 3.41	**67.95 ± 4.05**	46.22 ± 6.49	39.82 ± 1.74	43.11 ± 5.30	40.53 ± 2.82	45.33 ± 5.07	41.71 ± 0.52	55.09 ± 5.30
Lightning2	70.49 ± 3.84	64.92 ± 7.48	65.57 ± 4.34	58.03 ± 2.74	69.18 ± 5.84	71.04 ± 1.89	51.37 ± 4.13	63.93 ± 3.28	58.36 ± 5.26	68.85 ± 4.34	60.00 ± 3.77	**77.05 ± 4.18**
Lightning7	64.38 ± 5.97	66.03 ± 3.41	60.00 ± 4.37	60.27 ± 3.36	61.10 ± 9.76	74.43 ± 1.58	20.09 ± 4.81	65.75 ± 1.37	36.71 ± 3.41	**75.34 ± 2.37**	26.30 ± 5.76	66.85 ± 7.34
Mallat	68.22 ± 8.96	69.46 ± 5.32	**95.78 ± 0.57**	90.38 ± 0.87	87.49 ± 9.31	92.79 ± 0.83	66.92 ± 5.03	85.20 ± 1.21	79.12 ± 6.08	89.69 ± 0.98	25.09 ± 0.73	89.72 ± 1.50
Meat	53.00 ± 18.31	59.33 ± 6.08	85.00 ± 4.86	**87.00 ± 7.94**	60.00 ± 5.27	61.11 ± 4.19	68.89 ± 7.52	55.56 ± 19.25	55.33 ± 20.66	86.11 ± 7.88	38.67 ± 15.92	64.00 ± 6.08
MedicalImages	**56.61 ± 3.74**	54.08 ± 3.55	52.39 ± 5.17	34.84 ± 5.66	48.24 ± 3.52	46.27 ± 5.02	40.00 ± 3.41	45.09 ± 2.04	47.45 ± 1.06	47.72 ± 6.48	18.58 ± 8.91	54.68 ± 2.10
MelbournePedestrian	**84.60 ± 1.98**	81.55 ± 1.62	80.47 ± 2.59	30.74 ± 5.50	68.14 ± 1.82	73.08 ± 1.13	75.36 ± 1.75	78.01 ± 1.38	77.48 ± 0.93	76.77 ± 0.49	68.14 ± 1.82	70.88 ± 1.49
MiddlePhalanxOutlineAgeGroup	44.29 ± 12.74	54.68 ± 6.95	55.19 ± 4.97	54.16 ± 10.18	53.51 ± 5.82	**58.66 ± 1.35**	50.43 ± 7.47	54.11 ± 4.79	50.13 ± 8.45	56.49 ± 1.12	41.82 ± 18.66	51.04 ± 7.11
MiddlePhalanxOutlineCorrect	60.00 ± 5.84	**65.02 ± 2.98**	60.62 ± 1.86	54.23 ± 4.23	56.36 ± 2.69	50.86 ± 7.18	64.72 ± 6.49	54.98 ± 7.77	58.97 ± 7.04	52.23 ± 9.34	52.44 ± 10.59	59.18 ± 5.65
MiddlePhalanxTW	**49.74 ± 5.84**	48.05 ± 4.11	49.48 ± 2.12	45.71 ± 2.70	46.23 ± 6.05	46.10 ± 6.40	37.23 ± 5.86	42.64 ± 7.61	35.97 ± 2.62	41.34 ± 8.07	23.64 ± 8.52	44.42 ± 6.27
MixedShapesRegularTrain	67.65 ± 7.57	70.16 ± 5.08	72.53 ± 6.92	70.49 ± 2.59	73.35 ± 4.70	73.87 ± 1.26	24.40 ± 1.33	74.08 ± 2.13	28.76 ± 6.79	73.65 ± 2.23	63.69 ± 2.98	**79.60 ± 1.74**
MixedShapesSmallTrain	70.61 ± 10.30	69.46 ± 11.49	75.93 ± 3.55	73.56 ± 2.63	73.16 ± 8.70	78.08 ± 2.18	23.04 ± 1.80	77.03 ± 1.02	27.06 ± 6.78	77.29 ± 1.92	65.95 ± 4.04	**81.61 ± 2.47**
MoteStrain	89.38 ± 0.88	89.14 ± 3.04	90.18 ± 0.68	86.26 ± 0.20	86.58 ± 1.30	87.19 ± 1.69	77.40 ± 1.04	87.43 ± 0.48	85.32 ± 0.75	**90.65 ± 0.48**	67.41 ± 11.72	89.23 ± 1.53
NonInvasiveFetalECGThorax1	55.91 ± 3.81	50.28 ± 5.45	55.91 ± 3.81	**81.90 ± 0.59**	63.56 ± 4.40	77.13 ± 1.08	36.83 ± 0.46	78.18 ± 0.43	46.96 ± 3.44	75.71 ± 1.08	58.08 ± 1.54	76.84 ± 1.40
NonInvasiveFetalECGThorax2	62.50 ± 1.19	67.13 ± 2.71	62.50 ± 1.19	**86.79 ± 1.25**	75.21 ± 2.50	83.55 ± 0.48	57.79 ± 2.45	83.27 ± 1.00	69.23 ± 2.65	80.68 ± 0.66	70.22 ± 2.79	82.53 ± 1.33
OSULeaf	**72.64 ± 5.62**	69.26 ± 4.17	69.01 ± 3.75	34.88 ± 3.54	39.34 ± 3.85	46.42 ± 2.52	19.83 ± 4.60	40.50 ± 2.19	29.83 ± 2.82	37.19 ± 2.07	28.10 ± 5.12	59.42 ± 3.58
OliveOil	56.00 ± 14.79	40.00 ± 0.00	**66.67 ± 9.72**	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	45.56 ± 9.62	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00	40.00 ± 0.00
PLAID	28.04 ± 4.01	25.14 ± 3.42	22.46 ± 6.07	34.45 ± 1.13	54.53 ± 5.93	50.16 ± 2.34	32.59 ± 0.99	44.82 ± 3.36	37.77 ± 2.12	46.80 ± 0.84	54.53 ± 5.93	**59.89 ± 3.28**
PhalangesOutlinesCorrect	56.88 ± 6.98	56.74 ± 8.68	57.48 ± 7.09	54.27 ± 11.23	49.91 ± 6.62	**58.78 ± 4.01**	55.79 ± 3.74	55.36 ± 17.14	52.42 ± 4.71	54.23 ± 12.22	53.99 ± 12.03	55.73 ± 8.07
Phoneme	**28.10 ± 1.26**	25.96 ± 1.60	**28.10 ± 1.26**	8.58 ± 0.50	11.88 ± 0.54	13.82 ± 0.86	9.55 ± 1.16	11.71 ± 0.45	10.91 ± 1.85	10.88 ± 0.47	10.94 ± 0.56	21.95 ± 1.27
PickupGestureWiimoteZ	50.80 ± 15.01	68.40 ± 2.61	**73.60 ± 2.19**	51.60 ± 1.67	54.40 ± 6.84	56.67 ± 3.06	26.67 ± 1.15	54.67 ± 1.15	42.00 ± 5.10	60.67 ± 1.15	54.40 ± 6.84	70.40 ± 2.61
PigAirwayPressure	26.35 ± 2.05	12.88 ± 0.53	**50.58 ± 0.92**	4.81 ± 0.00	4.13 ± 0.43	4.81 ± 0.48	2.08 ± 1.00	6.25 ± 1.27	4.62 ± 0.87	6.73 ± 0.48	5.00 ± 0.65	9.62 ± 0.48
PigArtPressure	54.71 ± 1.57	25.19 ± 5.85	**69.81 ± 3.29**	8.27 ± 0.40	6.06 ± 2.62	8.97 ± 0.28	6.41 ± 1.00	9.78 ± 0.73	6.35 ± 0.92	8.65 ± 1.27	18.08 ± 2.06	31.73 ± 1.92
PigCVP	28.85 ± 2.68	19.62 ± 1.84	**61.35 ± 2.42**	7.12 ± 0.40	3.27 ± 1.61	7.37 ± 0.73	7.05 ± 1.21	8.33 ± 1.69	4.13 ± 0.80	5.93 ± 1.11	11.63 ± 1.24	19.52 ± 1.34
Plane	**100.00 ± 0.00**	99.43 ± 1.28	**100.00 ± 0.00**	96.57 ± 1.28	96.95 ± 1.24	98.10 ± 0.00	98.73 ± 0.55	97.14 ± 0.95	98.48 ± 0.52	97.14 ± 0.95	49.71 ± 5.36	99.24 ± 1.24
PowerCons	84.00 ± 3.74	85.67 ± 2.68	**88.11 ± 1.91**	73.22 ± 7.34	83.67 ± 3.78	85.93 ± 2.85	53.33 ± 0.56	86.30 ± 2.25	59.67 ± 5.39	84.63 ± 0.85	74.33 ± 14.06	82.33 ± 5.46
ProximalPhalanxOutlineAgeGroup	69.07 ± 5.10	80.49 ± 4.77	80.98 ± 2.76	80.98 ± 2.95	75.90 ± 8.78	81.30 ± 5.55	81.14 ± 3.66	**81.63 ± 4.89**	76.88 ± 9.31	77.56 ± 6.40	68.78 ± 16.36	79.80 ± 3.30
ProximalPhalanxOutlineCorrect	63.71 ± 11.74	64.60 ± 2.91	63.85 ± 3.21	56.01 ± 10.53	61.86 ± 4.47	64.60 ± 0.69	**66.55 ± 3.51**	64.26 ± 0.69	61.86 ± 6.47	65.98 ± 3.31	51.34 ± 15.85	60.07 ± 4.20
ProximalPhalanxTW	60.78 ± 11.91	70.05 ± 3.85	**72.39 ± 1.88**	51.12 ± 7.39	70.73 ± 1.42	64.07 ± 2.20	60.16 ± 2.69	65.53 ± 2.25	57.95 ± 5.18	59.84 ± 3.91	25.07 ± 15.12	69.66 ± 3.47
RefrigerationDevices	42.67 ± 7.00	44.27 ± 6.42	43.41 ± 5.31	34.72 ± 1.59	37.71 ± 6.85	44.00 ± 2.97	34.40 ± 1.33	34.76 ± 2.56	35.31 ± 4.05	**44.62 ± 2.22**	32.37 ± 1.67	41.23 ± 4.90
Rock	41.60 ± 6.84	34.40 ± 4.56	36.00 ± 1.41	81.60 ± 0.89	57.20 ± 9.01	82.67 ± 1.15	84.67 ± 6.11	**88.00 ± 0.00**	62.00 ± 6.16	86.67 ± 3.06	30.80 ± 4.82	76.40 ± 2.97
ScreenType	35.84 ± 5.22	36.48 ± 5.07	37.23 ± 5.64	36.37 ± 2.19	39.15 ± 5.32	**40.53 ± 1.87**	35.73 ± 2.12	36.09 ± 2.74	33.12 ± 2.51	39.82 ± 2.27	36.32 ± 3.91	37.12 ± 3.93
SemgHandGenderCh2	63.47 ± 5.58	**74.93 ± 6.24**	65.43 ± 6.18	67.83 ± 4.45	66.37 ± 2.41	62.89 ± 2.12	51.67 ± 5.43	65.78 ± 6.45	55.43 ± 2.94	60.72 ± 4.24	58.77 ± 7.13	63.80 ± 10.73
SemgHandMovementCh2	37.69 ± 5.41	36.27 ± 1.15	37.82 ± 1.58	37.42 ± 2.10	32.22 ± 2.74	30.67 ± 2.78	25.04 ± 0.56	34.52 ± 3.16	25.69 ± 1.85	33.04 ± 3.33	33.64 ± 4.19	**45.56 ± 2.93**

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Table 6: Per-dataset 10-shot classification results on 128 shared UCR datasets (continued).

Dataset	ResNet	TapNet	COSCO-ResNet	DLinear	PatchTST	TimesNet	Autoformer	Crossformer	FEDformer	Informer	GPT4TS	Ours
SemgHandSubjectCh2	43.56 ± 2.78	45.20 ± 5.06	40.22 ± 5.31	62.80 ± 4.54	54.13 ± 4.47	60.37 ± 0.71	34.22 ± 3.27	**68.07 ± 2.65**	39.96 ± 2.48	60.96 ± 1.10	38.18 ± 2.42	66.71 ± 4.45
ShakeGestureWiimoteZ	82.00 ± 6.16	91.20 ± 2.28	88.80 ± 1.10	55.60 ± 2.97	68.40 ± 7.67	60.00 ± 4.00	33.33 ± 3.06	62.67 ± 3.06	45.20 ± 7.29	56.67 ± 1.15	68.40 ± 7.67	**91.60 ± 2.97**
ShapeletSim	86.33 ± 6.80	**100.00 ± 0.00**	**100.00 ± 0.00**	48.78 ± 0.82	95.22 ± 4.26	59.07 ± 5.56	48.70 ± 1.40	50.19 ± 2.80	51.00 ± 1.98	49.81 ± 1.95	52.33 ± 4.64	96.67 ± 1.96
ShapesAll	68.60 ± 2.31	66.63 ± 1.82	68.60 ± 2.31	62.47 ± 0.34	65.47 ± 2.49	72.11 ± 0.69	12.89 ± 4.16	73.28 ± 0.38	31.00 ± 5.80	65.56 ± 1.08	70.00 ± 1.20	**83.13 ± 0.32**
SmallKitchenAppliances	**67.20 ± 8.07**	61.87 ± 4.21	64.69 ± 7.08	36.59 ± 1.57	60.00 ± 3.85	55.11 ± 4.59	41.51 ± 2.40	38.76 ± 2.56	47.25 ± 3.08	51.02 ± 1.54	41.33 ± 5.03	57.60 ± 5.56
SmoothSubspace	**92.00 ± 1.56**	87.07 ± 3.67	**92.00 ± 3.02**	38.13 ± 6.59	67.73 ± 10.31	79.56 ± 2.78	80.67 ± 0.67	60.00 ± 11.02	86.13 ± 3.38	90.67 ± 4.00	43.60 ± 9.96	89.47 ± 2.60
SonyAIBORobotSurface1	95.11 ± 2.42	**96.01 ± 0.87**	92.45 ± 1.40	42.96 ± 0.07	87.95 ± 1.92	64.28 ± 1.83	67.05 ± 1.01	76.43 ± 0.69	64.86 ± 0.43	67.83 ± 1.06	45.09 ± 4.84	86.09 ± 3.97
SonyAIBORobotSurface2	**96.73 ± 3.11**	92.82 ± 3.91	94.67 ± 2.40	77.50 ± 0.71	90.95 ± 2.30	81.25 ± 2.70	83.07 ± 2.21	86.08 ± 2.22	82.71 ± 1.50	85.73 ± 2.18	61.15 ± 12.88	84.11 ± 2.29
StarLightCurves	71.47 ± 14.56	82.56 ± 3.87	75.10 ± 5.06	74.16 ± 3.22	77.51 ± 4.09	75.29 ± 4.53	29.60 ± 9.11	76.31 ± 3.24	30.20 ± 4.95	74.55 ± 5.89	70.71 ± 14.11	**85.26 ± 3.54**
Strawberry	79.73 ± 8.80	76.70 ± 3.26	76.70 ± 4.55	60.81 ± 1.94	79.89 ± 6.30	61.98 ± 1.58	**80.63 ± 1.13**	59.10 ± 1.09	77.51 ± 4.36	68.11 ± 2.15	50.54 ± 10.66	75.84 ± 0.99
SwedishLeaf	**88.03 ± 1.39**	83.68 ± 1.26	**88.03 ± 1.39**	66.11 ± 1.36	73.47 ± 2.54	75.95 ± 1.36	60.69 ± 4.66	81.39 ± 2.49	68.80 ± 1.87	74.72 ± 1.82	28.96 ± 3.20	85.89 ± 1.27
Symbols	**93.83 ± 2.55**	89.89 ± 7.08	92.52 ± 1.13	75.94 ± 0.93	85.87 ± 1.87	83.32 ± 0.52	34.77 ± 0.30	85.96 ± 1.03	61.21 ± 7.54	82.51 ± 0.70	21.11 ± 2.93	89.33 ± 0.82
SyntheticControl	95.60 ± 1.14	**97.87 ± 0.61**	97.60 ± 0.43	59.07 ± 3.98	94.20 ± 1.64	90.78 ± 1.64	59.44 ± 1.95	79.44 ± 4.86	68.40 ± 3.43	90.11 ± 1.35	22.60 ± 7.62	97.33 ± 0.85
ToeSegmentation1	**93.25 ± 1.60**	89.56 ± 3.33	91.58 ± 1.68	55.53 ± 3.14	68.95 ± 9.85	56.73 ± 5.97	50.29 ± 6.71	55.85 ± 3.55	49.39 ± 2.18	55.70 ± 4.98	50.53 ± 1.40	80.53 ± 2.75
ToeSegmentation2	81.38 ± 10.25	**87.08 ± 4.53**	85.85 ± 2.64	50.15 ± 3.62	73.38 ± 4.70	68.21 ± 10.44	56.67 ± 7.86	58.46 ± 6.30	46.77 ± 3.41	61.54 ± 3.35	43.54 ± 14.75	80.31 ± 4.70
Trace	92.20 ± 4.60	**100.00 ± 0.00**	95.20 ± 5.07	55.80 ± 5.72	86.20 ± 10.35	72.67 ± 1.53	55.67 ± 4.51	61.33 ± 4.62	74.80 ± 3.27	78.00 ± 4.36	46.80 ± 8.17	**100.00 ± 0.00**
TwoLeadECG	**99.86 ± 0.08**	98.79 ± 1.25	97.56 ± 2.37	58.05 ± 1.68	91.89 ± 5.00	70.85 ± 3.05	75.62 ± 1.63	65.26 ± 1.91	78.86 ± 1.56	78.99 ± 2.23	54.01 ± 3.73	99.68 ± 0.24
TwoPatterns	70.89 ± 4.77	77.56 ± 2.96	**78.86 ± 3.59**	42.28 ± 1.08	74.73 ± 6.48	42.55 ± 1.37	26.22 ± 0.85	42.35 ± 1.82	29.82 ± 3.35	37.95 ± 1.72	25.60 ± 4.32	70.06 ± 0.70
UMD	92.78 ± 3.35	98.33 ± 1.05	**98.75 ± 0.58**	64.17 ± 3.05	91.53 ± 4.77	76.62 ± 2.44	75.93 ± 7.23	89.58 ± 4.22	93.06 ± 1.47	87.27 ± 3.21	47.50 ± 8.96	98.67 ± 7.42
UWaveGestureLibraryAll	48.45 ± 3.31	48.15 ± 2.07	49.06 ± 3.81	80.57 ± 3.71	68.35 ± 4.39	84.19 ± 2.90	27.54 ± 1.68	**84.63 ± 3.19**	42.77 ± 5.45	83.32 ± 3.39	81.20 ± 2.71	52.41 ± 1.70
UWaveGestureLibraryX	51.98 ± 1.26	57.08 ± 2.26	52.99 ± 2.93	60.99 ± 2.04	57.69 ± 1.76	**62.37 ± 1.85**	17.30 ± 0.19	62.30 ± 0.55	36.29 ± 8.88	61.26 ± 1.95	56.81 ± 2.90	53.50 ± 1.91
UWaveGestureLibraryY	45.58 ± 1.39	47.42 ± 0.67	46.02 ± 2.85	55.44 ± 1.29	53.52 ± 3.23	**56.83 ± 0.99**	23.04 ± 3.42	56.65 ± 1.12	32.16 ± 4.98	55.22 ± 1.56	46.61 ± 1.40	52.03 ± 0.97
UWaveGestureLibraryZ	48.34 ± 2.78	51.60 ± 2.75	44.60 ± 1.67	53.18 ± 1.58	53.45 ± 2.14	54.99 ± 0.25	20.01 ± 2.19	54.36 ± 2.47	34.40 ± 4.42	51.61 ± 2.72	46.80 ± 1.81	**60.65 ± 2.84**
Wafer	84.60 ± 6.19	78.03 ± 11.28	81.53 ± 4.78	69.44 ± 4.07	**84.77 ± 11.10**	82.09 ± 7.52	76.30 ± 12.19	83.57 ± 7.70	81.81 ± 10.33	81.67 ± 9.12	78.96 ± 10.53	82.53 ± 8.58
Wine	**68.15 ± 6.33**	52.59 ± 5.80	64.07 ± 8.03	50.74 ± 1.66	50.00 ± 0.00	58.64 ± 5.35	50.62 ± 1.07	50.00 ± 3.70	50.00 ± 0.00	62.96 ± 7.41	47.78 ± 4.97	49.26 ± 2.11
WordSynonyms	35.14 ± 2.03	30.72 ± 3.50	35.14 ± 2.03	35.86 ± 2.07	39.44 ± 4.12	51.78 ± 3.11	15.15 ± 0.48	51.10 ± 4.21	25.64 ± 4.61	49.01 ± 4.47	33.92 ± 1.82	**58.90 ± 2.04**
Worms	50.39 ± 8.29	**51.95 ± 5.35**	45.71 ± 7.71	23.38 ± 5.11	39.74 ± 4.27	23.38 ± 6.49	26.84 ± 3.97	27.27 ± 9.09	25.45 ± 3.74	25.97 ± 9.09	20.78 ± 5.35	48.57 ± 4.91
WormsTwoClass	**64.94 ± 5.51**	63.12 ± 4.37	61.30 ± 7.20	49.09 ± 4.63	47.27 ± 1.97	50.65 ± 6.49	48.05 ± 4.50	48.05 ± 4.68	48.05 ± 3.56	51.95 ± 4.50	46.49 ± 5.15	56.36 ± 3.85
Yoga	59.91 ± 2.90	**59.97 ± 4.98**	59.88 ± 3.11	55.72 ± 1.71	57.07 ± 1.49	58.79 ± 3.68	53.03 ± 3.90	59.02 ± 1.75	51.43 ± 2.65	59.87 ± 3.99	51.89 ± 2.29	57.73 ± 2.05
Avg. Acc.	66.24	64.93	67.18	55.43	62.72	62.79	46.04	61.94	51.50	63.13	46.88	**69.16**
Avg. Rank	4.83	5.03	4.55	8.00	6.23	5.75	9.87	5.73	8.87	5.64	9.58	**3.91**
#Best	25	15	28	6	5	7	3	5	1	8	1	**40**

Table 7: Key few-shot adaptation settings for DuoTSP.

Component	Setting
Base LLM	`meta-llama/Llama-3.2-1B`
Vision backbone	`facebook/dinov2-base`
Pseudo-image construction	`tivit_sqrt_overlap`
Patch length / stride	16 / 8
Hidden size	128
Warm-up / joint epochs	5 / 55
Batch size / eval batch size	8 / 8
Gradient accumulation	4
LR (encoder / projector / LoRA)	2×10^{-4} / 1×10^{-4} / 1×10^{-4}
Tokenizer training mode	`class_rows`
Decoding	constrained label-token decoding
Freeze vision backbone	true
Data augmentation	none

B Additional Experimental Details

We summarize the key implementation choices of DuoTSP for the few-shot benchmark in Table 7. These settings are fixed across the main benchmark reported in Section 4.

C Additional Plots and Future Ablations

Figure 2 visualizes the shot-scaling trend over the 128 shared UCR datasets. DuoTSP remains the strongest method on the averaged benchmark curve across the entire few-shot range, with the largest absolute advantage appearing at 10-shot. We do not include incomplete ablation results in the current appendix; once the planned branch, pretraining, and decoding ablations are fully completed, we will append their summary tables here while keeping the detailed per-dataset breakdown in the appendix only.

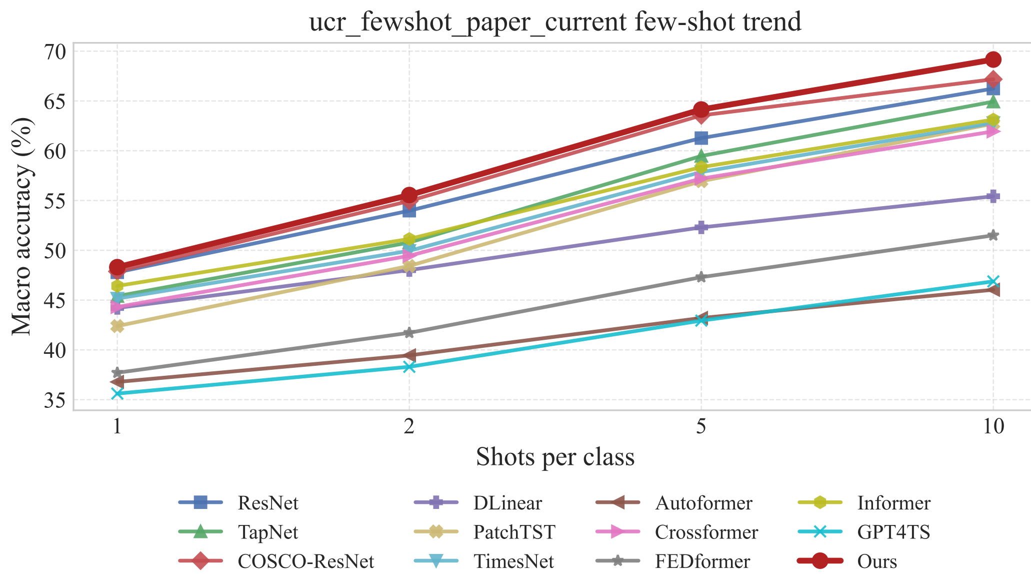

Figure 2: Shot-scaling trend on the 128 shared UCR datasets. Each point reports the mean accuracy over five runs. DuoTSP remains the strongest method across all four shot settings.

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